

UNIVERSITY OF
NOTRE DAME
2024-25 DBF DESIGN REPORT



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Acronyms and Symbols

α - Angle of Attack	M_1 - Mission 1 Flight Score
AR - Aspect Ratio	M_2 - Mission 2 Flight Score
AR_{VT} - Aspect Ratio of the Vertical Tail	M_3 - Mission 3 Flight Score
β - Sideslip Angle	MATLAB - Matrix Laboratory
b_W - Wing Span	M_{max} - Maximum Moment
b_{HT}, b_{VT} - Horizontal, Vertical Tail Spans	n - Load Factor
CA - Cyanoacrylate (glue)	N_c - Number of Aircraft Components
CAD - Computer Aided Design (Software)	NACA - National Advisory Committee for Aeronautics
CAM - Computer Aided Manufacturing	ND DBF - Notre Dame Design, Build, Fly
C_D - Coefficient of Drag	PCB - Printed Circuit Board
C_L - Coefficient of Lift	PID - Proportional Integral Derivative
$C_{L_{max}}$ - Maximum Coefficient of Lift	PWM - Pulse Width Modulation
$\bar{c}_W$ - Mean Aerodynamic Chord	Q&A - AIAA Rules Clarification Question and Answer
C_{HT}, C_{VT} - Horizontal, Vertical Tail Coefficients	RC - Radio Controlled
CF - Carbon Fiber	Re - Reynold's Number
CG - Center of Gravity	RK4 - 4th Order Runge-Kutta Method
C_m - Pitching Moment Coefficient	RPM - Revolutions Per Minute
C_{m_α} - Slope of the C_m vs α curve	S_W - Planform Area
CNC - Computerized Numerical Control	S_{HT} - Area of Horizontal Tail
D - Drag	S_{VT} - Area of Vertical Tail
DBF - Design, Build, Fly	SM - Static Margin
e - Oswald Efficiency Factor	T - Thrust
ElH - Engineering Innovation Hub	V - Airspeed
ESC - Electronic Speed Control	V_{climb} - Climb Velocity
FEA - Finite Element Analysis	V_{land} - Landing Velocity
FoM - Figures of Merit	V_s - Stall Speed
g - Acceleration Due to Gravity	V_{HT} - Horizontal Tail Volume Ratio
GM - Ground Mission	W - Weight
GM - Ground Mission Score	X_{CG} - x-location of Center of Gravity
GPS - Global Positioning System	X_{NP} - x-location of Neutral Point
I - Second Moment of Inertia	XFLR5 - Xfoil Low Reynolds Version 5
IMU - Inertial Measurement Unit	XPS - Extruded Polystyrene
L_{HT}, L_{VT} - Horizontal, Vertical Tail Moment Arm Distance	μ - Kinematic Viscosity
L - Lift	ϕ - Bank Angle
LiPo - Lithium Polymer	ρ - Density of Air
M1 - Mission 1	σ - Bending Stress
M2 - Mission 2	τ - Flap Effectiveness Parameter
M3 - Mission 3	$\Lambda_{t/c}$ - Sweep of the Maximum Thickness Over the Chord Line
	$\dot{\psi}$ - Turn Rate

1 Executive Summary

The following report outlines the process followed by the Notre Dame Design Build Fly team to prepare for entry into the 2025 American Institute of Aeronautics and Astronautics Design/Build/Fly Competition. The team has designed and built a radio-controlled aircraft and an X-1 test vehicle to complete three flight missions and one ground mission, as described in the AIAA competition rules [1]. To achieve this, the main aircraft was optimized to carry a large payload for Mission 2 and to have high endurance for Mission 3. The X-1 test vehicle was designed to execute a flight path to land in the maximum scoring zone.

The main aircraft features a low wing design with a puller configuration and tail dragger landing gear. The empennage consists of a conventional tail connected to a tapered section of the fuselage by a boom. The low wing design was selected to maximize maneuverability, while a dihedral wing angle was incorporated to improve stability. The single motor puller configuration was chosen for its efficiency and minimal interference with mission objectives. A tail dragger configuration was chosen for its reduced weight and drag, and the boom tail further minimizes weight and surface drag from the rear section of the aircraft. The fuselage cross section is a Norman window shape, providing both a simplified interface for external components and reduced drag with its curved roof. This configuration is shown in Figure 1. The fuselage and wings are primarily constructed from wood due to its lightweight properties and ease of manufacturing. Select structural components, such as the spars and boom tail, are made from carbon fiber, while the wingbox is machined from aluminum. Removable pylons can be attached to the wing tips to hold fuel tanks, which are loaded with a combination of concrete powder and steel weights. The X-1 test vehicle is attached to the bottom of the fuselage below the wingbox, with its vertical stabilizer passing through a bomb bay door in the fuselage floor. This location was chosen to minimize shifts in the aircraft's center of gravity when the vehicle is dropped. The X-1 test vehicle has a low wing configuration with a conventional tail, both chosen for their reliable stability and low weight, and is controlled by a custom system that rotates the wings to function as either ailerons or an elevator. Control inputs are informed by a GPS, an IMU, and a Pitot tube. The X-1 test vehicle is constructed from carbon fiber and foam to minimize weight.

Scoring analysis was conducted to determine the optimal aircraft configuration for maximizing payload weight, lap time for M2, and endurance for M3. Following scoring analysis and considering lessons learned from test flights, the team chose to carry a 8 lb payload and have a total flight time of 87 seconds for M2. For M3, the aircraft is designed to complete 9 laps, with the X-1 test vehicle weighing 0.29 lbs to aim for the 2.5-point bonus zone. The pylons and X-1 attachment mechanism were designed to minimize installation time for a higher GM score while remaining securely attached throughout the flight. To improve on detailed designs the team planned five flight tests of iterative aircraft. In addition, structural and propulsion tests were planned to inform performance of specific components. Early flight tests led to a major redesign to improve the landing performance of the aircraft. Subsequent tests have demonstrated that the current design track is able to perform and excel in all scoring missions.

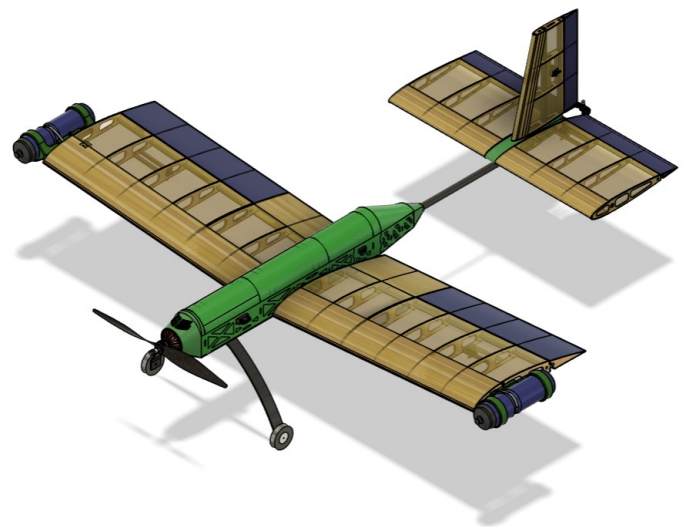


Figure 1: ND DBF 2025 Aircraft

2 Management Summary

2.1 Team Organization

The Notre Dame DBF team consists of approximately 25 undergraduate members, with about 15 returning and 10 new members. The overall organizational structure of the team is illustrated in Figure 2. The general membership of the team is divided into three main design squads: Aerodynamics, Structures and Manufacturing, and Payload. The Aerodynamics squad is responsible for the aerodynamic parameters of the plane, including lifting surface design, propulsion design, and stability analysis. The Structures and Manufacturing squad is responsible for the structural integrity of all components, fuselage design, and manufacturing process selection. The Payload squad is responsible for all payload systems,

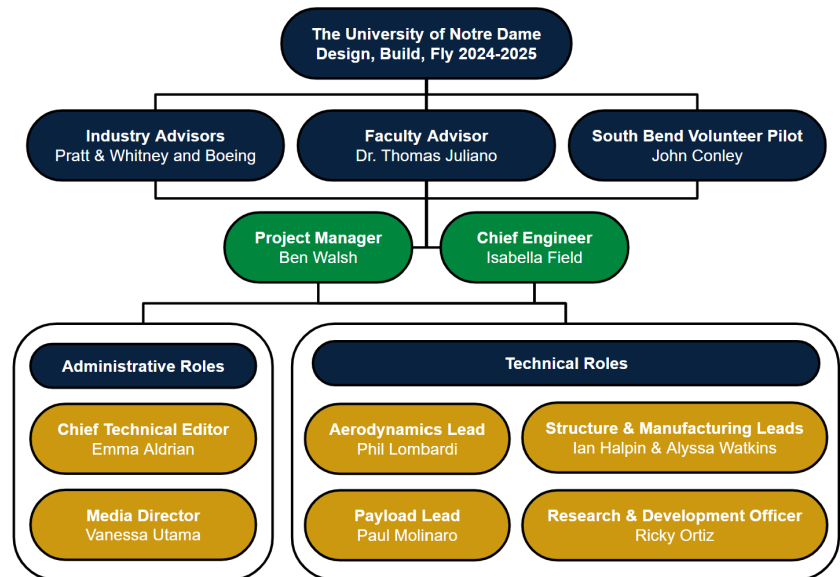


Figure 2: Team Organizational Structure

avionics, and mechanisms necessary for mission completion. It is notable to mention that the X-1 test vehicle design is shared between the three squads rather than being a payload-specific task. Each squad is run by a lead who is responsible for conducting meetings and ensuring that squad-specific design requirements are met in a timely manner. The Structures and Manufacturing squad is managed by co-leads to handle the large number of responsibilities and to provide mutual support and oversight. Each squad lead is supported by sub-leads, who act as assistants for leads instead of having specific tasks. Sub-leads help with both design and administrative tasks, ensuring the smooth operation of the squad and serving as reliable team members. All three squads are overseen by the Project Manager and the Chief Engineer. The Project Manager is responsible for keeping the team on schedule, within budget, and focused by setting team goals, managing budget constraints, organizing meetings, and purchasing supplies. The Chief Engineer oversees top-level requirements and makes all final design decisions. The Chief Engineer also provides design requirements to the squads, and the Project Manager ensures those requirements are efficiently completed. The whole team is advised by Industry and Faculty advisors during design reviews. Additionally, the team receives empirical feedback from a non-student pilot during test flights. The team also includes several auxiliary roles to address areas of past weaknesses. The Research and Development Officer is responsible for enhancing the team's performance potential by improving subsystems and processes. The Chief Technical Editor oversees the organization of all reports and proposals. The Media Editor manages the team's social media presence and recruits new members.

2.2 Milestones

The Project Manager holds the team to the schedule outlined in Figure 3. Squad tasks are completed during weekday meetings, and a full team meeting is held on Sundays to maintain communication between squads. The schedule has been adjusted throughout the year, mainly due to flight tests that resulted in minimal structural damage. This led to small repairs and flights being moved to the following week to collect more information.

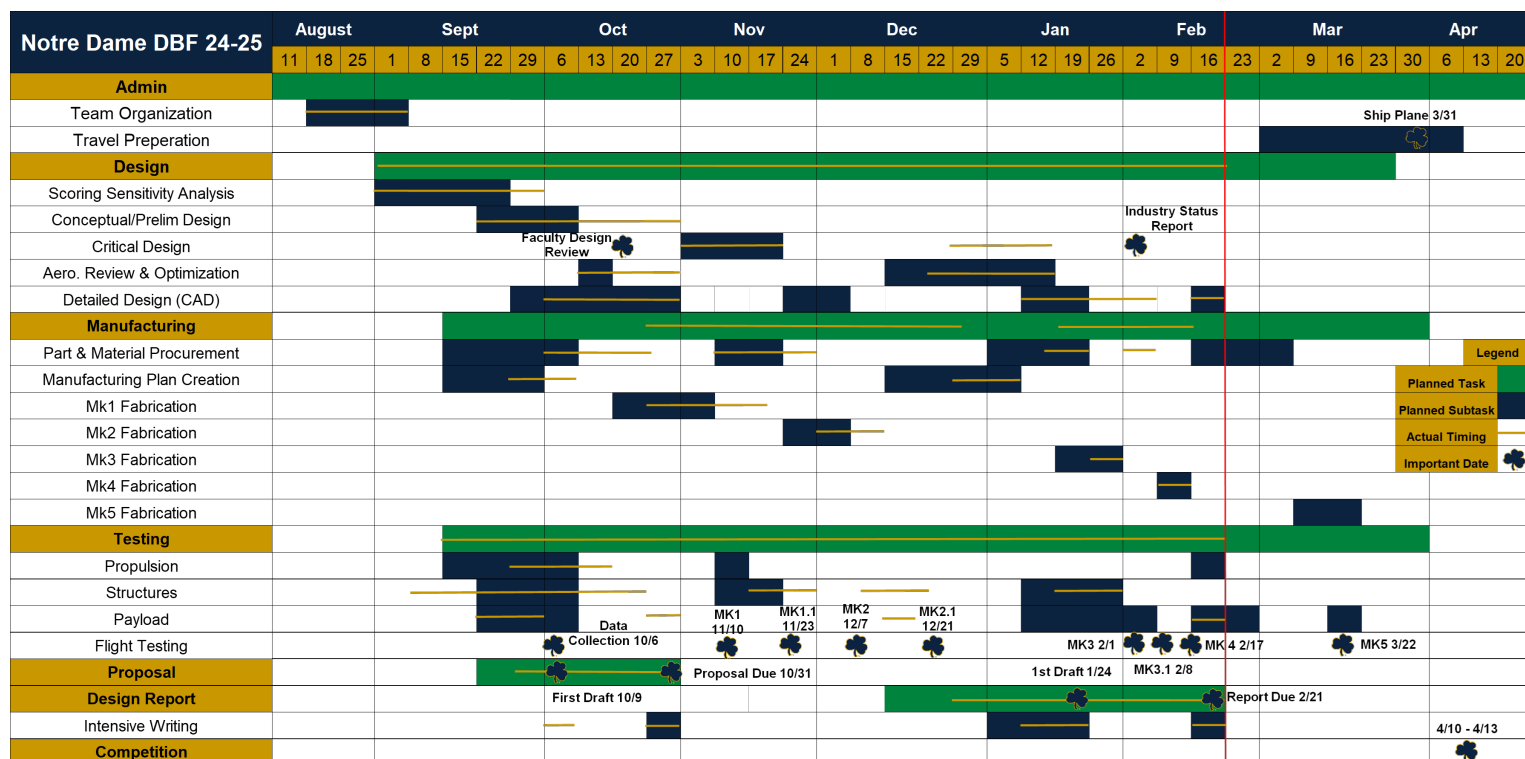


Figure 3: 2024-2025 Gantt Chart

3 Conceptual Design

3.1 Engineering Requirements

ND DBF's engineering requirements were developed using a careful analysis of mission requirements, obtained through a thorough review of the rules and Q&As released by AIAA.

3.1.1 Problem Statement

ND DBF must design, build, and fly an airplane to execute an X-1 Supersonic Flight Test Program, completing three assigned flight missions and a ground mission in order to ensure airworthiness, test payload capacity and speed, and demonstrate the ability to effectively deploy an X-1 test vehicle from the main aircraft.

3.1.2 General Aircraft Requirements and Restrictions

AIAA has provided comprehensive guidelines through the official rules and subsequent Q&A sessions. These guidelines outline design constraints, performance expectations, and payload regulations for the ND DBF team. Table 1 details the requirements that the team considers central to the aircraft design and their impact on each ND DBF squad.

Table 1: Mission Requirements and Squad Impact

Category	Requirement	Aerodynamics Squad	Struct. & Manu. Squad	Payload Squad
Aircraft Design	The maximum airplane wingspan is six feet.	Design tradeoff between wing area and aspect ratio.	The structure must endure a maximum span of six feet.	Minimal Implications.
	Aircraft must be capable of completing all three missions.	Plane aerodynamic design must allow completion of all three missions.	Plane structural design must allow completion of all three missions.	Payload design must allow completion of all three missions.
	Aircraft must be flown in same configuration for all three missions excluding payloads.	Aerodynamic design must complete all three missions without changes.	Structural design must complete all three missions without changes.	Payload design must complete all three missions without changes.
	No structure or components may be dropped from the plane during flight excluding the X-1 test vehicle in M3.	Minimal Implications.	No loose components are permitted.	X-1 test vehicle must be secured when attached to main aircraft.
	Commercially available propeller, battery, and motor must be used.	Only commercially available propulsion systems can be considered for the aerodynamic design.	Minimal Implications.	Minimal Implications.
	Aircraft must pass a wing tip test in the heaviest flight condition.	Minimal Implications.	Plane structural design must pass a wing tip test at maximum flight weight.	Minimal Implications.
	Aircraft can be any configuration other than rotary and lighter than air.	Aircraft must be designed with traditional fixed wings.	Minimal Implications.	Minimal Implications.
Assembly	Assembly work on the aircraft must be completed with the plane in the upright, parking position. The aircraft cannot be picked up, rotated, or put in any other position. The only exception to this is the ability to add blocks underneath the wheels of the landing gear to allow for access underneath the plane if need be.	Minimal Implications.	Structural members and hatches must be placed carefully to facilitate assembly and loading in the upright configuration.	Payload components must be designed and placed to facilitate assembly and loading in the upright configuration.
	Aircraft will enter the staging area in the parking configuration with the propulsion battery and payload removed. Aircraft assembly and payload installation must be completed in less than five minutes by one person.	Minimal Implications.	Structural members and hatches must be placed carefully to facilitate assembly in less than five minutes.	Payload components must be designed and placed to facilitate assembly and loading in less than five minutes.
Flight Performance	Aircraft must have a successful landing to receive a score for the mission.	Aircraft must be designed to land at a velocity that is controllable for the pilot	Structure must withstand the force of landing without failure	Minimal Implications
	The airplane will use ground rolling takeoff and landing.	Wing, propulsion system, and weights must be selected to allow for rolling takeoff. Must take off in length of testing runway.	Aircraft structures must withstand the forces of extended takeoff and landing.	Minimal Implications.
Payload	Fuel tanks can be any commercially available bottle as long as it has a screw-on lid. Bottle can be painted/decorated, but otherwise cannot be altered in any way. Bottle must have a minimum capacity of 16 oz. Bottle cannot be glass.	Bottle chosen must be as aerodynamic as possible.	Minimal Implications.	Fuel tank must be designed to meet the requirement.
	Fuel tank bottles can be filled with any amount of inert material to get the desired payload weight.	Design tradeoff between fuel tank weight and speed.	Minimal Implications.	Fuel tank must be strong enough to hold intended payload weight.
	At least two fuel tanks must be carried externally (minimum of 2 fuel tanks required). External fuel tanks must be identical. Internal fuel tank can differ from external. Internal fuel tank is optional.	Minimal Implications.	Fuselage and wings must be designed to withstand additional weight from fuel tanks.	Fuel tanks must be designed to be properly incorporated with aircraft, whether internal or external.
	External fuel tanks must be attached using detachable pylons. Pylons must hold fuel tank securely for all phases of flight.	Pylons should be as low-drag and light as possible since their weight will not affect mission score.	Minimal Implications.	Pylons must be secure enough to hold fuel tanks through all phases of flight. Also must design pylons to be detachable. Fuel tanks chosen must interface properly and securely with pylon.
	X-1 test vehicle must be attached to the plane and dropped at an altitude of 200-400 ft via command from the transmitter.	Minimal Implications.	Minimal Implications.	Must design a low-weight mechanism to hold and release the X-1 test vehicle on command/at altitude.
	Fuel tanks must be placed on either side of the X-1 test vehicle.	Minimal Implications.	Minimal Implications.	Fuel tank placement must be chosen properly to meet the requirement.
	After being dropped, the X-1 test vehicle must achieve stable flight after release from the airplane, execute a 180 degree turn, and then fly a descending pattern or orbit until landing on the ground.	X-1 test vehicle must be designed to achieve steady flight and required flight path.	Minimal Implications.	A mechanism for controlling the X-1 test vehicle must be designed to achieve flight requirements.
	X-1 test vehicle must have flashing lights/strobes which turn on only after release from plane and remain on through landing. Lights must be easily visible to flightline judge.	Minimal Implications.	X-1 test vehicle must protect necessary components for strobe light operation through landing.	X-1 test vehicle must be designed to meet this requirement.
	X-1 test vehicle cannot exceed 0.55 lbs in weight.	X-1 test vehicle aerodynamic surfaces must be light while still maintaining control authority.	Materials chosen for X-1 test vehicle must be light but strong enough to withstand flight loads.	X-1 test vehicle components and total weight must remain under 0.55 lbs in weight.

3.1.3 Mission Scoring Summary

AIAA has outlined the mission rules and scoring criteria. A brief description of the requirements for each mission, along with the associated scoring equations, is shown in Table 2.

Table 2: Mission Scoring Summary

Mission	Description	Scoring
GM	Record time to install pylons, X-1 test vehicle with attachment mechanism, and fuel tanks. Verify function of all control surfaces and X-1 test vehicle release function. Aircraft enters staging area in M1 configuration before timer begins.	$GM = \frac{\text{time}_{\min}}{\text{time}_{\text{team}}}$
M1	Delivery flight with no payload. Complete 3 laps of flight course within 5 minute time window. Complete a successful landing.	$M1 = 1.0$
M2	Captive carry flight with loaded fuel tanks and X-1 Vehicle payload. Complete 3 laps of flight course in shortest possible time. Carry X-1 and fuel tanks for duration of flight and successfully land.	$M2 = 1 + \frac{\left(\frac{\text{weight}_{\text{fuel}}}{\text{time}}\right)_{\text{team}}}{\left(\frac{\text{weight}_{\text{fuel}}}{\text{time}}\right)_{\text{max}}}$
M3	Launch flight with unloaded fuel tanks, X-1 test vehicle, and payload. Complete maximum number of laps in 5 minute window. At beginning of final lap drop X-1 test vehicle prior to the upwind turn. Once dropped X-1 test vehicle must complete a 180° turn before landing in bonus point area. X-1 test vehicle strobe light must be on during descent and after landing, but never before. Primary aircraft must complete a successful landing.	$M3 = 2 + \frac{\left(\text{laps flown} + \frac{\text{bonus box score}}{\text{X-1 weight}}\right)_{\text{team}}}{\left(\text{laps flown} + \frac{\text{bonus box score}}{\text{X-1 weight}}\right)_{\text{max}}}$

3.1.4 Competition Scoring Summary

The overall competition score is a combination of Mission Score, Report Score, and Participation Score, P , further detailed in Table 3. Of note is that the Total Mission Score is multiplied by the Report Score. The larger variance in Mission Score means that it has a larger effect, however, the Report Score acts as a multiplier on the performance of the team during the missions. The Total Report Score is split into components for the proposal and design report as described in Equation (2).

Table 3: Participation Scoring

Participation Score	Requirement
1	Attend Fly-off
2	Pass Tech Inspection
3	Attempt Flight Mission

$$\text{Competition Score} = \text{Total Report Score} \times \text{Total Mission Score} + P \quad (1)$$

$$\text{Total Report Score} = (0.15 \times \text{Proposal Score}) + (0.85 \times \text{Design Report Score}) \quad (2)$$

$$\text{Total Mission Score} = M_1 + M_2 + M_3 + GM \quad (3)$$

3.2 Scoring Sensitivity Analysis

After breaking down mission requirements into specific aircraft design criteria, the team analyzed this year's critical scoring parameters to optimize the wing and propulsion system for maximum performance. Using a MATLAB script, potential aircraft configurations were generated by fixing the wingspan at the maximum allowable 6 ft and iterating over the wing aspect ratio and payload weight [2]. Fixing the wingspan at 6 ft allowed for the design of efficient, high aspect ratio platforms, but was later altered to accommodate the payload fuel tanks being attached at the wingtips. To account for scoring dependence on landing location and weight, the X-1 test vehicle was treated as an additional 0.55 lbs in the total aircraft weight. The script estimated the structural and electronics weights based on previous DBF aircraft and selected a propulsion system using a custom database of manufacturer-provided data for motors, propellers, and batteries. Each configuration of aspect ratio, motor, propeller, and battery was required to achieve three primary goals:

1. Attain a minimum static thrust-to-weight ratio of 1.2.
2. Fly for at least 5.5 minutes.
3. Take off within 250 ft (the length of the team's test runway).

The thrust-to-weight ratio was chosen from a historical range of 0.8 to 1.2, with the upper limit selected to ensure sufficient thrust to offset the added drag from the fuel tanks. Configurations meeting these criteria were deemed minimally viable for the competition. Each viable configuration was mathematically simulated and scored using the M2 and M3 scoring equations, as well as performance equations from Thomas Corke's *Design of Aircraft* [3]. Figure 4 shows the flow of the scoring analysis script. The results, illustrated in Figures 5 and 6, show each point representing a viable aircraft and propulsion configuration.

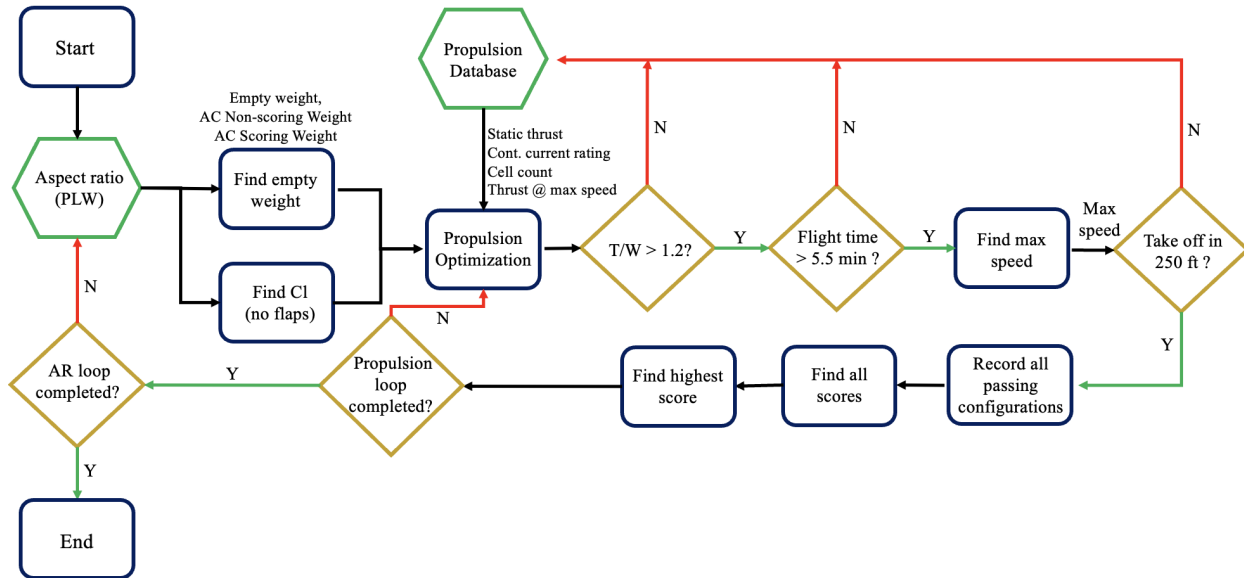


Figure 4: Flow Chart of ND DBF's Scoring Analysis Script

After preliminary testing flights with failures resulting from an excessively high required landing speed, it was determined that the original scoring strategy did not adequately take into account pilot controllability factors such as landing speed. Thus, the scoring analysis code was altered to implement penalties for high landing and takeoff speeds. This was done by requiring each configuration to have a maximum stall speed of 40 ft/s and capping the maximum wing area to 7.5 ft², both of which were determined through ND DBF's previous experience with successful flights, best manufacturing practices, and conventional wisdom regarding RC aircraft. Once these penalties were implemented to ensure the selected design would be flyable by a pilot, Figures 5 and 6 were created. Due to budget constraints, the team was unable to select a new motor for the aircraft and thus decided to map the current motor to a new, winning configuration. In both figures, each minimally viable aircraft is plotted by its maximum speed, determined by calculating the excess thrust of each aircraft and the speed at which it is minimized. For the selected aircraft, the maximum velocity for each mission is indicated by a red X. However, cruise speed values, defined as the flight speed when lift is equal to weight without any additional pitch control input, are generally much lower than the maximum speed. The cruise speeds for each mission were calculated using Equation (4), where lift is assumed to equal aircraft weight.

$$L = W = \frac{1}{2} \rho V^2 C_L S_W \quad (4)$$

These values are shown for the given aircraft in Figures 5 and 6 as a black X. These velocity values are low relative to the maximum velocity of the aircraft. However, based on previous year's flight data, the team has determined that, by taking advantage of pitch control and diving capability, the pilot can exceed this cruise value to roughly 95% of the maximum velocity of the aircraft. This value, marked by the golden star in Figures 5 and 6, is the predicted competition lap velocity and is used for a majority of the following analyses. The final results of the scoring analysis, including performance predictions and physical design parameters, are shown in Tables 4 and 5.

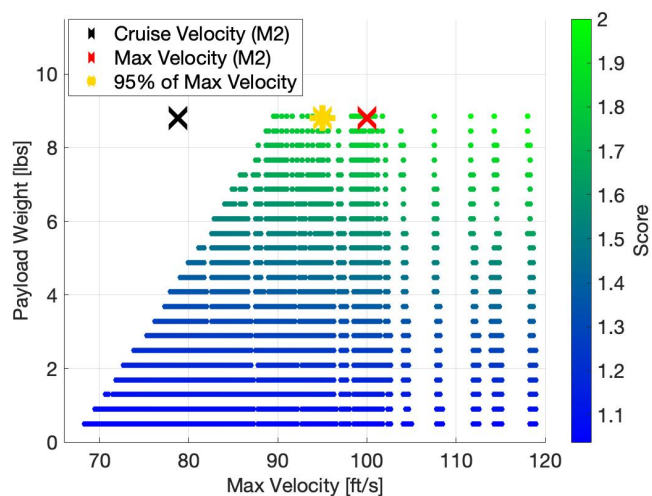


Figure 5: ND DBF M2 Scoring Analysis

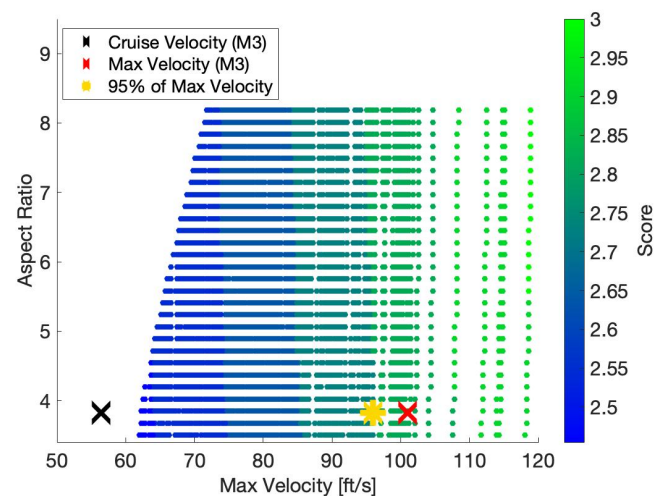


Figure 6: ND DBF M3 Scoring Analysis

Table 4: Preliminary Scoring Parameters

Parameter	Plane Weight (lbs)	Payload Weight (lbs)	Max Velocity (ft/s)	Cruise Velocity (ft/s)	Predicted Flight Velocity (ft/s)	Lap Time (s)
M1	9.09	0.00	101	53.0	96.0	24.5
M2	18.5	9.41	100	78.8	95.0	26.2
M3	9.59	0.55	101	56.5	96.0	24.6

Table 5: Preliminary Design Parameters

Aspect Ratio	Span (ft)	Motor	Propellor	Battery
3.82	5.33	BadAss 6245-215	BadAss 20x12-EW	MaxAmps 2250mAh 12S

3.3 Concepts Selection Process

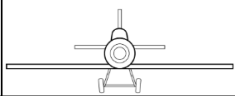
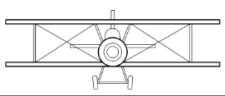
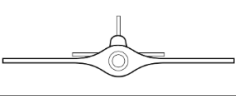
To select the overall aircraft configuration, including the wing and empennage designs, the team utilized weighted decision matrices to evaluate various options against Figures of Merit (FoM) determined by the design requirements. Every design option was rated on a scale of 1 to 5 for each FoM, with the highest total score, a sum of the individual FoM scores multiplied by their respective weights, representing the best choice. The assigned weight and scores were selected based on knowledge from the team's past airplane performance, additional research, and the performance of other teams' past aircraft, with the selected configuration for each decision matrix marked by a filled green box.

3.3.1 Overall Aircraft Configuration Trade Study

To ensure a successful design, it was crucial for the team to choose an overall configuration that was most suited for the challenges of this year's competition. This was accomplished using the trade study shown in Table 6. As such, out of the five FoM utilized in Table 6, mission capability (as determined by a configuration's ability to meet more specific design criteria outside of weight, drag, and maneuverability) was weighted the highest, followed closely by manufacturability to ensure prudent decisions were made about the team's capabilities. Three options were considered: a monoplane, a biplane, and a blended wing. While the blended wing performed marginally better in mission capability, it was deemed far more difficult to manufacture than other designs, limiting the team's ability for rapid prototyping. The biplane was far more comparable to the monoplane, but was not chosen due to its

lower drag and manufacturability scores. Instead, the monoplane configuration was chosen as a reliable, well-tested configuration that performed well across the board and lent itself more towards rapid prototyping opportunities with its greater manufacturability.

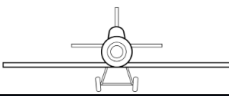
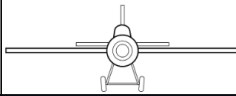
Table 6: Overall Aircraft Configuration

				
Figures of Merit	Weight (%)	Monoplane	Biplane	Blended Wing
Mission Capability	35%	2.0	1.0	3.0
Drag	15%	3.0	2.0	4.0
Weight	15%	4.0	4.0	5.0
Maneuverability	10%	3.0	4.0	2.0
Manufacturability	25%	5.0	3.0	1.0
Total:	100%	3.3	2.4	2.9

3.3.2 Wing Configuration

Table 7 includes the following wing configurations: high wing, low wing, and mid wing. Drag and weight were given the highest weightings in the FoM because of the effect of the external fuel tanks on the speed and flight capability of the plane in M2 and M3. Manufacturability was given equivalent importance to drag and weight because of the strict time considerations necessary to allow for frequent redesigns. The mid wing design was not selected because of manufacturability concerns and the additional weight it would require to provide support at the middle of the fuselage. While the high wing generally has a higher lift-to-drag ratio and better dynamic stability, the low wing was selected for its manufacturability and its greater maneuverability at high speeds.

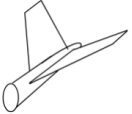
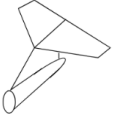
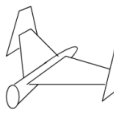
Table 7: Wing Configuration

				
Figures of Merit	Weight (%)	High Wing	Low Wing	Mid Wing
Mission Capability	15%	3.0	4.0	4.0
Drag	20%	4.0	4.0	5.0
Weight	20%	2.0	3.0	2.0
Maneuverability	10%	3.0	5.0	4.0
Stability	15%	5.0	3.0	4.0
Manufacturability	20%	5.0	5.0	4.0
Total:	100%	3.7	4.0	3.8

3.3.3 Empennage Configuration

Table 8 shows the five configurations considered for the main aircraft tail and the FoM used. The trade study considered five FoM, with mission capability, drag, and manufacturability weighted equally the highest. The V-tail had many benefits in terms of reducing drag, reducing weight, and increasing maneuverability, but, like the H-tail, there were concerns about the difficulties of manufacturing. Additionally, the cruciform, T-tail, and H-tail configurations would need more internal supports, and would therefore weigh more than a conventional tail. For these reasons, a conventional configuration was chosen for its high ease of manufacturability, mission capability, and reliable performance in stability and maneuverability.

Table 8: Empennage Configuration Trade Study

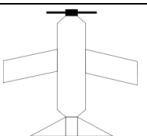
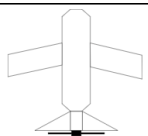
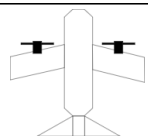
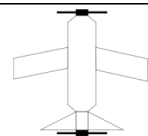
						
Figures of Merit	Weight (%)	V Tail	Cruciform	Conventional	T Tail	H Tail
Mission Capability	20%	4.0	3.0	4.0	4.0	3.0
Drag	20%	5.0	4.0	4.0	3.0	3.0
Weight	15%	5.0	2.0	3.0	2.0	2.0
Maneuverability	10%	5.0	3.0	4.0	3.0	3.0
Stability	15%	1.0	4.0	3.0	3.0	5.0
Manufacturability	20%	1.0	3.0	5.0	3.0	2.0
Total:	100%	3.4	3.2	3.9	3.1	3.0

3.4 Aircraft Component Selection, Processes, and Results

3.4.1 Propulsion

Four potential configurations, detailed in Table 9, were considered for the propulsion system. Five FoM were used to evaluate the configurations, with unique weightings attributed to each based on their importance to mission performance. A pusher configuration would weigh roughly the same as the puller, but the motor position would reduce the motor efficiency due to disturbed airflow from the fuselage and tail. Additionally, the pusher design came with concerns that a dropped X-1 test vehicle may collide with a propellor extending below the fuselage behind the X-1 test vehicle at high speeds. Additionally, the dual-motor and pusher-puller configurations introduce concerns with weight, as both would require a second motor with additional supports, as well as greater complexity in manufacturing which could result in added failure points during flights. Ultimately, a puller configuration was chosen due to its high efficiency, low weight, and ease of manufacturability. Although there were concerns with the induced drag of the design, these were outweighed by the aforementioned benefits.

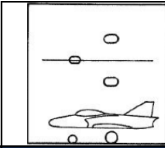
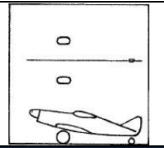
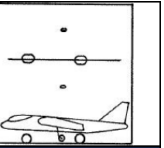
Table 9: Propulsion System Design Trade Study

					
Figures of Merit	Weight (%)	Puller	Pusher	Dual-Motor	Pusher-Puller
Mission Capability	20%	3.0	2.0	4.0	4.0
Drag	15%	2.0	4.0	3.5	3.5
Weight	15%	5.0	4.5	1.0	2.0
Efficiency	30%	5.0	2.0	5.0	3.0
Manufacturability	20%	4.0	3.0	2.0	3.0
Total:	100%	4.0	2.9	3.4	3.1

3.4.2 Landing Gear Configuration

Three landing gear configurations were considered for the main aircraft: tricycle, tail dragger, and bicycle. Initially, the tricycle configuration was chosen from the trade study due to its greater stability and the ease with which the payload could be mounted. Additionally, the idea of a retractable landing gear was pursued in order to reduce drag. However, due to difficulties with hard landings and the retracting mechanism, the team decided against the tricycle configuration and switched to a tail dragger. This configuration would weigh significantly less, offer reduced drag compared to a non-retractable tricycle gear, and protect the propeller from rough landings. All considerations that are reflected in the updated trade study are shown in Table 10, in which figures of merit have been reevaluated to reflect the team's findings.

Table 10: Landing Gear Configuration Trade Study

				
Figures of Merit	Weight (%)	Tricycle	Tail Dragger	Bicycle
Stability	25%	4.0	3.0	2.0
Drag	15%	2.0	4.0	4.0
Aircraft Integration	20%	3.0	4.5	2.0
Weight	20%	2.0	4.0	5.0
Payload Mounting	20%	5.0	2.0	3.0
Total:	100%	3.3	3.5	3.1

3.5 Payload Configuration Trade Studies

3.5.1 External Fuel Tank Properties

The team considered four shapes of bottles for the external fuel tanks shown in Table 11: standard ribbed, bullet, box, and tapered midsection. The FoM accounted for included aerodynamic efficiency, ease of attachment to the plane, and filling efficiency of each bottle. Of these, aerodynamic efficiency was determined to be of the greatest importance, with filling efficiency determined of the least importance. The standard ribbed bottle had average scores but did not perform the best in any category. The box bottle scored similarly; though its higher cross-sectional area resulted in a lower aerodynamic efficiency score. The bullet shaped bottle had high scores in aerodynamic efficiency and filling, but its smooth body limited interfacing ability with the pylon, causing its ease of attachment score to be lower. The tapered midsection bottle was ultimately selected due to its high aerodynamic efficiency score relative to the other bottles and because its tapered shape greatly increased its ease of attachment to the aircraft.

Table 11: External Fuel Tank Shape Trade Study

					
Figures of Merit	Weight (%)	Standard Ribbed	Bullet	Box	Tapered Midsection
Aerodynamic Efficiency	60%	3.5	5.0	3.0	4.5
Ease of Attachment	30%	4.0	3.0	3.5	5.0
Filling Efficiency	10%	3.5	5.0	4.0	3.5
Total:	100%	3.7	4.4	3.3	4.6

The bottle material was selected to be plastic instead of aluminum due to its lighter weight and greater commercial availability. The fuel tanks were filled with powdered concrete mix and steel weights evenly interspersed throughout. Initially, the team chose a design with the pylons on the underside of the wing at the wing tip. However, after further iteration and testing, a design that attaches to the wing tip, parallel to the chord line, was developed. This position was ultimately chosen as it greatly improves the ease with which the pylon can be removed.

3.5.2 X-1 Test Vehicle Control System

For the test vehicle control system used to initiate the strobe lights after release, perform a 180 degree turn, and land in the 2.5-point bonus box, the team considered an off-the-shelf autopilot module, a custom PCB and autopilot software, and an X-1 test vehicle with its control surfaces in a fixed position for a spiraling flight path, as shown in Table 12. Controllability, weight of the control system, manufacturability, and compatibility with the vehicle frame were the five FoM weighed against each system in

consideration. A custom PCB and software were chosen because they provided the most control over the vehicle's flight path, could be customized to fit any airframe, and the tools and technology required to manufacture and assemble the board were available at Notre Dame's EIH. While a fixed control surface aircraft would be the lightest solution, it promised no reliability with its flight path, with the risk of not completing the 180° turn requirement in the face of unpredictable wind gusts. The off-the-shelf autopilot module was not chosen because it would require the team to build the vehicle around the module's geometry, potentially adding weight. The lack of customizability also presents itself within the autopilot program, as the team would be limited by the provider's intended use of the autopilot. The custom PCB and software were ultimately chosen due to their customizability with the physical system and autopilot, despite the added complexity of fabrication and assembly.

Table 12: X-1 Test Vehicle Control System Trade Study

Figures of Merit	Weight (%)	Off-The-Shelf Autopilot Module	Custom PCB & Software	Fixed Control Surfaces
Controlability	25%	4.0	5.0	1.0
Weight	30%	3.0	4.0	5.0
Manufacturability	25%	4.0	2.5	5.0
Compatability	20%	2.0	5.0	5.0
Total	100%	3.3	4.1	4.0

3.5.3 X-1 Test Vehicle Control Surfaces

The team evaluated the possibility of equipping the X-1 test vehicle with a variety of control surface configurations, detailed below in Table 13. These configurations were judged based on three FoM: weight minimization, control provided, and manufacturability. Minimizing weight was given the highest priority due to its direct contribution to scoring. It was followed closely by control, to ensure the X-1 test vehicle could properly satisfy mission requirements for a 180 degree turn, alongside manufacturability to ensure the design was feasible for the team to construct. Utilizing only full wing control surfaces allow for significantly less weight than other configurations by reducing the number of servos required for control surface actuation. A configuration using ailerons and an elevator would yield the greatest relevant control, providing authority over pitching and rolling movements that contribute directly to the vehicle's ability to turn and control its descent. While a rudder would allow the vehicle to compensate for adverse yaw experienced during the turn, this was deemed a relatively low effect for the scale and duration the X-1 test vehicle is operating over. Utilizing only wing control surfaces as ailerons and flaps would only provide control over roll, as well as some pitch control if the center of gravity is positioned significantly ahead of the wing. This configuration, while providing less control, offers the greatest manufacturability by using the fewest control surfaces. Since the X-1 test vehicle does not generate thrust, pitch and yaw control do not significantly impact the orbiting motion of the vehicle to achieve the 180 degree turn. Therefore, ailerons were deemed the only significant control surface to perform the turn, and reduced pitch control from moving the wings symmetrically was deemed effective in this application given a sufficiently nose-heavy vehicle. With these considerations, the team chose to implement only main wing control surfaces acting as both ailerons and flaps, favoring the low weight and simpler manufacturability.

Table 13: X-1 Control Surface Trade Study

Figures of Merit	Weight (%)	Only Aileron/Flaps	Ailerons and Elevator	Aileron and Rudder
Weight	40%	5.0	1.0	2.0
Control Provided	30%	3.0	5.0	4.0
Manufacturability	30%	5.0	3.0	4.0
Total:	100%	4.4	2.8	3.2

3.6 Final Conceptual Design

The final conceptual design of the plane is of a low rectangular wing monoplane with a 5.33-foot wingspan, an aspect ratio of 3.82, and a single propeller propulsion system. The plane has a conventional tail attached by a boom to the fuselage, and tail dragger landing gear. The design decisions determined by the trade studies, including lighter landing gear and a rectangular wing, allow for a plane that is more maneuverable and capable of completing the year's missions. Figure 7 models the team's aircraft conceptual design.

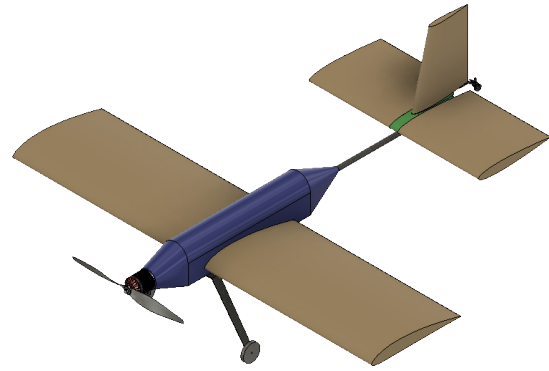


Figure 7: ND DBF Aircraft Conceptual Design

4 Preliminary Design

The preliminary design phase focused on generating, collecting, and analyzing data to guide design decisions. The team used a variety of software tools and applied theoretical equations from existing literature to support the iterative design process.

4.1 Design Methodology

The team's design methodology is detailed in the flow chart below. Beginning with the conceptual design, the team used MATLAB R2024b [2] to create a scoring analysis code, which is described in Section 3.2 and outputs basic parameters for the highest-scoring aircraft out of all viable competition configurations. This scoring analysis code provided a basis for the preliminary design as well as a guide for which optimization parameters yield the highest scores. Because of this, the scoring analysis code was revised and revisited several times throughout the design process to inform optimization and guide design changes. Given the initial parameters output from the code, component trade studies were conducted as detailed in Section 3.3. This analysis led to the initial conceptual design. These semi-qualitative decisions were modeled in the preliminary design phase using XFLR5 [4] and OpenVSP [5], software packages that generate advanced aerodynamic and performance analyses of the aircraft, including component placement, CG analysis, and stability calculations, which are detailed in the following section.

Next, the team entered the detailed design phase, which included CAD of the aircraft components and assembly, as well as structural analyses of critical points on the aircraft using FEA. Once the aircraft was modeled and analyzed to ensure structural integrity, the team manufactured a prototype of the aircraft using the various methods described in Section 6.2. Team members performed various tests on the prototype, including flight tests to simulate competition missions and component tests, all of which are described in Section 7. Finally, the team conducted a thorough analysis of all collected data to gain insight into the performance of the aircraft, discussed in Section 8. The team repeated this process to improve each subsequent iteration, referencing and correcting the scoring analysis code to inform design decisions, until a final design was reached. Figure 8 illustrates the flow of this process. To date, the team has completed four prototypes and improved the design with each version.

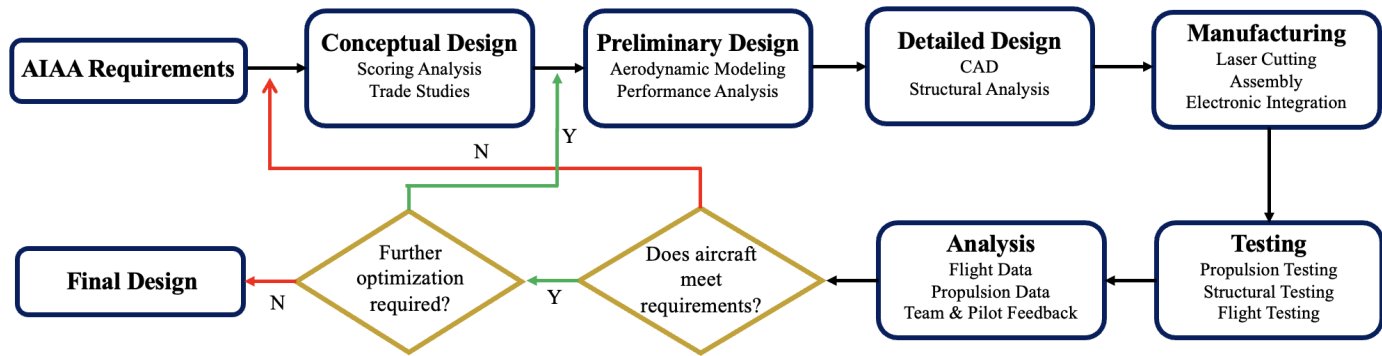


Figure 8: ND DBF Design Methodology

4.1.1 Aerodynamic Critical Parameters

4.1.1.1 Wing Span & Aspect Ratio

The wingspan must fit within the 6 ft maximum, as per AIAA rules. Additional constraints on the maximum wingspan include the positioning of fuel tanks on the wingtips, which the team balanced against the aircraft's aspect ratio, to achieve greater lift and increase aircraft efficiency. Higher aspect ratios, which increase with longer wingspans, also result in greater stability at the cost of maneuverability, a feature deemed to be a necessary tradeoff to ensure safe flight while loaded with high payload weights.

4.1.1.2 Wing Area

To carry a heavier fuel tank payload, which directly improves M2 performance, the team desired a high wing area to ensure greater lift forces. However, a greater wing area can also lead to high skin friction drag and structural concerns as lift forces increase. As discussed in Section 4.2.1 and Section 4.4.5, the team balanced a multitude of constraints on wing area to determine an optimal sizing that directly related to the team's goals for payload weights.

4.1.1.3 Airfoil

Several airfoils were considered for the main wing. Each of these were analyzed in XFLR5 to gain accurate predictions of the airfoils' properties, including coefficients of lift and drag as well as glide slope predictions and stability parameters. Detailed explanations of the team's considerations and decisions can be found in Section 4.2.4.

4.1.1.4 Component Location & Stability

Given the impact that payload weight has on the location of the aircraft's center of gravity, the team positioned payload components as close to the desired center of gravity as possible to ensure minimal adjustments in center of gravity between missions. The team used XFLR5 to determine the static margin and various stability metrics and inform component placement decisions.

4.1.2 Control Surface Critical Parameters

4.1.2.1 Flaps and Ailerons

The team chose not to implement independent flaps due to the lack of restriction on takeoff distances for this year's competition, allowing for reduced aircraft weight and system simplification by minimizing the required number of servos for control surface actuation. Instead, the sole control surface on the wings will be the ailerons, which will be crucial for providing control over the aircraft's roll. These ailerons were placed starting from the wingtips to maximize the rotational moment generated, which was also dependent on the total area of the ailerons and their angle of deflection.

4.1.2.2 Elevator

The elevator's effectiveness is dependent on having a large enough surface area and actuation angle to produce a significant pitching moment as it alters the effective camber of the horizontal stabilizer, allowing pilot input to control the pitch of the aircraft around the lateral axis. This pitching moment is also dependent on the distance from the elevator to the center of gravity.

4.1.2.3 Rudder

Rudder effectiveness depends on total area, maximum deflection angles, and distance from the aircraft's center of gravity. Rudder deflection enables the pilot to adjust the aircraft's yaw about the vertical axis, allowing for increased directional stability. Linking rudder movement to tail wheel movement will allow for steering during taxi, takeoff, and landing using only one servo.

4.1.3 Propulsion Critical Parameters

4.1.3.1 Motor

Motors were analyzed based on several factors, primarily their static thrust capacity and their efficiency. A high static thrust capacity is essential for generating enough thrust to fly with heavier payload weights. High efficiency is necessary to ensure the aircraft can fly for as many laps as possible while maintaining high speeds, ensuring that the battery capacity is properly utilized. Additionally, the motor must meet constraints placed by the AIAA rules for the competition.

4.1.3.2 Main Battery

The team considered a number of batteries that fell within the set of AIAA restrictions, with the goal of choosing a battery that would provide sufficient power to the motor for propulsion. The battery's capacity and performance are impacted by a number of factors, which are analyzed in greater detail in Section 4.2.7.

4.1.3.3 Electronic Speed Controller

The ESC is essential for controlling the flow of power from the battery to the motor, as well as facilitating pilot input to control the speed of the motor. To achieve this, it must be able to handle the battery's maximum output current without taking up excessive space or weight in the fuselage. Additionally, the team uses the ESC to collect experimental data during test flights in order to better inform further design choices.

4.1.4 Flight Course Breakdown

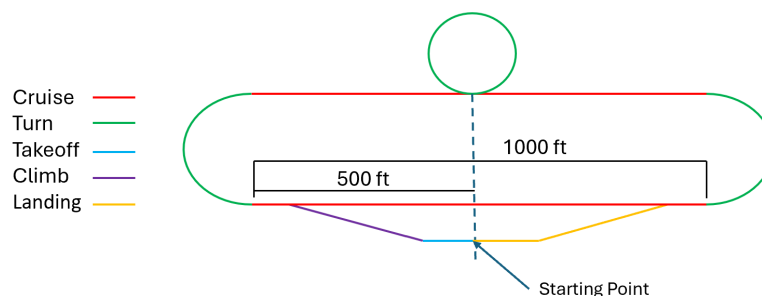


Figure 9: AIAA Standard Lap Components

4.1.4.1 Takeoff

Since the competition has no specified limit on takeoff distance, the only requirement the team considered was a maximum distance of 250 feet, the distance of ND DBF's testing runway.

4.1.4.2 Climb

Immediately after takeoff, a period of climb is required in order to fly at safe altitudes without obstruction or disturbances from the ground, as well as to reach the required altitude for dropping the X-1 test vehicle in M3. For predictive purposes, the team will model this period of climb using Equation (5), where V_{climb} is the aircraft's velocity, T is thrust, D is drag, and W is weight. In this equation, the excess thrust is shown as $(T - D)$, which produces the necessary force for the climb.

$$\text{Rate of Climb} = \frac{V_{climb}(T - D)}{W} \quad (5)$$

4.1.4.3 Cruise

Given the 1,000 ft and 500 ft straightaways that make up a significant portion of the flight course, a comprehensive understanding of the aircraft's performance at cruise is necessary to ensure accurate predictions and optimal mission performance. A detailed analysis of cruise performance is discussed in Section 4.4.3. It should be noted that the team's practical findings, discussed in Section 3.2, indicate that cruise speed is not necessarily indicative of flight speed during a competition lap.

4.1.4.4 Turn

Faster flight times rely heavily on quick turns with short turn radii, which are directly affected by factors such as aileron effectiveness and wing loading. These factors are analyzed in Section 4.4.4 to gain accurate predictions for performance across all flight missions.

4.1.4.5 Landing

Given the high payload weights involved in this year's design, ensuring a safe landing was of particular concern this year for scoring points in any mission. The team went through a number of design changes to improve the aircraft's ability to land safely; these considerations are outlined primarily in Section 4.4.5.

4.1.5 Assumptions & Uncertainties

The design process of ND DBF featured two separate types of uncertainties, analytical and environmental. Analytical uncertainties stem from standard mathematical models for aircraft not applying exactly to the aircraft as built. These can be caused by differences in the final manufactured product when compared to the model analyzed in XFLR5 or OpenVSP, or by equations and rules of thumb meant for full size aircraft design failing to translate on the scale of ND DBF's work. These errors can be accounted for by designing to a large factor of safety. Environmental uncertainties are presented in Table 14. These are the differences in weather between the competition site in April and ND DBF's testing site and season.

Table 14: Environmental Uncertainties

Uncertainty	Tucson International Modelplex Park Association	South Bend Radio Control Club, Bremen IN
Season	Spring	Late Fall/Winter/Early Spring
Average Temperature (High/Low/Mean)	83°/52°/68° F	44°/30°/36° F
Average Wind Speed	12.7 ft/s	17.3 ft/s
Elevation (above sea level)	2,643 ft	854 ft
Average Air Pressure	2111 lbs/ft ²	2128 lbs/ft ²
Air Density	0.0023 slugs/ft ³	0.0025 slugs/ft ³

ND DBF's testing site in Bremen, Indiana has several clear environmental differences when compared to the fly-off site in Tucson. Firstly, because the air in Bremen is slightly denser than that in Tucson, the lift generated by the aircraft will be slightly higher

during test flights. However, this difference in humidity creates differences in lift on a magnitude of less than Additionally, the fly-off site is significantly hotter which could potentially lead to overheating of electrical components, such as the ESC, that would not be experienced in the Bremen winter and early spring. To combat this, the team should add additional airflow to those components, or use components rated well beyond the expected current draw of the system.

4.2 Design and Sizing Considerations

The following section describes the methodology and techniques used to translate conceptual and preliminary designs into the qualitative and quantitative design parameters that define the detailed design. Sizes of lifting and control surfaces were initially estimated using the aerodynamic equations outlined in the following sections. The team then verified and adjusted these parameters through analyses using MATLAB programs [2] and the XFLR5 analysis software. XFLR5 determines coefficients of lift, drag, and other aerodynamic parameters for different lifting surfaces and mass placements through Vortex Lattice Method [4]. XFLR5 is limited due to its inability to accurately simulate fuselage geometry, so further calculations were made using OpenVSP [5], primarily to gain estimates of parasite drag in Section 4.4.1.

4.2.1 Wing Geometry

Because the wing is the primary source of lift, the team optimized its design parameters to maximize mission scoring. As detailed in Section 3.2, the wing was preliminarily sized in the scoring analysis simulation script. This section details the most important parameters that were considered to reach the preliminary wing geometry. The span was first set to 6 ft, the highest span allowed by AIAA, to maximize aspect ratio for potential wing areas and maximize efficiency through the wing lift-to-drag ratio. This allows for optimal trade-off between cruise speed and payload capacity. Scoring analysis script set the initial wing area at 3.85 square feet, resulting in a mean aerodynamic chord of 9.43 in. A taper ratio of 0.6, optimized to minimize profile drag without over-sizing the root chord, was applied to the wing to mitigate induced drag and further optimize the wing's lift-to-drag ratio.

The slope for the 3D lift coefficient was calculated using Equation (6), where β is a function of the effective Mach number. Using the predicted flight speed for the configuration, aircraft weight, zero lift angle of attack, and $\frac{dC_L}{d\alpha}$, the aircraft trim angle for steady, level flight could be determined using Equation (4) stated in Section 3.2.

$$\frac{dC_L}{d\alpha} = \frac{2\pi AR}{2 + \sqrt{4 + (AR\beta)^2 \left(1 + \frac{\tan^2(\Lambda_{t/c})}{\beta^2}\right)}} \quad (6)$$

After multiple test flights, the team determined that the aircraft's glide slope was too steep for a safe landing, resulting in crashes on landing. The wing geometry was reevaluated, as the team found that higher lift was required for softer landings. The aircraft's stall speed, the minimum speed required to maintain level flight, was calculated and used as a relative estimate of the aircraft's landing speed, as discussed in Section 4.4.5. Using these stall speed calculations, a wing area was calculated taking into account the team's goal for the total plane weight. To maximize the aspect ratio, the new wingspan was set to 64 in to allow for space at the wingtips to fit the fuel tanks within the 6 ft limit. To improve roll stability, the team implemented a dihedral angle of 5 degrees, as discussed in Section 4.3.2. The team determined basic metrics for the aircraft's performance with the new wing, which are listed in Table 15 using the selected airfoil geometries detailed in Section 4.2.4. These were used for further analysis on the new wing geometry, which was conducted using XFLR5 to determine flight and stability characteristics in Section 4.4. As discussed in Section 3.2, these cruise characteristics are not necessarily the average flight speed of the aircraft in a competition lap.

Table 15: Cruise Characteristics for All 3 Missions

Figures of Merit	M1	M2	M3
Cruise Speed [ft/s]	53.0	78.8	56.5
Lift Coefficient	0.376	0.336	0.350
Cruise α [deg]	0.55	-0.06	0.54
Lift at $0^\circ \alpha$ [lbs]	17.6	17	17
Lift at Cruise α [lbs]	9.354	18.500	9.904

4.2.2 Fuselage Design and Sizing

The implementation of a boom tail meant that minimizing fuselage length would also minimize overall drag, since a larger portion of the aircraft would have lower surface area and therefore produce less surface drag. As such, the cross-sectional shape of the fuselage needed to allow for the dense placement of components into a relatively short length of the fuselage, leading the team to lean towards a rectangular cross-section that maximized internal area relative to the mostly rectangular components being used (such as the battery and ESC). However, the shape of the fuselage's cross-section also played a significant role in drag minimization, favoring a more circular shape with rounded edges. To balance these requirements, the team chose to use a Norman window cross-section for the fuselage, tapered at both ends and sized to incorporate all components with optimal locations to achieve an ideal center of gravity.

4.2.3 Tail Design and Sizing

Through the trade study found in Section 3.3.3, the team settled on a conventional design for the overall configuration of the vertical and horizontal stabilizers. The team chose to implement a symmetrical NACA 0012 airfoil for both the vertical and horizontal stabilizers due to its high lift-to-drag ratio relative to other symmetrical airfoils as well as its reliability in previous years as a tail airfoil. When sizing the tail, the team began by using a combination of tail sizing equations and conventional assumptions to provide an initial estimate for sufficient tail sizing parameters. Volume coefficients for the horizontal and vertical tails (V_{HT} and V_{VT}) were determined to be 0.5 and 0.075, respectively, with the vertical tail's coefficient being increased marginally from the recommended 0.06 due to prior experience [6]. Using these, along with the main wing's chord $\bar{c}_W$, area S_W , and span b_W , the team utilized Equations (7) through (12) to determine appropriate ranges for the tail areas, S_{HT} and S_{VT} , as well as the distances from the wing, l_{HT} and l_{VT} . Additional assumptions, made from previous experience and verified through listed sources, were that the aspect ratio of the horizontal tail, AR_{HT} , should be around half that of the wing, and the vertical tail's aspect ratio AR_{VT} should be around 150% that of the wing [7]. Additionally, the taper ratio λ was set to 0.6 to minimize induced drag and maximize structural efficiency. The tip chord of the vertical tail is offset in order to maintain a constant trailing edge.

$$V_{HT} = \frac{l_{HT} S_{HT}}{\bar{c}_W S_W} \quad (7)$$

$$AR_{HT} = \frac{b_{HT}}{c_{HT}} \quad (9)$$

$$AR_{VT} = \frac{2b_{VT}}{c_{r,VT}(1 + \lambda_{VT})} \quad (11)$$

$$V_{VT} = \frac{l_{VT} S_{VT}}{\bar{c}_W S_W} \quad (8)$$

$$AR_{VT} = \frac{b_{VT}^2}{S_{VT}} \quad (10)$$

$$\lambda = \frac{c_t}{c_r} \quad (12)$$

With these sizing estimates in mind, the team simulated tail designs in XFLR5 to find a tail that provided proper stability in relation to the predetermined wings and the constraints on component placements within the fuselage. This process was used iteratively throughout the year in response to necessary changes in the wing. Detailed parameters for the tail design are found in Table 16.

Table 16: Empennage Design Parameters

Design Parameter	Horizontal Tail	Vertical Tail
Area (sq. in)	390.0	102.0
Aspect Ratio	2.31	5.65
Moment Arm (in)	33.6	33.3
Incidence ($^\circ$)	0	0
Volume Coefficient	0.5	0.075
Sweep ($^\circ$)	0	10.62

4.2.4 Airfoil Design and Sizing

Aerodynamic design and analysis began with the airfoil selection for the main wing. The Airfoil Tools database [8] and XFLR5 were used to analyze airfoil characteristics of potential airfoils at the cruise Reynolds number, computed using Equation (13).

$$Re = \frac{\rho u \bar{c}_w}{\mu} \quad (13)$$

With a 16.75 in chord, and an estimated cruise velocity of 77 ft/s, the Reynolds number was computed to be 700,000. Using this Reynolds number, the team looked to narrow down the list of potential airfoils. Airfoils with both a high coefficient of lift and high lift-to-drag ratio were prioritized to maximize payload weight and maximum cruise speed, respectively. Furthermore, only airfoils with flat bottoms or symmetric geometries were considered in order to ensure ease of manufacturing. After analyzing the database, referring to previous experiences, and discussing with technical advisors and prior DBF members, the team narrowed its airfoil options to the three shown in Table 17.

Table 17: Airfoil Comparison

Aerodynamic Properties			
	Clark Y	NACA 2213	N-22
Cl, max	1.46	1.45	1.56
Cd at Cl, max	0.053	0.040	0.025
Cl, cruise	0.424	0.195	0.656
Cd at Cl, cruise	0.006	0.007	0.008
Max L/D	104.7	84.1	99.2
α [deg] at Cl, max	16.0	15.0	10.5

From Table 17, the Clark-Y and N-22 airfoils have a higher lift-to-drag ratio than the NACA 2213 airfoil, allowing for a faster cruise speed. Between the Clark-Y and N-22, the N-22 has a higher maximum coefficient of lift and cruise coefficient of lift, as well as a lower coefficient of drag at the maximum coefficient of lift. The N-22 was selected as the main wing airfoil to optimize payload capacity and cruise speed.

4.2.5 Control Surface Design and Sizing

4.2.5.1 Elevator

The elevator was sized to trim the aircraft to a minimal angle of attack using pitching moment data from XFLR5 as well as other geometric parameters. Equations (14) and (15) below display the process for computing the flap effectiveness parameter τ :

$$C_m = C_{m_0} + (C_{m,\alpha})\alpha + (C_{m,\delta_e})\delta_e = 0 \quad (14)$$

$$\tau = \frac{-(C_{m,\delta_e})}{\eta_{HT} V_{HT} (C_{L,\alpha_{HT}})} \quad (15)$$

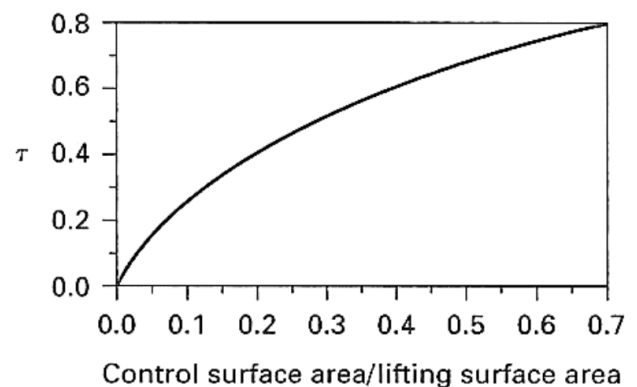


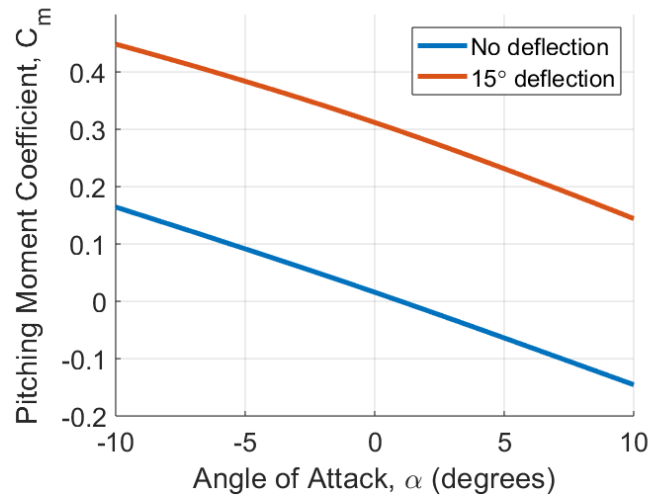
Figure 10: Flap Effectiveness Parameter, τ [9]

where C_{m_0} is the zero angle of attack pitching moment coefficient, C_{m,δ_e} is the slope of the C_m vs δ_e curve, δ_e is the elevator deflection angle, η_{HT} is the horizontal tail efficiency, and $C_{L,\alpha_{HT}}$ is the slope of C_L vs α_{HT} curve. After computing τ , the control surface area to lifting surface area ratio could be determined from Figure 10 above. From this, the elevator area could be determined, yielding the dimensions found below in Table 18.

Table 18: Elevator Sizing Parameters

Elevator Deflection (°)	Flap Effectiveness Parameter	Control Surface Area/ Lifting Surface Area	Elevator Area (sq. in)	Elevator Span (in)	Elevator Chord (in)
15	0.266	0.10	75	30	2.5

The actual as-built elevator was designed to be able to exceed a 15 degree deflection angle to provide more authority if necessary to account for any inefficiencies due to manufacturing processes. The elevator was implemented into the XFLR5 aircraft model and Figure 11 was generated, further demonstrating the ability of the elevator to trim the aircraft to high angles of attack. Finally, flight testing played a large role in finalizing the size of the elevator. The team's initial sizing decisions were confirmed after receiving positive feedback affirming the elevator's sufficient control authority from the team's pilot.

**Figure 11:** Pitching Moment Coefficient vs Angle of Attack

4.2.5.2 Rudder

The rudder was sized for the aircraft to trim a 20 ft/s crosswind using yawing moment data from XFLR5 as well as other geometric parameters. Equations (16) and (17) below display the process for computing the flap effectiveness parameter τ :

$$C_n = C_{no} + (C_{n,\beta})\beta + (C_{n,\delta_r})\delta_r \quad (16)$$

$$\tau = \frac{-(C_{n,\delta_r})}{\eta_{VT}V_{VT}(C_{L,\alpha_{VT}})} \quad (17)$$

where C_n is the yawing moment coefficient, C_{no} is the zero angle of attack yawing moment coefficient, $C_{n,\beta}$ is the slope of the C_n vs β curve, C_{n,δ_r} is the slope of the C_n vs δ_r curve, δ_r is the rudder deflection angle, η_{VT} is vertical tail efficiency, and $C_{L,\alpha_{VT}}$ is the slope of the C_L vs α_{VT} curve. After computing τ , the control surface area to lifting surface area ratio could be determined from Figure 10 above. From this, the rudder area could be determined, yielding the dimensions found in Table 19.

Table 19: Rudder Sizing Parameters

Rudder Deflection (°)	Flap Effectiveness Parameter	Control Surface Area/ Lifting Surface Area	Rudder Area (sq. in)	Rudder Span (in)	Rudder Chord (in)
15	0.557	0.35	36	12	3

Like the elevator, the actual as-built rudder was designed to be able to exceed a 15 degree deflection angle to provide more authority if necessary to account for any inefficiencies due to manufacturing processes. The rudder was implemented into the XFLR5 aircraft model and Figure 12 was generated, further demonstrating the ability of the rudder to trim the aircraft during side slipping. As before, flight testing played a large role in finalizing the size of the rudder. After testing the design in flight, the team also confirmed with the pilot that this rudder sizing provided effective control.

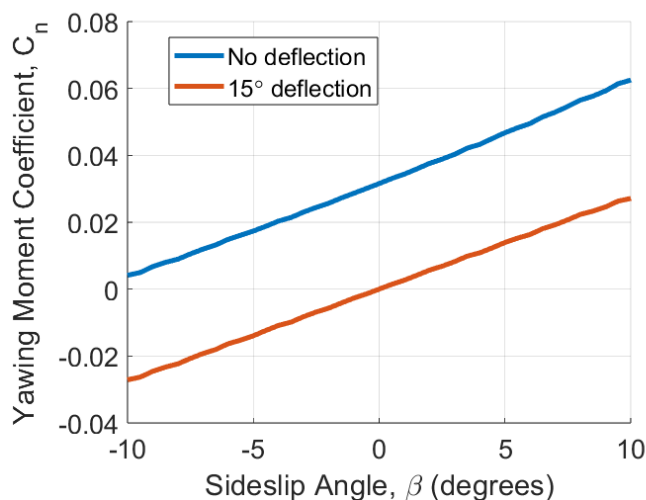


Figure 12: Yawing Moment Coefficient vs Sideslip Angle

4.2.5.3 Ailerons

The ailerons were sized to provide adequate roll authority for the aircraft using rolling moment data from XFLR5 as well as other geometric parameters. Equations (18) and (19) below display the process for computing the flap effectiveness parameter τ :

$$C_l = (C_{l,\beta})\beta + (C_{l,\delta_a})\delta_a \quad (18)$$

$$\tau = \frac{b_W(C_{l,\delta_a})}{2(C_{L,\alpha_W})} \quad (19)$$

where C_l is the roll moment coefficient, $C_{l,\beta}$ is the slope of the C_l vs β curve, C_{l,δ_a} is the slope of the C_l vs δ_a curve, δ_a is the aileron deflection angle, and C_{L,α_W} is the slope of the C_L vs α_W curve. After computing τ , the control surface area to lifting surface area ratio could be determined from Figure 10 above. From this, the aileron area could be determined, yielding the dimensions found below in Table 20.

Table 20: Aileron Sizing Parameters

Aileron Deflection (°)	Flap Effectiveness Parameter	Control Surface Area/ Lifting Surface Area	Aileron Area (sq. in)	Aileron Span (in)	Aileron Chord (in)
15	0.196	0.07	107.2	20	5.36

The actual as-built ailerons were designed to be able to exceed a 15 degree deflection angle to provide more roll authority if necessary to account for any inefficiencies due to manufacturing processes. The ailerons were implemented into the XFLR5 aircraft model and Figure 13 was generated, further demonstrating the roll authority generated by the ailerons. As with the other control surfaces, flight testing played a large role in finalizing the size of the ailerons, and the team used pilot feedback from test flights to confirm that the ailerons provided sufficient roll authority.

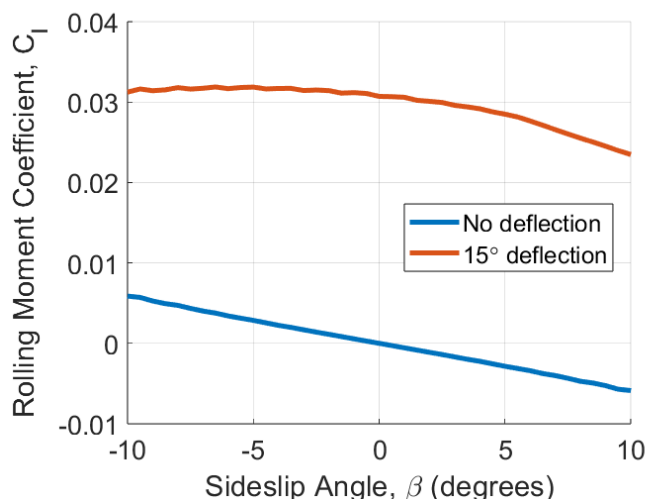


Figure 13: Rolling Moment Coefficient vs Angle of Attack

4.2.6 Servo Torque Sizing

The team calculated the torque required to fully actuate each control surface from an online RC Airplane Torque Calculator [10]. Table 21 details the required and recommended torque for each control surface. The recommended torque is calculated using a safety factor of 1.3.

Table 21: Servo Torque Requirements for Each Control Surface

Control Surface		Rudder	Elevator	Ailerons
Torque (oz-in):	Required	12.7	16.1	63.1
	Recommended	16.5	20.9	82.0

The team elected to use KST X10 Mini Digital Servos which provide 94 oz-in of torque because of their small form factor and large torque capacity. Although the maximum recommended torque is nearly reached on the ailerons, the factor of safety remains 1.1 above the required torque, which was deemed adequate for Mk 1.

4.2.7 Propulsion Design and Sizing

The team used the scoring optimization analysis performed at the beginning of the year to design the propulsion system. In this analysis, the team analyzed motor and propeller configuration performance data charts provided by the manufacturers such as BadAss [11], Cobra [12], and Tempest [13]. A range of battery manufacturers was also analyzed from brands such as Liperior [14] and MaxAmps [15]. These brands were selected because of their high-quality ratings, easily accessible data charts, and diversity of motor and battery size. The performance charts were key pieces of information used to determine the propulsion design, including static thrust, current draw, RPM, and optimal propeller size. As outlined in Section 3.2, a prospective propulsion configuration must meet several baseline requirements. First, a motor/propeller that provides a minimum thrust-to-weight ratio of 1.2 must be found on the performance data sheets. This value was selected from previous team DBF experience, advice from experienced RC pilots, and the knowledge that one must produce extra thrust to counteract the drag from the payload bottles. Second, an available battery must have enough power to allow 5.5 minutes of continuous flight. This was calculated using Equation (21). Third, a motor/propeller configuration must provide enough static thrust to enable the aircraft to take off in less than the length of the team's testing runway, 250 ft. These calculations are covered in more detail in Section 4.4.2.

This detailed analysis provided the team with multiple high-scoring propulsion configurations for the aircraft design. The team then purchased one motor that met the team's budget requirements and multiple propellers for further testing. More detailed theoretical analysis of dynamic thrust was completed on each propeller using data provided by Advanced Precision Composites (APC) [16] provided data and calculated using blade element theory [3]. Given that the maximum RPM of the motor is roughly 7000 RPM's, dynamic thrust values from a propeller of the same diameter and pitch as the one selected by the scoring analysis code were selected from the APC data base, and plotted with velocity. This data is further described in Section 4.4.3. The components selected during the preliminary design stage are summarized in Table 22.

$$\frac{T}{W} \geq 1.2 = \frac{\text{Maximum Motor Static Thrust}}{\text{Aircraft Mission Weight}} \quad (20)$$

$$\text{Flight Time} = \frac{\text{Battery Amp Hours} \times 60}{0.85 \times \text{Motor Static Current Draw}} \quad (21)$$

Table 22: Detailed Propulsion Design

Motor	ESC	Battery	Propellor
Badass 6245 180kV	Phoenix Edge HVF 160	MaxAmps 12S 2250mAh Li-Po	Badass 20x12

4.3 Stability and Control

4.3.1 Static Stability

The team began stability and control analyses by determining the aircraft's static margin, as defined in Equation (22). This varied slightly between missions for the unloaded aircraft in M1, the aircraft loaded with fuel tanks in M2, and the aircraft loaded with the x-1 test vehicle in M3. The team determined about 20% to be an ideal static margin based on past designs and conventional stability assumptions for RC aircraft of this scale. Table 23 displays the static margin for each mission, which all remain within 2% of the goal of 20%.

$$SM = \frac{X_{NP} - X_{CG}}{\bar{c}_W} \quad (22)$$

Table 23: Static Margin Parameters

Static Margin Parameters	M1	M2	M3
Center of Gravity Position (in)	15.95	15.28	15.89
Neutral Point Position (in)	19.19	18.92	18.92
Static Margin	19.34%	21.73%	18.10%

To further ensure static stability, XFLR5 was used to generate values for the pitching moment coefficient, C_m , at various angles of attack, and the yawing moment coefficient, C_n , at various sideslip angles. The negative C_m vs α slope indicates a slight forward pitching tendency at low angles of attack with an intercept just over the origin at 0.017, indicating a statically stable aircraft with the desired tendency to slightly pitch forward. This is particularly important for preventing stalling at high critical angles of attack, as the forward pitching moment naturally forces the aircraft back into safe angles of attack. The positive C_n vs β slope passing through the origin indicates desired lateral stability. Slopes from these analyses yielded static stability derivatives, detailed in Table 24.

Table 24: Lateral and Longitudinal Static Stability Derivatives

Stability Classification	Symbol	Derivative		
		M1	M2	M3
Static Longitudinal Stability	Cm,a	-0.8780	-0.8865	-0.8941
Static Lateral Stability	Cn,b	0.1559	0.1559	0.1564

4.3.2 Dynamic Stability

After verifying static stability, the dynamic stability of the aircraft was analyzed in XFLR5 for all three missions. A crucial component of designing for dynamic stability was the implementation of a dihedral wing angle of 5 degrees. With this design, a lateral rolling disturbance results in a lift vector that offsets the disturbance, thereby increasing roll stability and dampening spiral divergence. Figure 14 displays the root locus plots for the lateral and longitudinal dynamic stability modes.

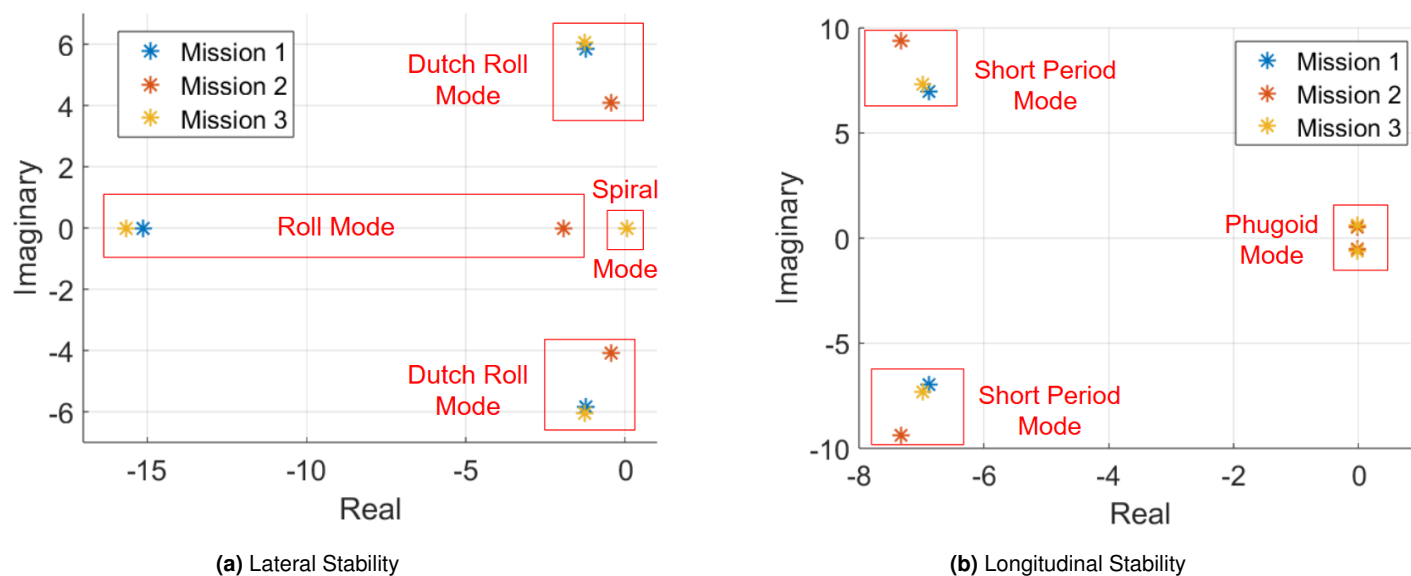


Figure 14: Root Locus Plots

The real part of each root determines the stability of its associated mode. A negative real part corresponds to a stable response, whereas a positive real part corresponds to an unstable response. As seen in Figure 14, each mode has a negative real part and thus a stable response except for the spiral mode, which has a slightly positive real part. This indicates a divergent spiral mode; however, this is common in many aircraft and can be corrected with pilot input given a slow growth rate. The fastest time to double for a spiral mode perturbation is over 19 seconds in M2. Flight tests have demonstrated that this is more than enough time for the pilot to correct a spiral mode disturbance. Figures 15 and 16 depict time response analyses, performed in XFLR5, for both longitudinal and lateral stability modes.

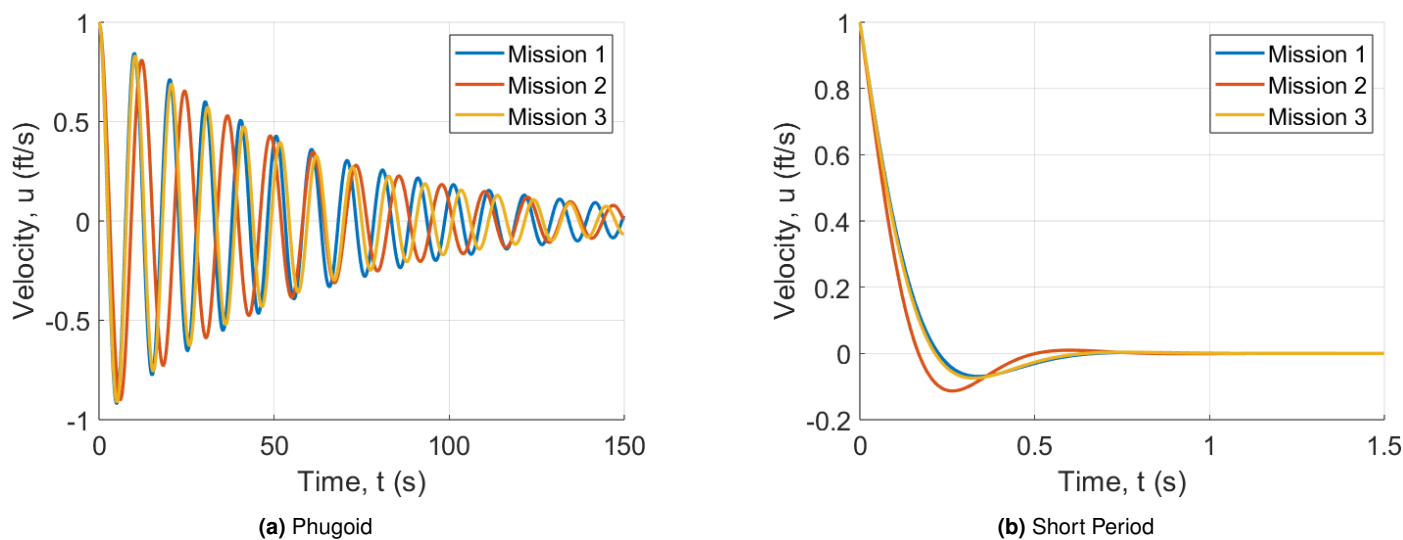


Figure 15: Time response plots for longitudinal stability modes.

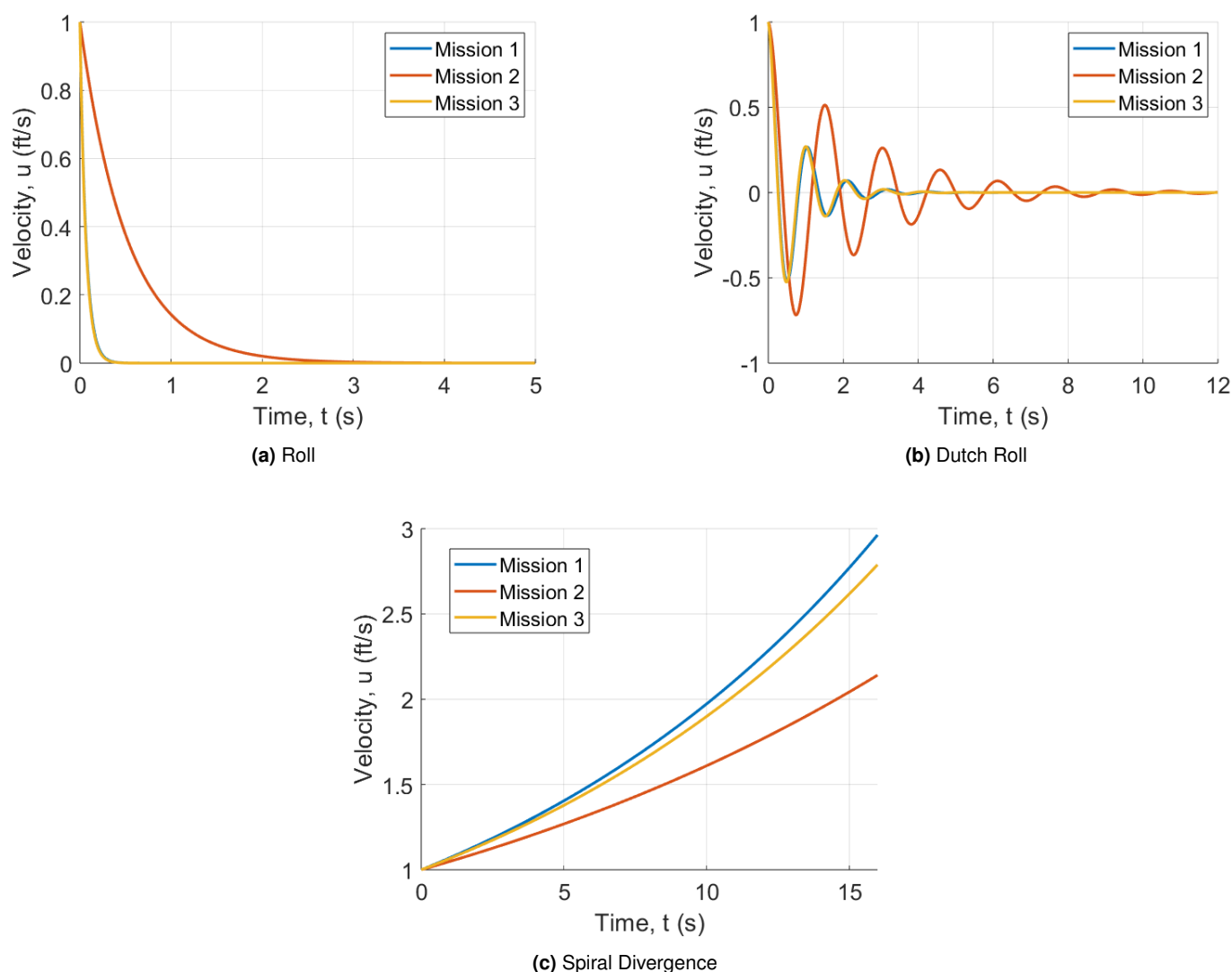


Figure 16: Time response plots for lateral stability modes.

Each of these plots display sound dynamic stability according to Sadraey [6]. The damping in the phugoid mode, which halves in roughly 10-15 seconds for each mission, is within the standard time to half for phugoid mode of 30-120 seconds [3]. Short period and dutch roll modes also decay rapidly, and for this reason, will not cause controllability issues. The roll response, although an unstable mode, takes 20 seconds in M2 to double, which is more than enough time for the pilot to react and correct for the spiral mode, as shown in Table 25.

Table 25: Dynamic Stability Analysis for All 3 Missions

(a) Times to half/double for longitudinal and lateral modes.

Mission	Longitudinal		Lateral		
	Short Period	Phugoid	Roll	Dutch Roll	Spiral
M1	0.100s to half	41.3s to half	0.046s to half	0.554s to half	10.2s to double
M2	0.095s to half	39.6s to half	0.356s to half	1.57s to half	14.6 to double
M3	0.099s to half	37.9s to half	0.044s to half	0.541s to half	10.8 to double

(b) Dynamic stability derivatives for longitudinal and lateral modes.

Mission	Dynamic Longitudinal Stability (C_m, q)	Dynamic Lateral Stability (C_n, r)
M1	-9.191	-0.2126
M2	-9.195	-0.2126
M3	-9.229	-0.2131

4.4 Performance Analysis

4.4.1 Lift and Drag

The team utilized XFLR5 to gain accurate representations of the lift and drag forces, outlined in Figures 17a and 17b. Local values of C_L are plotted across the span of the wing and horizontal stabilizer in Figure 17a, and total values of C_L are plotted over various angles of attack α in Figure 17b. The y-intercept of this plot indicates a positive C_L of 0.29 at a 0° angle of attack, which increases by 0.07 for each degree increase in α .

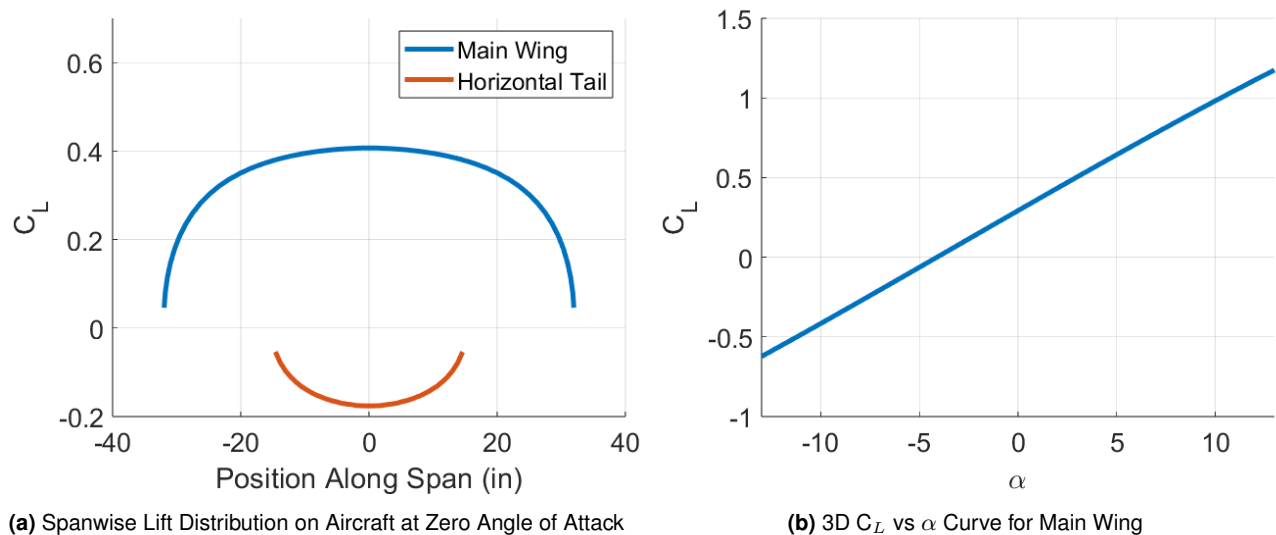


Figure 17: Primary Aerodynamic Characteristics of Aircraft

ND DBF used the OpenVSP [5] parasite drag analysis tool to calculate the total parasite drag of the aircraft. The tool uses linear inviscid models to calculate the parasite drag of each component. The induced drag C_{Di} was calculated from the aspect ratio, AR , Oswald efficiency factor, e , and the coefficient of lift, C_L , in cruise, shown in Equation (23) below.

$$C_{Di} = \frac{C_L^2}{\pi e AR} \quad (23)$$

Figures 18 and 19 illustrates the force of drag calculated for each component and total drag force sum above each bar. These values were obtained using Equation (24) below, where ρ is air density, V is velocity, and S_W is cross sectional area.

$$D = \frac{1}{2} \rho V^2 C_D S_W \quad (24)$$

The components which contributed most to the overall drag were the main wing, with the largest wetted area, the horizontal tail, with the second largest wetted area, and the landing gear, with the highest form factor. As expected, the induced drag is highest for M2, where the total aircraft weight is the highest.

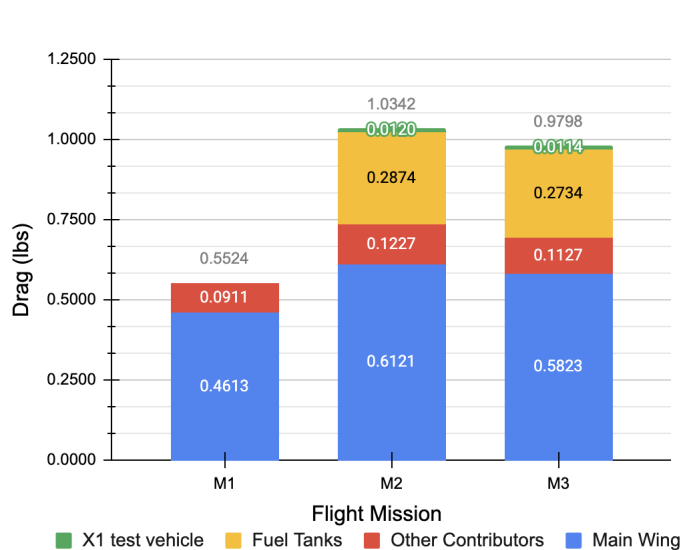


Figure 18: Aircraft Wing Drags for Each Flight Mission

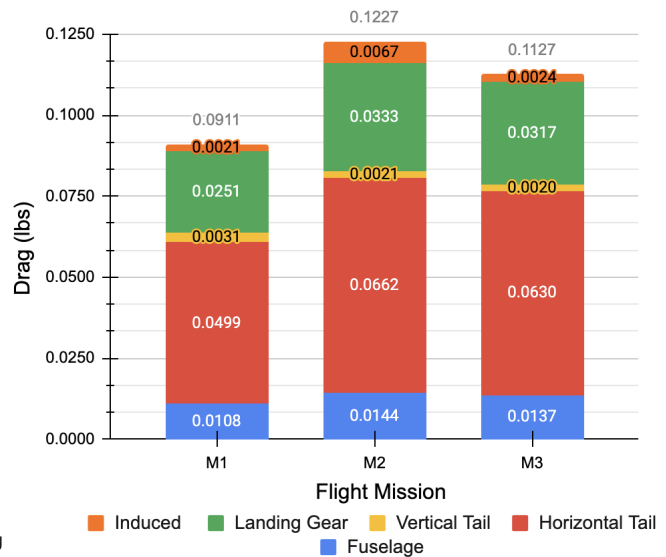


Figure 19: Aircraft Component Drags for Each Flight Mission

4.4.2 Takeoff Distance

To ensure the aircraft could takeoff within the length of the testing runway, a fourth order Runge-Kutta (RK4) simulation was implemented. This program increments through time until the aircraft achieves enough lift for takeoff. Figure 20 illustrates the takeoff runs for M1, M2, and M3, respectively. Each line's termination point represents the position and time at which the aircraft generates enough lift to takeoff. As can be seen, the aircraft takes off in both loaded and unloaded cases without any gusts or flaps implemented.

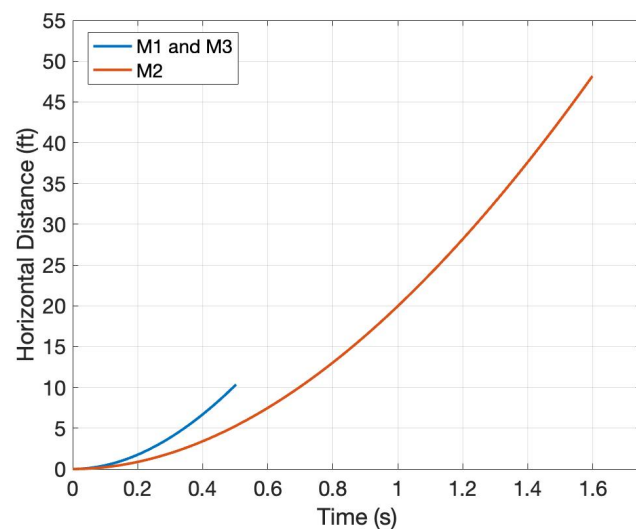


Figure 20: Takeoff Runs for Multiple Missions

4.4.3 Maximum Cruise Velocity

The maximum cruise speed for each mission significantly influences the mission score. The team found this speed by identifying the intersection point of the dynamic thrust and thrust required curves, as discussed in previous sections (4.4.1 for drag or thrust required). Figures 21, 22, and 23 visually represent the excess thrust at various velocities throughout each mission, as well as the terminal cruising velocity where the two curves intersect.

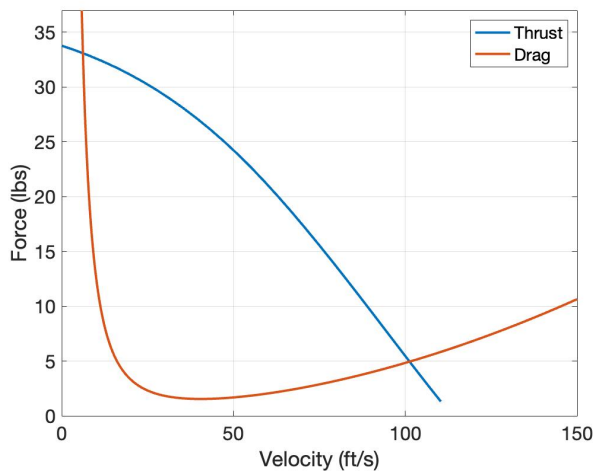


Figure 21: M1 Excess Thrust

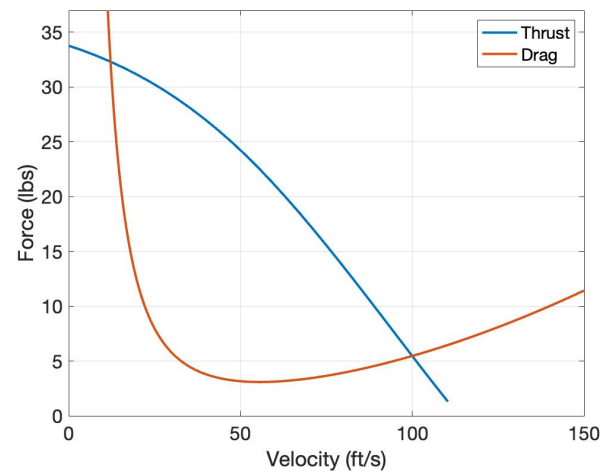


Figure 22: M2 Excess Thrust

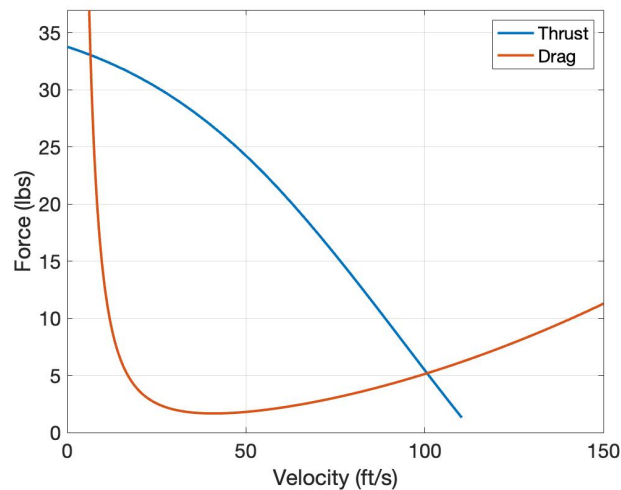


Figure 23: M3 Excess Thrust

The dynamic thrust in these figures corresponds to a 20x12 propeller spinning at 7000 RPM. The drag curve represents the combined parasite and induced drag for the mission. Table 4 in Section 3.2 lists the exact maximum flight speed for each mission. While these speeds are all similar, M2 flight speed is the lowest, as expected due to the higher payload weight in this mission.

4.4.4 Turning Analysis

The aircraft's level turning performance is limited by structural and aerodynamic design, as well as available power. The limiting load factor determines the maximum bank angle that the vehicle can maintain in level flight without exceeding the airplane's structural limitations or stalling. Equations (25) and (26) were used to approximate turn radius in a sustained turn, which would produce the highest load.

$$n = \sqrt{\frac{q}{k \cdot \frac{W}{S}} \cdot \left(\frac{T}{W} - \frac{q \cdot C_{D0}}{\frac{W}{S}} \right)} \quad (25)$$

$$\dot{\psi} = \frac{g\sqrt{n^2 - 1}}{V} = \frac{V}{R} \quad (26)$$

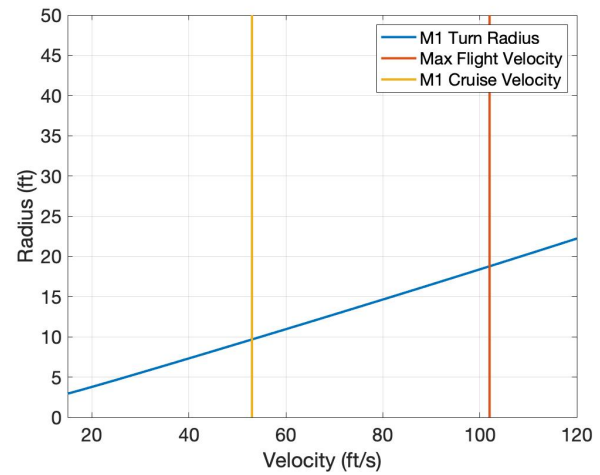


Figure 24: M1 Turn Radius as a Function of Turn Velocity

In these equation, V is the turn velocity, R is the turn radius, $\frac{W}{S}$ is wing loading, $\frac{T}{W}$ is thrust-to-weight ratio, q is dynamic pressure, k is $\frac{1}{\pi e A R}$, C_{D0} is the base coefficient of drag, and $\dot{\psi}$ is the turn rate in radians per second. Figures 24, 25, and 26 illustrate the turn radius as a function of velocity. For each mission the cruise speed and maximum speed are plotted, revealing a turn radius of about 19 ft in M1, 37 ft in M2, and 20 ft in M3, if the aircraft maintains top speed throughout the turn.

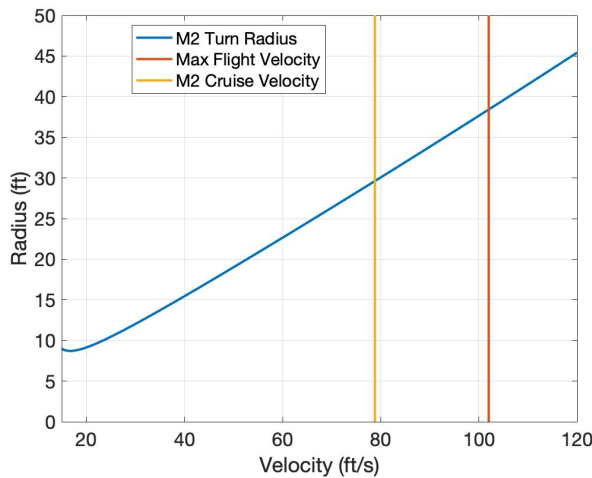


Figure 25: M2 Turn Radius as a Function of Turn Velocity

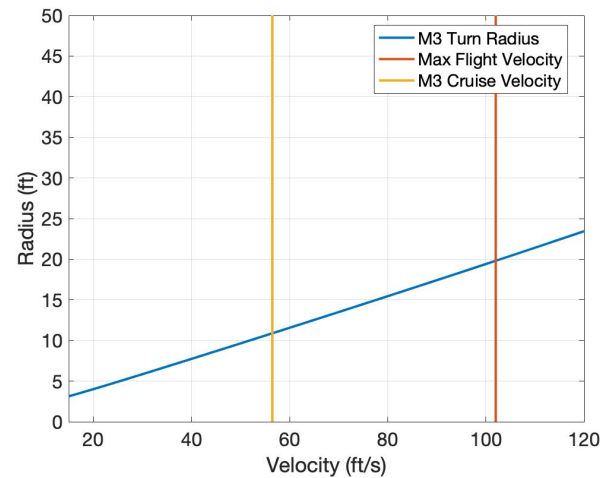


Figure 26: M3 Turn Radius as a Function of Turn Velocity

4.4.5 Landing Speed

As mentioned in Section 4.2.1, the team used stall speed calculations to comparatively estimate the aircraft's landing speed. The stall speed V_S was found using Equation (27), where W is the total aircraft weight, ρ is air density, S_W is the wing area, and $C_{L_{max}}$ is the maximum coefficient of lift for the N-22 airfoil at Reynolds numbers of up to 500,000 [17]. This value was compared to stall speeds of previous aircrafts and found to be significantly higher; using previous aircrafts and pilot input as a guide, the team determined a safe region of stall speeds to be under 40 ft/s. With these parameters, the team determined approximate wing areas in relation to total aircraft weights. Based on historical precedents concerning structural and manufacturing issues, the team chose to cap the maximum wing area at 7.5 square feet to ensure a feasible design. This area correlated with a total plane weight of 18.5 pounds.

$$V_s = \sqrt{\frac{2W}{\rho S_W C_{L_{max}}}} \quad (27)$$

4.5 Overall Predicted Mission Performance

The team derived estimates for the aircraft's overall performance from the conceptual design outlined above. These parameters were found using XFLR5 alongside previously mentioned aerodynamic equations computed using MATLAB scripts. These factors were mainly dependent on the wing area, 1072 in², the rectangular planform with 0-degree sweep and incidence angles, the wing aspect ratio of 3.82, and the mean aerodynamic chord of 16.75 in. Calculated values for these performance parameters are outlined in Table 26.

Table 26: Mission Performance Parameters

Performance Parameters	Mission 1	Mission 2	Mission 3
Takeoff Weight [lb]	9.35	18.50	9.90
α (Cruise) [deg]	0.00	0.00	0.00
W/S [lb/sq. ft]	1.26	2.49	1.33
Max CL	1.30	1.30	1.30
Cruise CL	0.34	0.32	0.33
Cruise CD	0.02	0.02	0.02
Cruise (L/D)	17.0	16.0	16.5
Cruise Speed [ft/s]	53.0	78.8	56.5
Stall Speed [ft/s]	28.4	39.9	29.2
Takeoff Ground Roll [ft]	11.2	50.0	11.2
Cruise Turn Radius [ft]	10.0	30.0	11.0
Lap Time [s]	26.4	29.6	26.6
Mission Time [min]	1.35	1.42	5.00

5 Detailed Design

The team conducted multiple design iterations in the conceptual and preliminary design phases. Subsystems saw changes between every iteration to optimize performance, decrease weight, and increase efficiency. Table 27 shows the dimensions and specifications of all the competition aircraft's main subsystems.

Table 27: Aircraft Dimensions and Component Specifications

Wing		Vertical Tail		Propulsion	
Span	64 in	Span	12 in	Motor Model	BadAss 6245
Chord Length	16.75 in	Tip Chord	7 in	Motor KV	180
Airfoil	N-22	Root Chord	10 in	Motor Weight	2.42 lb
Aspect Ratio	3.82	Airfoil	NACA 0012	Propeller	BA 20 x 12 Wood Electric
Static Margin	19.06%	Weight	0.25 lb	Propulsion Battery	MaxAmps 12s 2250mAh Li-Po
Weight	0.984 lb per wing	Stabilizer Area	102 sq. in	Battery Capacity	2250 mah
Wing Area	1072 sq. in	Aspect Ratio	5.65	Battery Weight	1.440 lb
Horizontal Tail		Fuselage		Max Continuous Current	67.5 A
Span	30 in	Weight	0.912 lb	ESC	
Chord Length	13 in	Total Length	32.625 in	Model	Castle Creations Phoenix Edge HV 160
Airfoil	NACA 0012	Boom Length	25.12 in	Voltage Rating	50.4 V
Weight	0.6 lb	Cross Sectional Area	17.284 sq. in	Max Continuous Current	160 A
Stabilizer Area	390 sq. in			Weight	0.602 lb
Aspect Ratio	2.31			Controls	
				Transmitter	Spektrum iX12
				Receiver	Spektrum AR8020T
				Battery Weight	0.194 lb
				Servos	KST X10

5.1 Structural Characteristics and Capabilities

5.1.1 Structural Limits of Main Wing Spar

The maximum load factors experienced by the aircraft for each mission were computed using Equation (28), with a maximum velocity, V_{max} , of 95 ft/s, a maximum lift coefficient, $C_{L_{max}}$, of 1.19, and the total aircraft weight for each mission. These load factors are shown in Table 28.

$$n_{max} = \frac{L_{max}}{W} = \frac{\frac{1}{2}\rho V_{max}^2 S_W C_{L_{max}}}{W} \quad (28)$$

$$n_{limit} = \frac{2M_{max}}{lL^2} \quad (29)$$

Table 28: Maximum Load Factors

M1	6.84
M2	3.29
M3	6.48

To ensure the spar could withstand these loads, the maximum structural load factor, n_{limit} , was calculated based on the maximum moment, M_{max} , on the spar (modeled as a cantilever beam) which was calculated using FEA. Equation (29) was used to determine $n_{limit} = 12.4$.

5.1.2 Shear and Bending Analysis of Main Wing Spar

The maximum stress was calculated using Equation (30), with c being the distance from the neutral axis to the farthest point of the spar's cross-section. The value c was iterated over as different spar geometries were evaluated. I is the second moment of inertia, which was calculated for the hollow spar geometry using Equation (31). The value r represents the outer spar radius and r_1 is the inner radius.

$$\sigma = \frac{cM_{max}}{I} \quad (30)$$

$$I = \frac{2\pi r^4}{64} - \frac{2\pi r_1^4}{64} \quad (31)$$

The stress was then compared to the ultimate tensile strength of carbon fiber, which was estimated to be 500,000 psi from the lowest value found [18]. A hollow circle cross section was output as optimal with a resulting safety factor of 2.35. Using the calculated lift values on each wing rib, shear and moment diagrams were generated to act as visuals of how the wing load is distributed across the spar and are displayed in Figure 27.

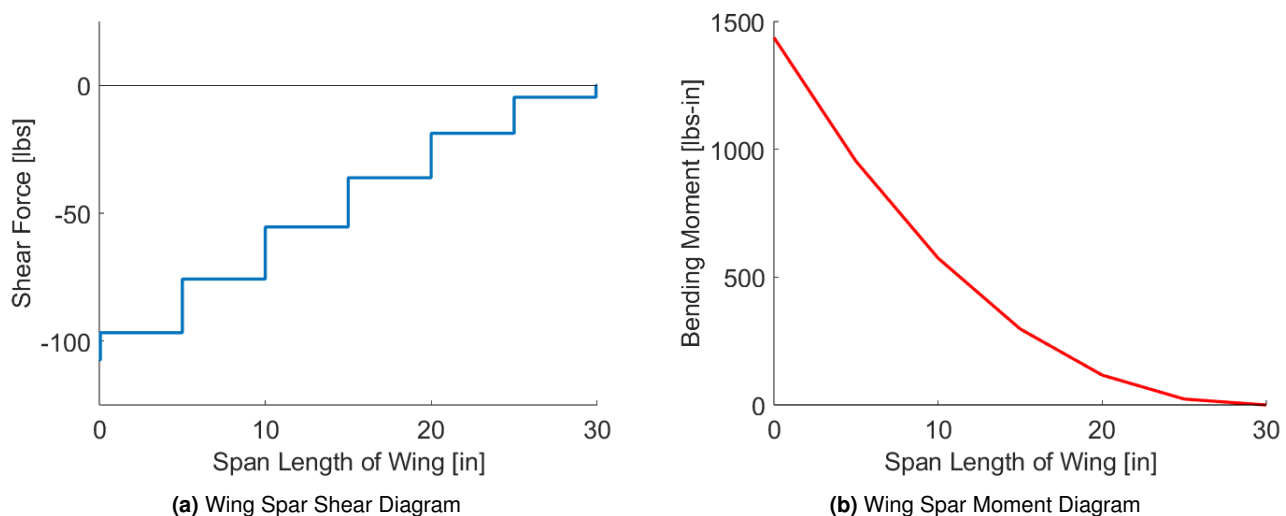


Figure 27: Shear and Moment Analysis of Spar During Maximum Loading

5.2 Subsystem Design

5.2.1 Fuselage Design

The fuselage is 32.5 in long with a Norman window cross section. The curved top was manufactured using balsa sheeting to optimize aerodynamics and weight. The side panels, floor, and platforms in the fuselage are constructed from 0.125 in lite plywood, which is stronger and less flexible than balsa. The fuselage is composed of three main compartments, with the front two compartments containing two levels and top hatches, while the third is one level and accessible by side hatch. The last seven inches of the fuselage tapers from a Norman window shape to the carbon-fiber boom tail. The front of the fuselage features a 3D printed nose cone that houses the motor and minimizes drag. The back of the fuselage has a similar cone to transition between the fuselage and boom tail.

5.2.2 X-1 Test Vehicle

5.2.2.1 Structural Design

The X-1 test vehicle, shown in Figure 28, is constructed using two carbon fiber plates, the top being 5.9 in x 1.4 in x 0.079 in and the bottom being 6.3 in x 1.4 in x 0.079 in. The plates are vertically connected and separated by four 0.4375 in threaded hex stand-offs, which are secured with 0.625 in head screws. Carbon fiber was selected as the construction material of the vehicle because of its superior durability and minimal weight, ensuring the vehicle is able to safely reach the designated landing zone and withstand landing impact. The top plate features two water jet-cut incisions for the release mechanism as well as a 1 in x 0.13 in horizontal incision to allow for the attachment of a magnetic proximity sensor for release confirmation. The pitch and roll of the test vehicle are controlled using two foam wings. A low wing was chosen to ensure the quarter inch clearance from the main aircraft wings. These wings feature a semi-rectangular planform with lifted wingtips, actuated at the quarter cord by two X08H KST Servos. The wings move symmetrically to control the pitch and asymmetrically to control the roll. The vehicle also includes a foam static horizontal and vertical stabilizer to improve flight stability. The vehicle features a custom-made printed circuit board (PCB), designed in Fusion and situated 0.598 in above the bottom plate for the autopilot system.

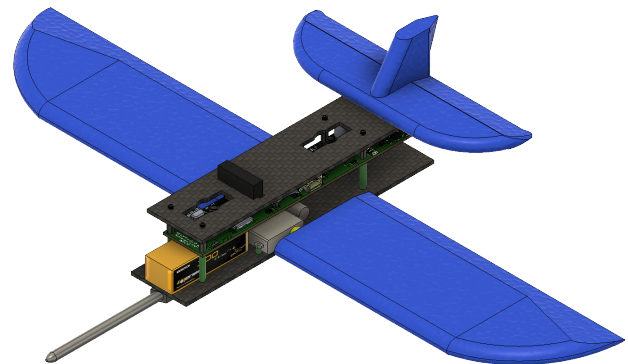


Figure 28: X-1 Test Vehicle CAD

5.2.2.2 Printed Circuit Board Design

The PCB consists of an inertial measurement unit (BNO085), a GPS (NEO-M9N), and a differential pressure sensor (ABP2) to gather rotational, groundspeed, and airspeed data, respectively. A magnet is mounted to the bottom of the fuselage so that a magnetic proximity sensor connected to the PCB can be used to detect the release of the X-1 test vehicle. A proportional-integral-derivative (PID) program was uploaded to the microcontroller that uses the sensors' data to turn on the strobe lights after the release, execute a 180 degree turn, and control rotational movement to autopilot the X-1 test vehicle to the 2.5-point bonus box. An SD card was included on the PCB to gather data during testing to be used to optimize the PID program.

The ESP32 Pico V3 microcontroller was selected due to its lightweight package, dual-core processor, and 160 MHz CPU. The package also has built-in flash memory and a crystal oscillator, reducing the area required for its peripherals. The PID program takes advantage of the two cores and the CPU speed by integrating a real-time operating system (RTOS). This allows data to be collected on one core and commands to be sent on another at a considerably high speed; this reduces latency and eliminates

any command blocking. Sensors send data to the ESP32 on an I2C bus, while an SD card logs the data on a separate SPI bus, also to reduce latency and command blocking. The PCB can be powered by a 5-volt USB-C connection during programming or by a two-cell LiPo battery during flight. A power multiplexor (TPS2121) selects the higher input voltage to power the board so that switching between power sources is automatic. Regardless of the input voltage used, a buck converter (TPS62142) steps down the input to 3.3V to power all sensors, the microcontroller, and data logging components. A 5V boost converter (MIC2875) steps up 3.3V to 5V to power the addressable LEDs used as strobe lights. A secondary boost converter (TPS55340) steps up the input to 8.4V to power the servos. Figure 29 shows the board layout and component grouping, and Table 29 describes the functions of each major component. The board has four copper layers. The first and fourth layers are used for signal traces, the second layer is used as the ground plane, and the third as a 3.3V plane since that is the most commonly used voltage on the board.

Table 29: Component Functions

Subsystem	Component	Function
Programming	CP2102	Converts USB-UART serial communication from programming device (laptop) to ESP-32.
	ESP32-PICO-V3	Microcontroller to collect and process data from sensors and control servos and LEDs.
Power	TPS2121	Power Multiplexor that selects the power input between the USB and Battery to power the board.
	TPS62142	Buck converter that steps down input voltage to 3.3V for all sensor and ESP32 logic/power.
	MIC2875	Boost converter that steps up 3.3V to 5V for addressable LED (strobe light) logic/power.
	TPS55340	Boost converter that steps up input voltage to 8.4V to power control surface servos.
	Lipierior 2S 300mAh	Two cell LiPo that powers the board during the mission.
Data	BNO085	Intertial Measurement Unit provides rotational movement data.
	NEO-M9N	GPS provides location and groundspeed data.
	ABP2	Differential pressure sensor that provides airspeed data when pitot tube is implemented.
	SD Card	Data logger for all sensor and control outputs.
Mission Compliance	WS2812B	Addressable LEDs used as strobe lights.
	Littelfuse 59140-030	Magnetic proximity sensor used to detect release of the glider.
Flight Control	KST X08H	Servos used to actuate wings.

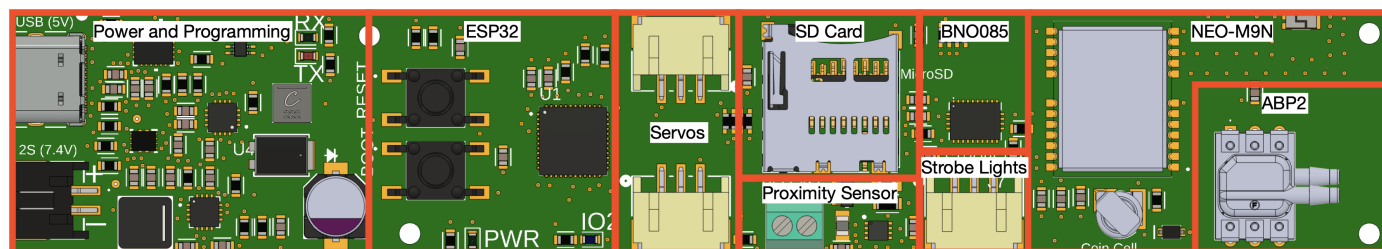


Figure 29: X-1 Test Vehicle Printed Circuit Board

The estimated current draw of the programming modules, sensors, LEDs, and servos is 1.3A; each power component has a current rating of greater than 3A to provide sufficient margin. Similarly, the LiPo battery has a continuous discharge rating of 19.5A. The passive components, such as resistors and capacitors, are all sized at 0603 since they use a relatively small surface area compared to all the active components, reducing the required board area and weight. Similarly, the GPS uses a ceramic chip antenna to connect to satellites because it is surface-mounted with a small package size, eliminating the need for a heavy, wired antenna. JST-PH connectors for the battery, servos, and strobe lights were selected, as they are sturdy enough to remain plugged in during all stages of M2 and M3.

5.2.2.3 Control Software Design

The flight control software running on the ESP32 microcontroller uses FreeRTOS tasks to maximize system performance by leveraging parallelism across both cores and is outlined in Figure 30. The system first performs an initialization routine that involves verifying that each of the ESP32's peripherals are functioning correctly. This includes the serial I/O, I2C bus for each

sensor (GPS, IMU, and airspeed), SD Card (SPI bus) data logging, strobe LED (initialized off), and both wing actuation servos. Servo actuation tests confirm the servo range of travel and allow for verification of the servo trim settings necessary to maintain a neutral (level wings) position. After initialization is complete, the microcontroller enters light sleep and suspends both CPUs until a high to low transition is detected on GPIO pin 4 to which the magnetic release detection sensor is attached.

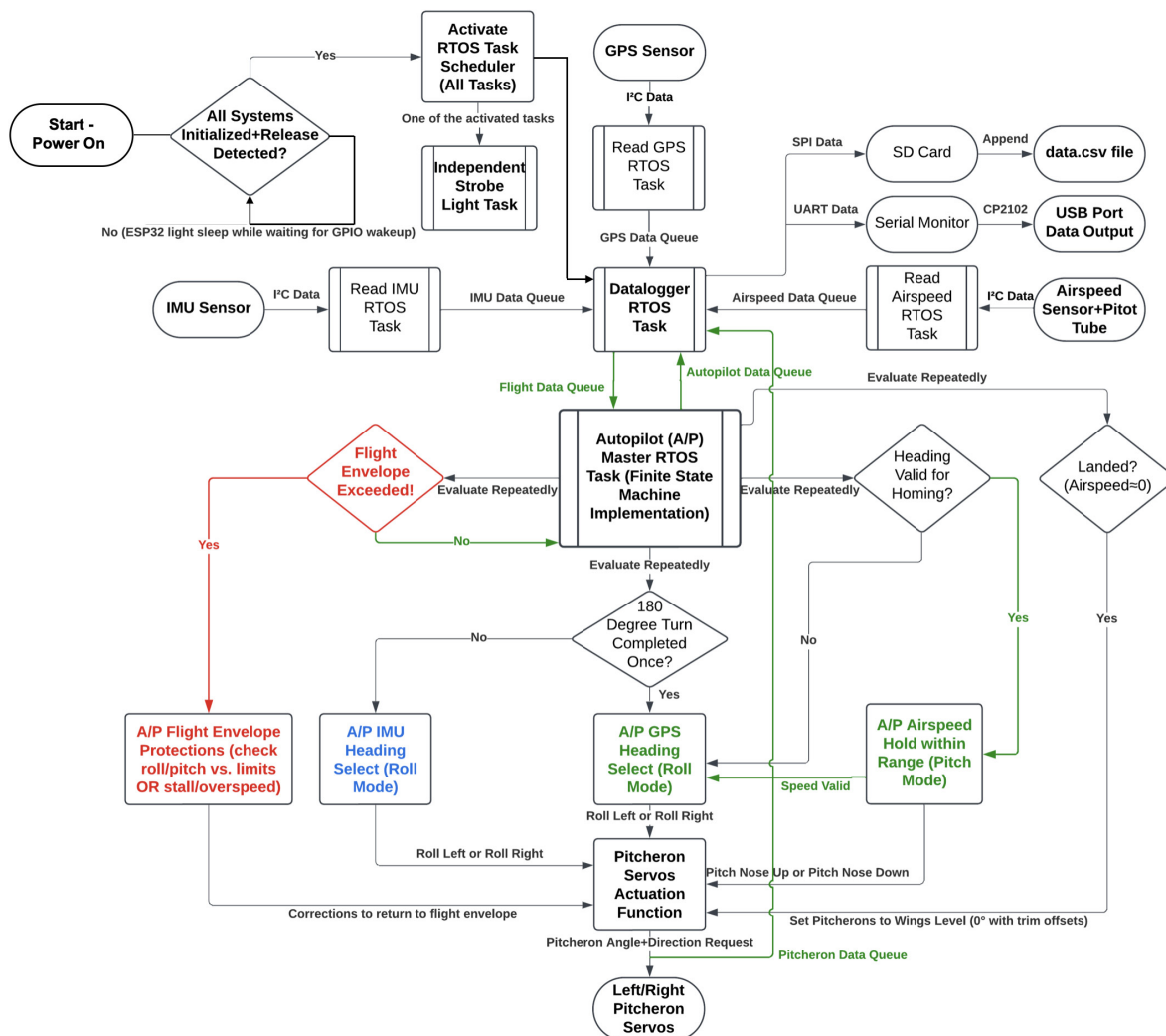


Figure 30: PID Control Logic Flowchart

Once release is detected, the following tasks are created and run: GPS reading, IMU reading, airspeed reading, datalogger, autopilot, and strobe light. The strobe light task operates independently and continues indefinitely after release is detected to ensure that the strobe LED continues flashing periodically after the glider has landed. Using FreeRTOS queues, sensor data is transferred to a central datalogger task that manages all data logging to the SD card (in a .csv file) and/or the serial monitor. It also routes all relevant data to the autopilot task. All sensors share an I2C mutex that only allows one sensor to access the I2C bus at a time to prevent conflicts. Additionally, after running once, each task is suspended until all other tasks have also run to prevent bottlenecks. After successful datalogging, all tasks are resumed before the next execution loop (with an optional delay).

The most important data collected from the IMU are roll, pitch, and yaw as Euler angles (normalized to 0-360 degrees). The differential pressure sensor data is used to calculate airspeed, v , according to Equation 32, where ΔP is the difference between dynamic and static pressure, and ρ is the air density updated before every flight based on local pressure, temperature, and relative humidity. The GPS provides current latitude, longitude, and heading data used in the autopilot for homing. It also provides ground

speed data that can be compared to airspeed data during post-flight analysis.

$$v = \sqrt{\frac{2\Delta P}{\rho}} \quad (32)$$

The autopilot master task is implemented as a finite state machine that uses the flight data during each execution loop to determine the current state of the aircraft. These states can be either "performing 180 degree turn," "GPS homing," "speed-pitch control," or "landed." Each state is associated with one of the following autopilot modes: IMU heading select, GPS heading select, or speed descent. In addition, all flight states (except for "landed") are associated with emergency flight envelope protection autopilot modes for roll, pitch, and airspeed. These have the highest priority and always activate whenever a flight envelope exceedance condition is met (regardless of state). They ensure that corrective action is taken to avoid exceeding a bank angle limit, pitch up/down limits, and stall and overspeed protections that use pitch adjustments to increase or decrease airspeed as required. The autopilot also sends data back to the datalogger task regarding the autopilot/flight modes, current wing angles, and target roll/pitch/bearing. Time since release detection in seconds and a line number are also logged for easier reference during post-flight data analysis.

The IMU heading select and GPS heading select modes are roll control modes that use the current bearing and other flight data to determine the roll angle required to achieve the target bearing. The IMU heading select mode uses the initial yaw angle detected on release and subtracts 180 degrees from it to determine the target yaw angle. The target roll angle is proportionally related to the difference between the current and target yaw angles. The wing servo actuation angles are also proportionally related to the difference between the current and target roll angles by adjusting the duty cycle of the PWM pulses. Once the 180 degree turn is complete and roll angle is near 0, the autopilot's GPS heading select mode is activated. This mode operates very similarly to the IMU heading select mode but uses GPS heading instead of IMU yaw angle. It continuously recalculates the target bearing, θ , with the current and target GPS coordinates using Equation 33. Equation 33 is based on the spherical law of cosines, where φ and λ represent latitude and longitude respectively, so that (φ_2, λ_2) and (φ_1, λ_1) are the current and target GPS coordinates respectively. The arctan function variant used, atan2, outputs a result that lies in the first two quadrants only (i.e. in the -180 degree to +180 degree range). The target bearing is normalized to the 0 to 360 degree range to match the GPS heading conventions used to determine the current heading. The only edge case is when the current and target coordinates are exactly equal, in which case the target bearing is made equal to the current bearing.

$$\theta = \arctan \left(\frac{\sin(\lambda_2 - \lambda_1) \cdot \cos(\varphi_2)}{\cos(\varphi_1) \cdot \sin(\varphi_2) - \sin(\varphi_1) \cdot \cos(\varphi_1) \cdot \cos(\lambda_2 - \lambda_1)} \right) \quad (33)$$

The only regular pitch control autopilot mode is the speed descent mode, which only activates after the glider is on the correct trajectory towards the landing zone and within roll angle limits. It is always overridden by GPS heading select whenever the glider goes off-course, or by flight envelope protections, and has the lowest priority. This mode actuates both wings in the same direction with an angle proportional to the pitch error, which is itself proportional to the airspeed error. For instance, if the target airspeed is higher than the current airspeed, the target pitch will be negative (nose down) and the wings will tilt up proportionally to achieve this pitch angle. The proportionality constants and flight envelope limits are fine-tuned based on flight testing results. Lastly the autopilot's landed state is activated when an airspeed of 0 ft/s is detected and simply returns both wings to their neutral angles (wings level), disabling the autopilot to prevent unwanted actuations from draining the battery after landing. Strobe light blinking and data collection continue until the system is powered off or reset.

5.2.2.4 Attachment and Release

The X-1 test vehicle is attached to the main aircraft internally by two 3D-printed ABS plastic servo horns, screwed into two servos attached to the underside of a flat piece of wood. This entire platform and servo release mechanism system, shown in Figure 31, is removable, in compliance with AIAA rules. Each servo horn consists of a 0.4 in diameter, 1.287 in long cylinder with four 0.268 in x 0.3 in x 0.15 in tabs that clamp around the top plate of the X-1 test vehicle to secure it in place. When the X-1 needs to be released, the servos rotate the horns 90 degrees and the cutout on the top plate of the X-1 vehicle aligns with the shape of the servo horn, releasing the vehicle. When the release mechanism is inserted into the plane, the servo horns slot into cutouts on a battery platform placed 1.45 in above the floor of the plane's fuselage, and protrude through holes on the floor of the fuselage, allowing the glider to sit external to the plane. There is also a slot made for the vertical stabilizer of the X-1 test vehicle that allows it to protrude 0.3 in into the bottom of the fuselage. For M1 and the start of the ground mission, these holes are covered by pieces of wood hinged to the floor of the fuselage, acting as bomb bay doors to prevent open holes in the fuselage. This release mechanism design was selected because the servo horns are lightweight, minimally impede the airflow underneath the main aircraft, and facilitate a secure attachment of the test vehicle to the main aircraft. The release mechanism was strategically placed near the quarter chord of the wing to ensure that once the X-1 test vehicle was dropped, the CG of the aircraft would be minimally affected.

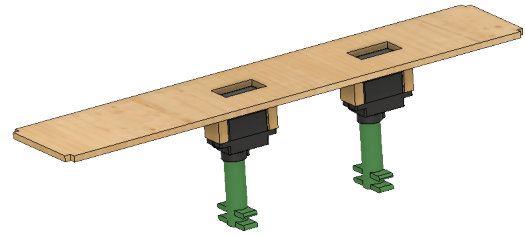


Figure 31: Release Mechanism System

5.2.3 Fuel Tanks

Two external fuel tanks carry a payload weight of 4 lbs each. Each fuel tank is a 24 fl-oz, commercially available HydraPak Insulated Polar Sport plastic bottle with height of 10.5 in and diameter of 2.75 in. For the fuel tank loading material, a commercially available material with a small particle size was needed to allow for uniform filling of the bottle. A density of approximately 4.96 slugs/ft³ was desired to satisfy the total target payload weight of 8 lbs. Type S Masonry Cement Mix, as a fine and dense material, was an initial option, but its density of approximately 2.53 slugs/ft³, as calculated by the team, is less than necessary. Thus, a combination of 1/2 oz and 1/4 oz steel weights was uniformly interspersed throughout the concrete to achieve the target weight. The team decided that this composite mix would allow for the most uniform fill and stability while still reaching the desired payload weight. To prevent material and weight shifting during flight, the bottles are filled completely.

5.2.4 Pylons Design and Finite Element Analysis

The pylons, shown in Figure 32, consist of a 3D-printed PLA with infused CF plastic part that encircles the fuel tank, a releasable zip tie that encircles the fuel tank at the point of its lowest diameter, and a 4.5 in carbon fiber rod that inserts into the main wing spar. The 3D-printed piece consists of two rings of 3.05 in inner diameter, 1 in width, and 0.2 in thickness that are connected on the outside by a 7.75 in long beam. The rings are spaced apart by 5.95 in at their closest point and fit around the body of the fuel tank, completely encircling it. The pylon is attached to the main aircraft by a 4.5-in long rod centered on the pylon's beam, which inserts into the aircraft's

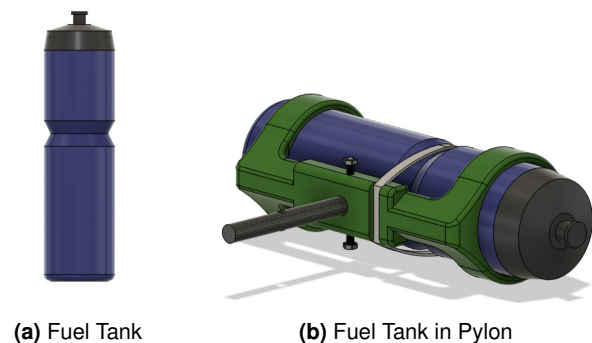


Figure 32: External Fuel Tank System

main wing spar. The rod and spar each have a hole, into which a 0.1875 in steel clevis pin is placed to secure the pylon to the wing, prevent rotation, and comply with the removability requirement of the pylon. A 3D-printed ABS alignment piece is glued to the main wing spar to assist with the clevis pin placement during M2 assembly. FEA was conducted to ensure that the pylons would not break or become damaged under the payload weight of 4 lbs per pylon, with the results shown in Figure 33. The safety factor from the study was 19.3.

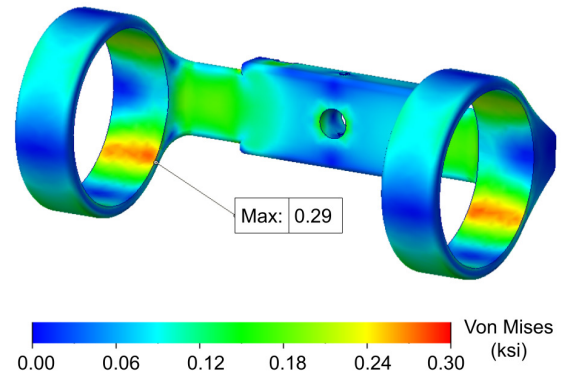


Figure 33: Pylon FEA Under 4 lb Load

5.2.5 Wing Design

The wing has a span of 64 in and a chord of 16.75 in. The wing is built of ribs spaced 5 in apart and wrapped in MonoKote, which acts as the skin of the wing. The ribs and MonoKote transfer flight loads onto the spar, as seen graphically in Section 5.1.1. A MATLAB script [2] was used to evaluate different spar shapes and dimensions, returning options that would withstand the given moments from the lift force and the payload weight. The lightest commercially available spar that still met cost requirements was selected. The chosen wing spar is a twill carbon fiber tube with an outer diameter of 0.591 in and a wall thickness of 0.059 in.

5.2.5.1 Wing Spar Finite Element Analysis

FEA was conducted on the spar to further ensure its structural capability for the missions. The mode of failure for the spar was selected to be where the composite begins to lose stiffness, which generally occurs at 77% of its ultimate tensile strength [19], which results in a yield strength of 231 ksi given an ultimate tensile strength of 300 ksi [20]. A static study is shown in Figures 34 and 35, with the spar fixed at the root for the depth it extrudes into the wingbox. The maximum downward deflection occurs when the aircraft is on the ground for Mission 2, with a downward force of 4 lb at the wingtip, modeling the fuel tank and pylon weights. The resulting safety factor from this study is 11.3, with a deformation of 0.567 in at the wing tip. The maximum upward deflection occurs when the plane is in a sustained turn for Missions 1 and 3, with a vertical pressure across the spar length at each of the wing ribs, modeling the force from lift. The maximum von Mises stress occurs at the wing root and is 57.9 ksi, yielding a safety factor of 3.9.

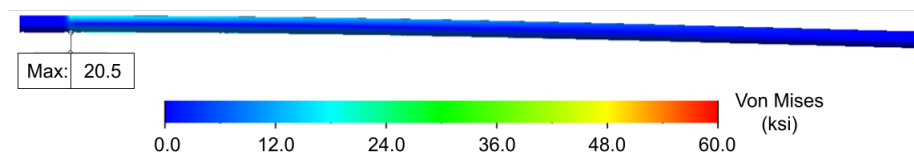


Figure 34: Spar Deformation During Maximum Downward Deflection

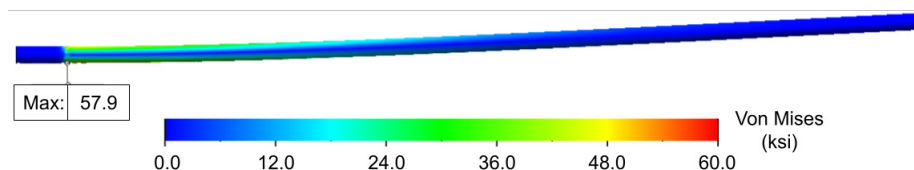


Figure 35: Spar Deformation During Maximum Upward Deflection

5.2.5.2 Wingbox Design and Finite Element Analysis

The wingbox is a CNC machined piece of aluminum with holes on each side for the spars. To keep the spar from sliding out of the wingbox axially, 3D printed spacers were glued to the spars and locked into place within the wingbox using steel bolts inserted from the bottom and top. Figure 36 contains a visual of this mechanism. The wingbox also includes a five degree angle to create

the dihedral in the wings. This angle is reflected in the 3D printed spacer to ensure a tight fit with the bolts. FEA was also conducted for this component. To model the payload weight on the ground, an upward pressure of 4.0 psi was applied to the top of the spar hole on each side. Independently, a pressure of 149.7 psi was applied to the bottom half of the spar hole to model the lift wing load in a sustained turn. The two simulations, which were modeled at extremes had safety factors greater than 7.6.

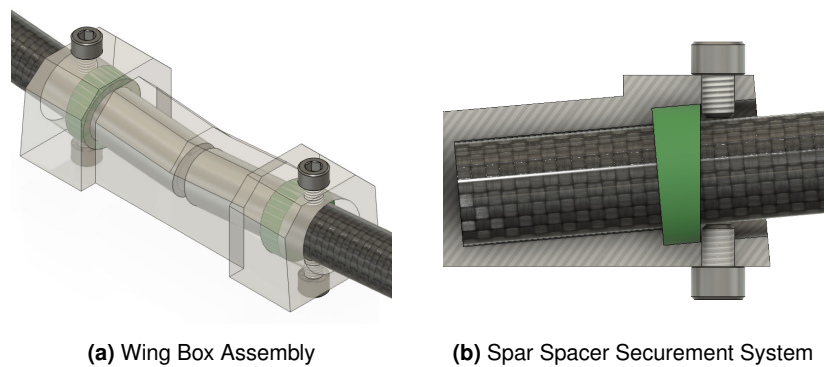


Figure 36: Wing Box

5.2.6 Empennage Mounting and Design

The empennage connection is a 3D printed part that connects the carbon fiber boom to the horizontal and vertical tails. The connection has a tube interface at its bottom, allowing the boom to slide in. It is then locked into place using two steel bolts that connect it to the empennage. The horizontal tail is built around the connector with carbon fiber and wood spars spanning the full stabilizer. The vertical tail is glued onto the top surface of the empennage and its spars extend 0.75 in into the connection, which allows the tail components to properly interface and remain supported during flight.

5.2.7 Boom Tail FEA

FEA was conducted on the boom tail to ensure that it would not break or excessively deform upon landing, when the maximum force is expected. Since the data acquired from the test flights did not output with a small enough time interval, the team was unable to approximate the force experience on landing. Instead, an overestimate of 100 lbs was used to simulate the system, which was then distributed across the area of the boom tail which houses the empennage. The maximum stress yields a safety factor of 2.7. The results are shown in Figure 37, while Figures 38 and 39 compare the empennage CAD to the as-built version.

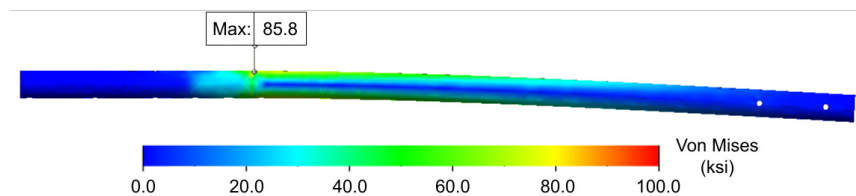


Figure 37: Boom Tail FEA Under 100 lb Load

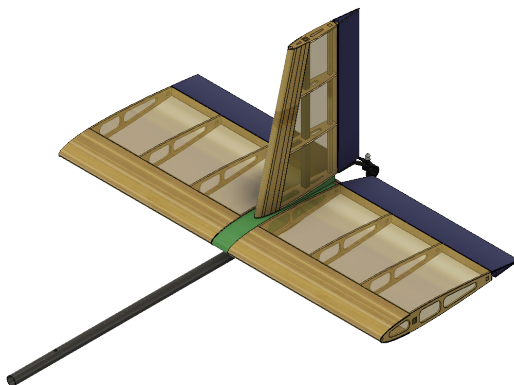


Figure 38: CAD of Empennage

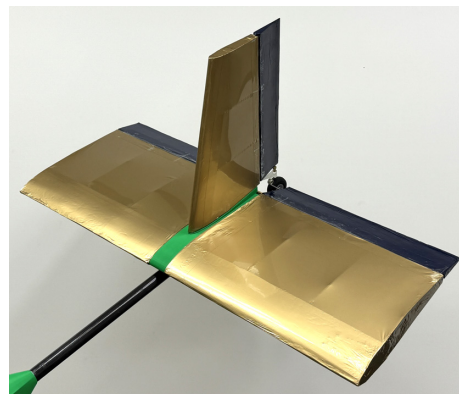


Figure 39: As-Built Empennage

5.2.8 Motor Mounting

The firewall is composed of two 0.08 in carbon fiber plates glued together, both slotted and glued into the fuselage. Carbon fiber was chosen over wood to ensure the integrity of the firewall under high load. The motor is secured to the firewall by four 0.25 in diameter bolts with lock nuts and washers on either side. Given the large thrust due to the motor, FEA was conducted on the firewall to ensure that the motor did not tear through or damage the plates. The deformation as a result of the maximum thrust of 32.2 lbs on the firewall is displayed in Figure 40 and has a safety factor greater than 30.

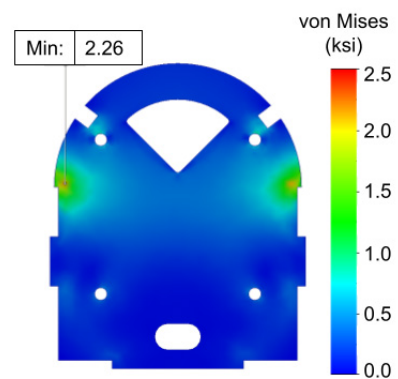


Figure 40: Firewall Stress

5.2.9 Landing Gear

The tail-dragger landing gear design consists of two components: the main front bow and the tail wheel. The team manufactured a carbon fiber front landing gear with a width of 20 in and a height of 10.5 in. It is located 8.5 in aft of the nose. The commercially available tail-dragger consists of a small wheel attached to a large spring that minimizes the impact on landing. The rear wheel assembly is attached to the rudder, allowing the plane to be steered on the ground without the need for an extra servo. FEA was conducted on the front landing gear to ensure structural integrity. The overestimate landing force of 100 lbs was distributed across both sides of the landing gear in the FEA simulation shown in Figure 41. This yielded a maximum stress of 16.8 ksi, which correlates to a safety factor of 13.75.

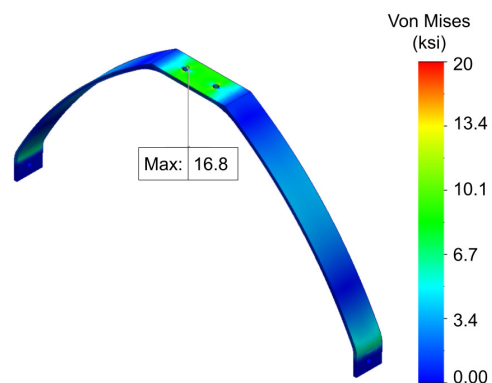


Figure 41: Landing Gear FEA under 100 lb load

5.2.10 Propulsion

Based on the scoring analysis results, a BadAss 6245 180KV motor was chosen with a BadAss 20x12 propeller and a MaxAmps 12s 2250mAh Li-Po battery. This configuration is rated to draw 67.5 amps of continuous current. During flight tests, the motor drew 51 amps continuous current, prompting the team to use a Castle Creations Phoenix Edge HV 160 electronic speed controller (ESC) that has a 160 amp continuous current rating, a mounted cooling fan, and a heat sink, which will help maintain the ESC's operating temperature when exposed to the hot conditions of the Arizona desert. A summary of the propulsion configuration is shown in Table 30.

Table 30: Detailed Propulsion Design

Motor	ESC	Battery	Propellor
Badass 6245 180kV	Phoenix Edge HVF 160	MaxAmps 12S 2250mAh Li-Po	Badass 20x12

5.2.11 Avionics

The Spektrum IX12 transmitter and the Spektrum AR8020T receiver were used because of their reliability and were already purchased in previous years. The 8-channel capability of the receiver also allows for the inclusion of the X-1 test vehicle release mechanism on this setup without the use of external modules. The KST X10 servos were selected for all control surfaces as they have a 150 oz-in torque rating, meeting the 82 oz-in torque requirement as outlined in section 4.2.6. KST DS113MG V6.0 servos were selected for the X-1 test vehicle release mechanism as they are relatively inexpensive, lightweight, and can withstand the maximum torque requirements experienced during flight. A GensAce 2s2p 2500 mAh 7.4V battery was selected to power the receiver and all servos because it meets the 7.4V requirement of our servos, lasts the duration of the flights, and minimizes weight.

Battery voltage, current, and propeller RPM are all measured by the ESC's internal sensors. To test endurance and preserve battery health, the ESC is programmed to have a "soft cutoff" when each battery cell reaches 3.5V. To validate the aerodynamic performance calculated in the preliminary design, a data collection module was integrated into the aircraft to gather airspeed, groundspeed, and rotational data during test flights. The team repurposed an extra X-1 test vehicle PCB since it had the necessary hardware for the data logging mentioned and a lightweight design. This module is powered by a two-cell 350mAh LiPo. A summary of the avionics is outlined in Table 31.

Table 31: Avionics

Transmitter	Receiver/Servo Battery	Control Surface Servos	Release Mechanism Servos
Spektrum IX12	GensAce 2s2p 2500 mAh 7.4V	KST X10	KST DS113MG V6.0
Receiver	Voltage, Current, RPM	Airspeed, Groundspeed, Rotational Movement	
Spektrum AR8020T	Phoenix Edge HV 160	Repurposed Custom X-1 Test Vehicle PCB	

5.2.12 Wiring

Power from the receiver battery is delivered to the servos from the built-in power delivery circuitry in the receiver using ribbon wires. There is a switch between the receiver battery and the receiver to comply with the rules. The receiver also transmits pulse-width modulation (PWM) signals through the same ribbon wire connection to control the servos and ESC based on inputs from the pilot's transmitter. The motor battery is connected to an inline 100A fuse and then to the ESC, the latter delivering a 3-phase signal to the motor based on the PWM signal from the receiver to control propeller spin speed. To reduce voltage drops and joule heating in the cables, 10 gauge wires were used from the propulsion battery to the ESC, and components were generally placed as close to each other as possible to reduce wire length. Figure 42 shows the physical connections of the flight-dependent avionics. Red and black lines represent power delivery wires (power and ground). Yellow lines indicate a PWM signal wire. The green lines correspond to a 3-phase power signal from the ESC to the motor. Two servos are used for the X-1 test vehicle release mechanism. Four servos are used for the control surfaces; one for each aileron, the elevator, and the rudder.

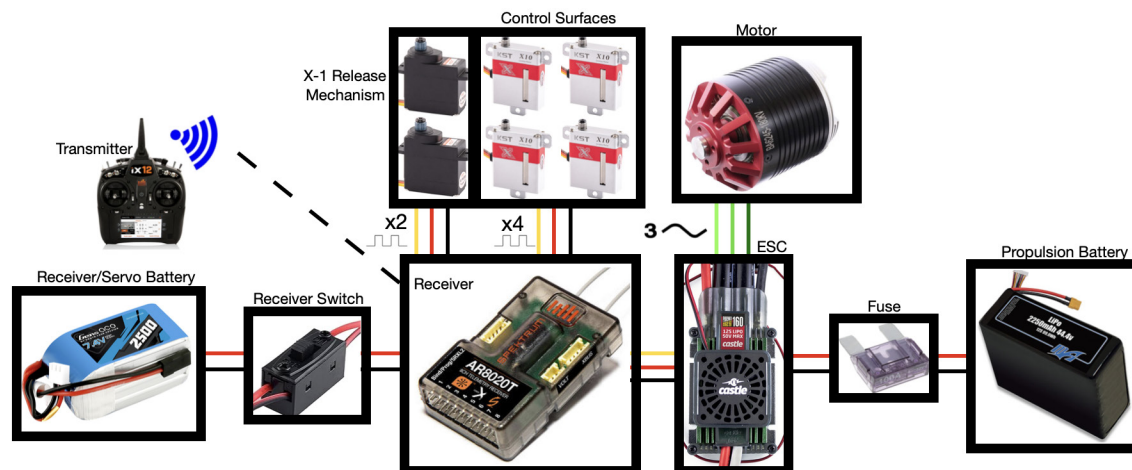


Figure 42: Wire Routing Diagram

5.3 Weight and Balance

Using Equation (34), values for the center of gravity across all missions were determined mathematically and verified using XFLR5 analysis. This equation sums the product of the weight W_i and position x_i of each component i for N_c components, then divides this sum by the total weight W to determine the position of the center of gravity X_{cg} . The x-axis aligns with the longitudinal axis, which is defined as the axis that runs from the propeller to the tail, with the front most tip of the plane marking the origin.

$$X_{cg} = \frac{\sum_{i=1}^{N_c} W_i x_i}{W} \quad (34)$$

The location of the center of gravity is crucial to the stability of the plane. The team's general goal was to position the center of gravity slightly ahead of the neutral point to ensure slight forward pitching moments at low angles of attack, as discussed in Section 4.3. This was achieved by using CAD to easily move components around in various fuselage designs until an optimal placement was achieved that still allowed for minimal fuselage size. The team then confirmed these locations would be feasible given various constraints, such as wire lengths and hatch placements, before finalizing this design. The component placements for each mission are shown in Table 32, along with component masses and each mission configuration's center of gravity.

Table 32: Weight and Balance

Component	Mass (lbs)	Placement along X-Axis (in.)		
		M1 (Empty)	M2	M3
Wing	1.60	10.40	10.40	10.40
HTail	0.35	62.00	62.00	62.00
VTail	0.15	65.00	65.00	65.00
Prop	0.16	1.00	1.00	1.00
Motor	2.70	4.50	4.50	4.50
ESC	0.57	10.50	10.50	10.50
Propulsion Battery	1.44	21.10	21.10	21.10
Servo Battery	0.20	24.50	24.50	24.50
Receiver	0.04	24.50	24.50	24.50
Fuselage and Boom Tail	1.20	19.00	19.00	19.00
Front Landing Gear	0.70	10.50	10.50	10.50
Wing Box	0.50	14.59	14.59	14.59
Payload Pylons	0.65	14.59	14.59	14.59
Payload Fuel Tanks	7.70	N/A	14.59	N/A
X-1 Test Vehicle	0.55	N/A	14.79	14.785
Total Weight (lbs):		9.60	18.50	10.80
CoG (in.) :		15.95	15.28	15.89

5.4 Final Flight and Mission Performance Parameters

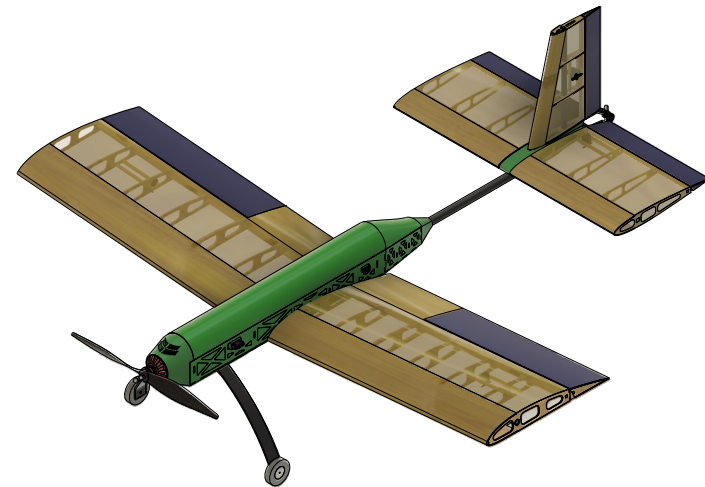
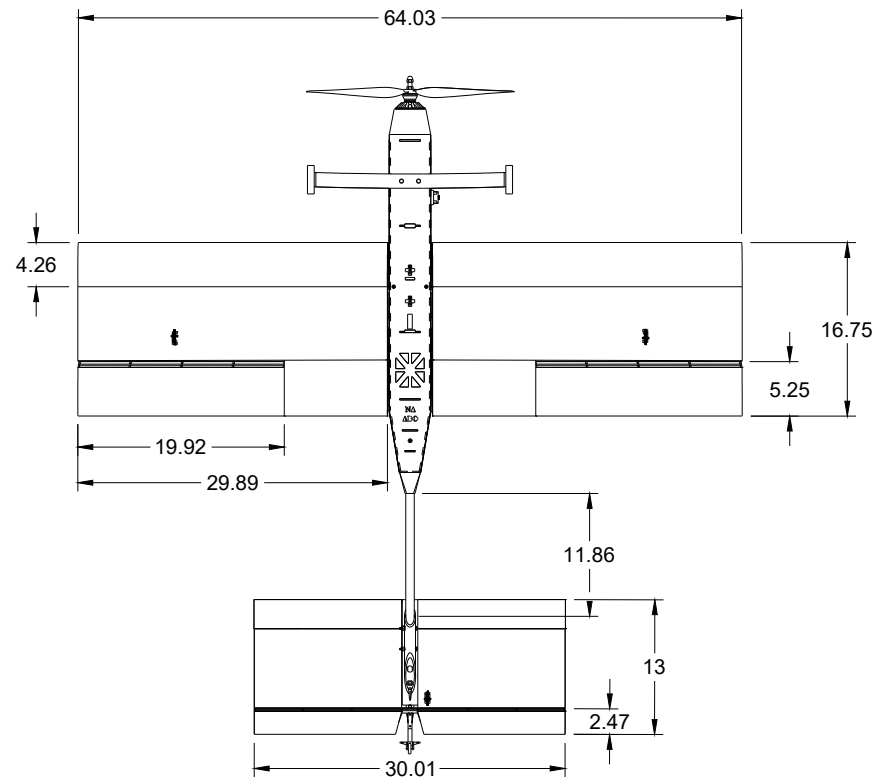
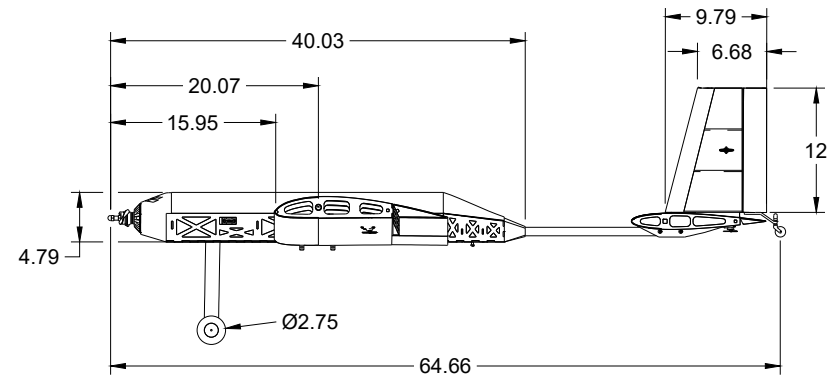
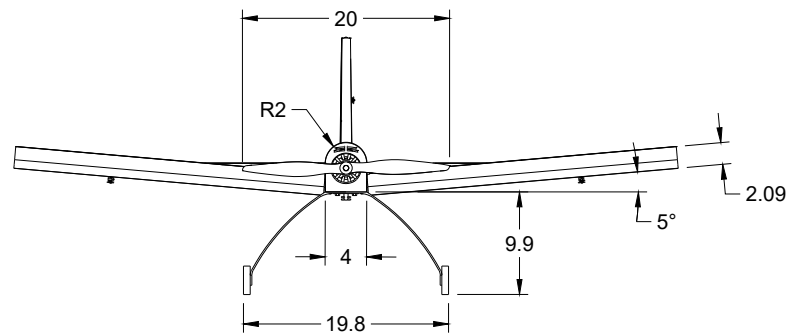
Table 33 lists the predicted performance of the aircraft's final design. These values were collected using a custom MATLAB performance code and the aforementioned scoring analysis code from Section 3.2.

Table 33: Final Scoring Parameters

Parameter	Plane Weight (lbs)	Payload Weight (lbs)	Max Velocity (ft/s)	Cruise Velocity (ft/s)	Predicted Flight Velocity (ft/s)	Lap Time (s)
M1	9.09	0.00	101	53.0	96.0	24.5
M2	18.5	9.41	100	78.8	95.0	26.2
M3	9.59	0.55	101	56.5	96.0	24.6

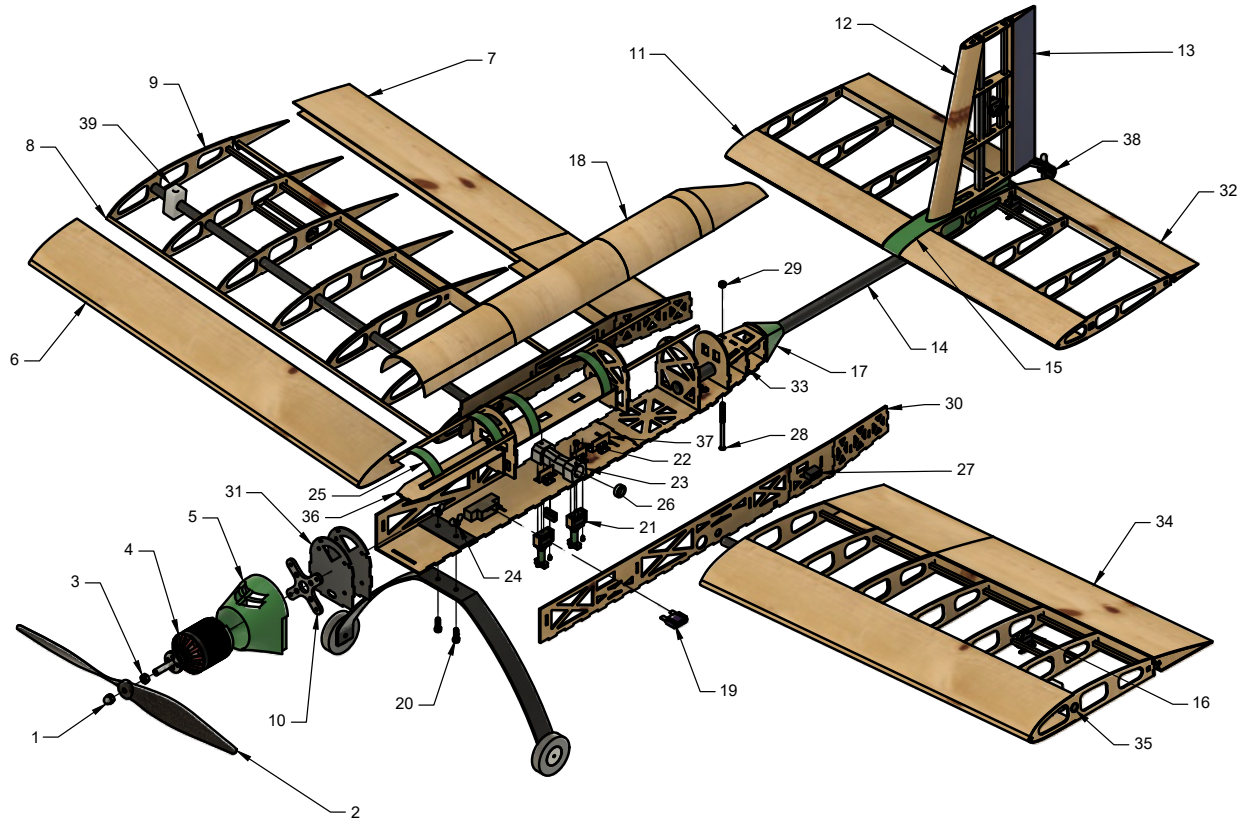
5.5 Drawing Package

The team used Fusion CAD software to model the full aircraft design as presented in the following drawing package. The first drawing displays major dimensions of components from three standard viewing angles. Drawing 2 shows subsystems in an exploded view of the full aircraft. Drawing 3 details all mission payloads configurations, including both fuel tank pylons and the X-1 test vehicle. Drawing 4 contains detail views of payload systems.

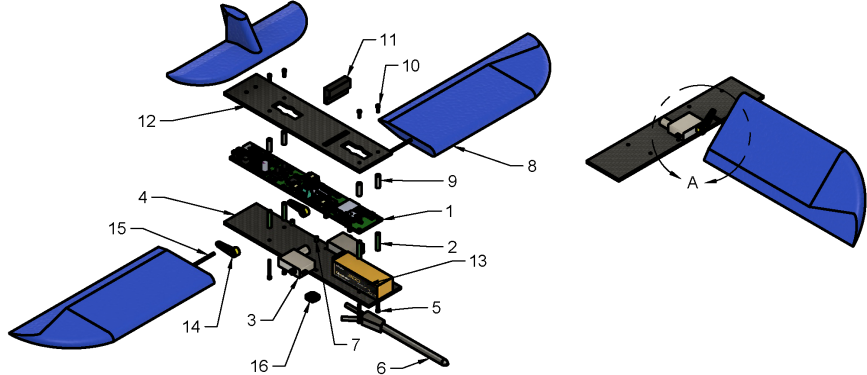


		All Dimensions in Inches	
		University of Notre Dame Design	
		Build Fly 2025	
		Configuration Drawing	
Size B	Drawing Package		
Scale 1: 12		Page: 1 of 4	

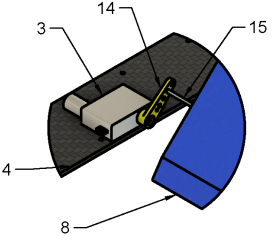
#	PART NAME	QTY	DESCRIPTION
1	Motor Cap Nut	1	Steel
2	Propellor	1	BA 20x12 EW
3	Motor Hex Nut	1	Nylon Locknut
4	Propulsion Motor	1	BadAss 6245-180Kv
5	Nose Cone	1	ABS
6	Leading Edge Sheeting	2	Balsa
7	Trailing Edge Sheeting	2	Balsa
8	Leading Edge Spar	2	Bass
9	Wing Rib	16	Lite Ply
10	Cross Motor Mount	1	Aluminum
11	Horizontal Stabilizer	1	CF/Lite Ply/Balsa
12	Vertical Stabilizer	1	CF/Lite Ply/Balsa
13	Rudder	1	Balsa/Lite Ply
14	Boom Tail	1	Carbon Fiber
15	Empennage Adapter	1	ABS
16	Servo	4	KST X10 Mini
17	Fuselage-Boom Taper	1	ABS
18	Fuselage Sheeting	5	Balsa
19	Fuse	1	Maxi Blade Fuse
20	Landing Gear Bolt	2	Aluminum
21	X1 Release Mechanism	2	ABS/KST DS113MG
22	Wing Box Bolt	2	Steel
23	Wing Box	1	Aluminum
24	Landing Gear Wingnut	2	Aluminum
25	Hatch Supports	4	ABS
26	Wing Spar Spacer	2	ABS
27	Switch	1	Plastic
28	Boom Bolt	1	Steel
29	Boom Nut	1	Steel
30	Fuselage Wall	6	Lite Ply
31	Firewall	2	Carbon Fiber
32	Elevator	1	Balsa/Lite Ply
33	Fuselage Bulkhead	6	Lite Ply
34	Aileron	2	Balsa/Lite Ply
35	Main Wing Spar	2	Carbon Fiber
36	Electronics Platform	2	Lite Ply
37	Bomb Bay Doors	3	Lite Ply/Nylon
38	Taildragger Landing Gear	1	Micro Steerable Tail Wheel
39	Pin Alignment Piece	2	ABS



	University of Notre Dame Design	
	Build Fly 2025	
	Exploded View	
	Size B	Drawing Package
Scale 1:8		Page: 2 of 4

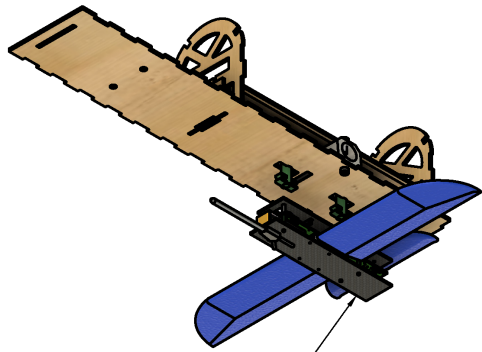


SCALE 1:4

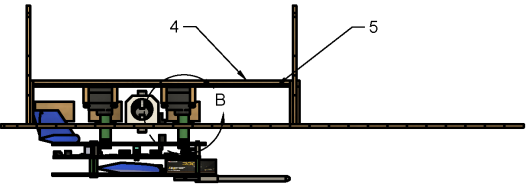


DETAIL A
SCALE 1:2

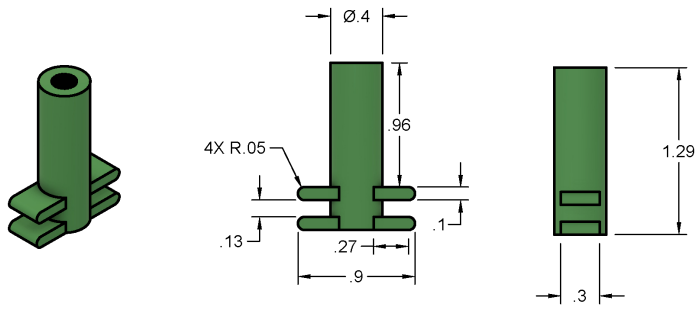
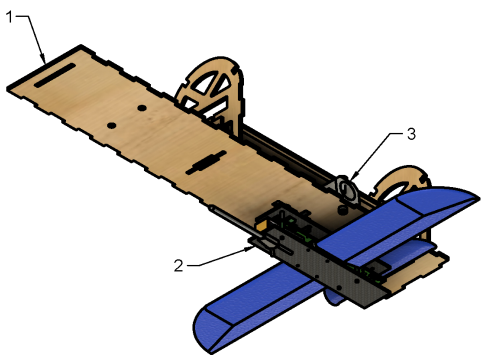
#	PART NAME	QTY.	DESCRIPTION
1	Circuit Board	1	PCB
2	Bottom Standoff	4	ABS
3	Servo	2	X08H KST
4	X-1 Bottom Plate	1	CF
5	Bottom Plate Screw	4	Steel
6	Pitot Tube	1	Aluminum
7	Servo Stand	8	ABS
8	X-1 Wing	2	Foam
9	Top Standoff	4	ABS
10	Top Plate Screw	4	Steel
11	Magnetic Proximity Sensor	1	Plastic and Steel
12	X-1 Top Plate	1	CF
13	X-1 Tail	1	ABS
14	Servo Arm	2	Plastic
15	Servo Rod	2	Aluminum
16	Strobe Light	1	WS2812B LED



Released X-1 Test Vehicle



DETAIL B
SCALE 1:2

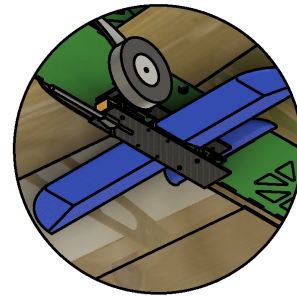
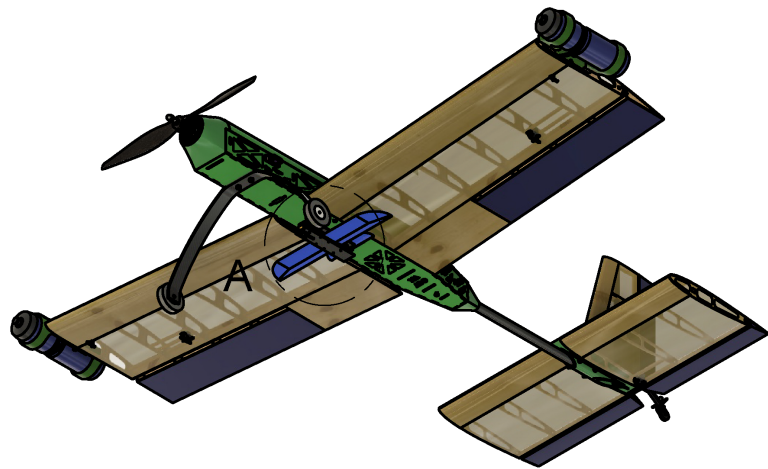


SCALE 1:1

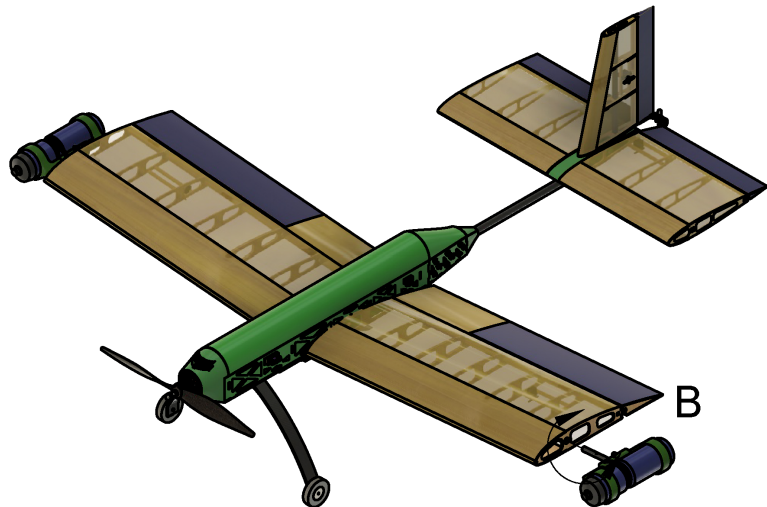
#	PART NAME	QTY.	DESCRIPTION
1	Bottom of Fuselage	1	Lite Ply
2	X-1 Test Vehicle	1	See Bespoke Drawing
3	Wing Box	1	Aluminum
4	Removable Wing Box Platform	1	Lite Ply
5	Permanent Wing Box Platform	1	Lite Ply
6	Servo	2	KST DS113MG
7	Wooden Standoff	4	Lite Ply
8	Servo Horn	2	ABS
9	Bomb Bay Door Hinge	3	Lite Ply



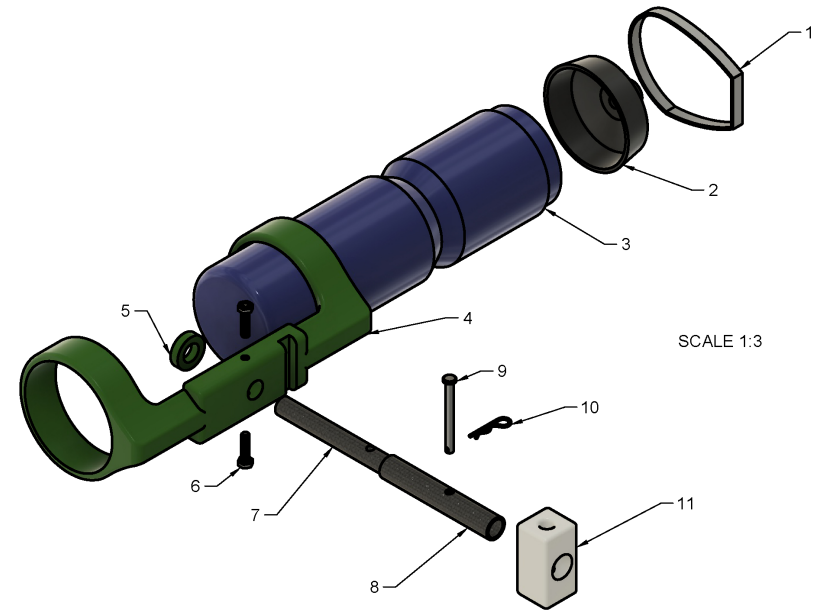
All Dimensions in Inches	
University of Notre Dame Design Build Fly 2025	
Payload Accommodation	
Size B	Drawing Package
Scale 1:5	Page: 3 of 4



DETAIL A
SCALE 1:5



DETAIL B
SCALE 1:5



SCALE 1:3

#	PART NAME	QTY.	DESCRIPTION
1	Zip Tie	1	Releasable, Plastic
2	Bottle Lid	1	Low-Density Polyethylene
3	Fuel Tank Bottle	1	24 fl oz HydraPak Insulated Polar Sport
4	Pylon	1	PLA W / CF
5	Spacer	1	PLA W / CF
6	10-32 Set Screw	2	Grade 8 Steel Hex Head
7	MECCANIXITY Rod	1	CF
8	Main Wing Spar	1	CF
9	Clevis Pin	1	18-8 Stainless Steel
10	Cotter Pin	1	Zinc-Plated Spring Steel
11	Pin Alignment Piece	1	ABS



University of Notre Dame Design
Build Fly 2025

Mission Configurations

Size B Drawing Package

Scale 1:12 Page: 4 of 4

6 Manufacturing Plan

Multiple manufacturing materials and methods were considered for each sub system and component of the aircraft. The evaluation and selection process for each major component is detailed below.

6.1 Manufacturing Processes Investigated

6.1.1 Laser Cutting

Laser cutting allows for precise machining of wood panels in two dimensions. The team chose to use plywood as the fuselage material for its sturdiness. The manufacturing process entails laser cutting sheets of plywood to obtain panels that easily interface with each other, and then attaching them with CA glue. The ability to precisely design parts is a strength of laser cutting, a versatile process that allows the team to achieve structural support with easily customizable pieces.

6.1.2 3D Printing

3D printing provides a method of creating complex shapes and geometries specified to exact measurements for each need. It is advantageous for creating task-specific items for the aircraft to interface with surfaces that may not be geometrically uniform, allowing the team to design a 3D printed prototype around exact measured interfaces instead of custom ordering a part from an outside source. The EIH offers a variety of 3D printer materials such as Acrylonitrile butadiene styrene (ABS), Polylactic acid (PLA) infused with CF, and PLA Aero. However, due to their weakness in relation to other materials, 3D-printed parts were not used for load-bearing parts of the plane.

6.1.3 CNC Milling

CNC milling is a high-precision machining process that can manufacture parts from a selection of metals, plastics, and foams. The mills offer versatility in their ability to machine complex geometries where other processes are limited in subtractive direction, material thickness, and precision. The EIH has a 5-Axis CNC Mill which allowed the team to manufacture the wingbox that includes the 5 degree dihedral. The CNC machining process for the wingbox is shown in Figure 43.

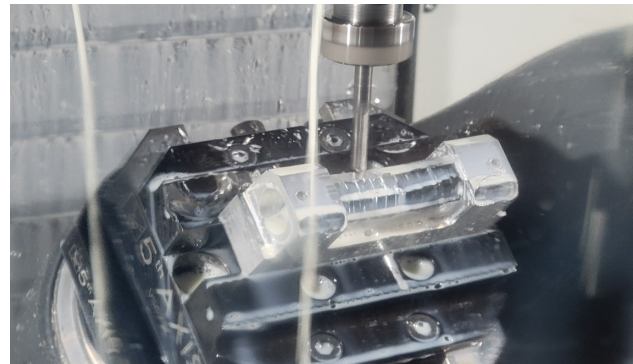


Figure 43: CNC Wingbox Machining

6.1.4 Water Jet Cutting

Water jet cutting is a type of manufacturing process that employs a stream of highly pressurized water to guide abrasive garnet sand to cut through different materials. Water jet cutters are useful for cutting high-strength materials in two dimensions with extreme precision, such as metals, ceramics, and carbon fiber. Since the EIH contains a water jet cutter, the firewall, body panels of the X-1 test vehicle, and front landing gear support plate were all water jet cut.

6.1.5 Composite Layup

Composite layup is a manufacturing process that creates composite materials by laying down alternating sheets of resin and material, such as carbon fiber, into a mold to create a composite structure. This process is repeated until the resulting material

has obtained the desired strength. The mold is 3D-printed from ABS plastic in order to guarantee the desired shape for the final part, and a polyvinyl alcohol coating is applied to the mold to ensure easy removal of the part from the mold. The mold and layup are then placed in a vacuum bag to guarantee proper adhesion and eliminate impurities with higher pressure. Although this process is labor intensive with many points of failure, which often causes defects in the product, a finished part can be custom designed and is high-strength and lightweight; ideal for a structure like the landing gear of the plane, which was created with this method of manufacturing.

6.1.6 MonoKote

MonoKote, a lightweight film that can be melded to a structure with heat, covering a frame to create an aerodynamic surface. It is attached with a hot iron and smoothed with a heat gun, which removes wrinkles by shrinking the film. The aerodynamic properties, portability, and quick application process of MonoKote are strengths of the material, making quick fixes to a plane easily accessible in case of rips or tears in its exterior coating.

6.1.7 Printed Circuit Board

Custom PCBs can be fabricated by JLC PCB. The surface-mount components are assembled onto the PCB using a reflow solder process and through-hole components are hand-soldered to the board. The PCB is assembled with tools available at Notre Dame's EIH. The reflow soldering process involves applying solder paste to the individual pads using a stencil purchased from JLC PCB and a stencil printer. A manual pick-and-place machine is then used to position the surface-mount components on their respective pads. Finally, the assembled board is placed into a reflow oven, where the soldering process was completed, securing the components in place.

6.2 Manufacturing Processes Selected

6.2.1 Wings

The wings are manufactured from commercially available carbon fiber spars, laser cut plywood ribs, balsa sheeting, and a MonoKote skin. This was chosen due to the teams confidence that this method would result in a wing close to the design target model. The wing structure is composed of seven 1/8 in laser-cut plywood ribs with 5 in spacing between each rib. Each rib has a 0.591 in diameter hole to fit the spar securely and is glued onto the circular main carbon fiber spar. Balsa sheeting was used to cover the leading edge up to the quarter chord on the top and bottom of each rib. Water was gently applied to the balsa wood to curve it around the leading edge to ensure a smooth airflow. An aileron was attached to each wing by applying packing tape securely to the top and bottom at the farthest possible actuation angles. Servo motors contained between the ribs connect a rod from the servo to the aileron. Finally, a layer of MonoKote covered the wings and ailerons.

6.2.2 Fuselage

The fuselage is made of laser cut plywood panels due to the ease of integration with wings, empennage, and mission payloads while ensuring adequate strength to sustain normal flight loads. The fuselage side panels, floor, and platforms were designed with puzzle-piece fits to prevent misalignment during plane assembly. The team used CA glue to connect the pieces together, assembling them tactically such that team members could hold them together as they quickly dried. Once the main structure of the fuselage was glued together, a sheet of balsa was cut to fit the top of the fuselage and curved over the top by dampening the wood. CA glue was applied to the connection points between the plywood structure and top sheet, which was then held down

as the glue dried to complete the main structure. The team finalized the fuselage structure by installing three hatches – one on the side and two on the top – before covering the properly assembled fuselage in a layer of MonoKote. The curvature of these hatches was supported by 3D printed ABS plastic arches glued to the top hatch.

6.2.3 Landing Gear

The front landing gear bow was custom built using a carbon fiber wet-layup due to the unavailability of commercial landing gear that met the weight and clearance requirements of the design. The landing gear is made out of twill weave carbon fiber and foam tires. The carbon fiber frame is 9 in tall and 18.5 in wide. Based on the FEA performed, it was determined that 15 layers of carbon fiber at the connecting surface between the landing gear and the fuselage and 7 layers at the wheel locations were required to withstand the impacts of landing with maximum payload. Two aluminum shear bolts are used to secure the landing gear to the bottom of the fuselage. In the event of a crash, the bolts shear to diminish the damage to the fuselage and landing gear. Two plastic wheels with foam tires add 0.75 in to the overall height of the landing gear, bringing the height to 9.75 in. Steel wheel shafts and shaft collars are used to mount the wheels to the carbon fiber frame.

6.2.4 Boom Tail

The boom tail consists of a long carbon fiber cylindrical tube with 0.82 in outer diameter and 0.75 in inner diameter. It is aligned using bulkheads on the interior of the fuselage and secured with a bolt and nylon lock nut. At the aft portion of the boom tail, a 3D-printed empennage mounting mechanism is installed to provide support and attachment for the tail assembly. 3D-printing was chosen because it allows the design to exactly match the airfoil at the interface with the vertical and horizontal stabilizers. It is secured to the boom tail by two bolts with nylon lock nuts.

6.2.5 Wingbox

The wingbox is made of aluminum. The complex geometry, caused particularly by the 5 degree wing dihedral, was designed using CAD to minimize weight while still structurally able to support the main wing spars, and then CAM was used to generate the machining plan. The CAM file was transferred to a 5-axis CNC mill where the wingbox was machined. The wingbox was adhered to the fuselage using CA glue and assisted by the bolts that secure the spar spacers.

6.2.6 Vertical Stabilizer

The vertical stabilizer is constructed using a rib-and-spar design, consisting of four equally spaced, laser-cut lite plywood ribs shaped according to the NACA 0012 airfoil. The main structural support for the vertical stabilizer is a 10 mm x 10 mm carbon fiber spar that runs through all the ribs and extends into the boom tail. There is a 0.25 in x 0.25 in balsa spar both on the leading edge and the rear of the ribs, to ensure a strong connection. To hold the shape of the airfoil across the length of the stabilizer, balsa sheeting is glued on the leading edge.

6.2.7 Horizontal Stabilizer

The horizontal stabilizer consists of two symmetrically constructed sections utilizing a rib-and-spar design. The ribs are equally spaced along a 10 mm x 10 mm carbon fiber spar, providing primary structural support. A 0.25 in x 0.25 in balsa spar is positioned at the leading edge to connect the ribs, while a 0.25 in x 0.50 in balsa spar is mounted at the rear of the ribs. To maintain the airfoil shape along the length of the stabilizer, balsa sheeting is glued to the leading edge.

6.2.8 Control Surfaces

Control surfaces are manufactured from laser cut plywood ribs, balsa sheeting, 1/4 in by 1/4 in bass spars, and MonoKote. The ribs are attached to the spar with CA glue and sheeting is placed on the top and bottom surface. The full part is MonoKoted and attached to the end of lifting surfaces with packing tape. The exception is the rudder, which additionally uses nylon hinges glued inside the spar to better withstand forces during landing since the tail wheel is attached to the rudder to simplify steering.

6.2.9 X-1 Test Vehicle

The top and bottom plates of the X-1 test vehicle were cut from a larger 0.079 in thick CF plate using a water jet cutter. Carbon fiber was selected as the vehicle material, over XPS foam, because of its superior durability, minimal weight, and reliable manufacturability, ensuring the vehicle is able to safely reach the designated landing zone. The wings and tail were repurposed from a purchased foam X-1 test vehicle to fit the controllability needs of the vehicle, and the four X-1 test vehicle stands were 3D-printed using ABS plastic. A control system was implemented on a custom designed and manufactured PCB, and all components were joined together with specifically purchased hardware. The final assembled test vehicle is in Figure 44.

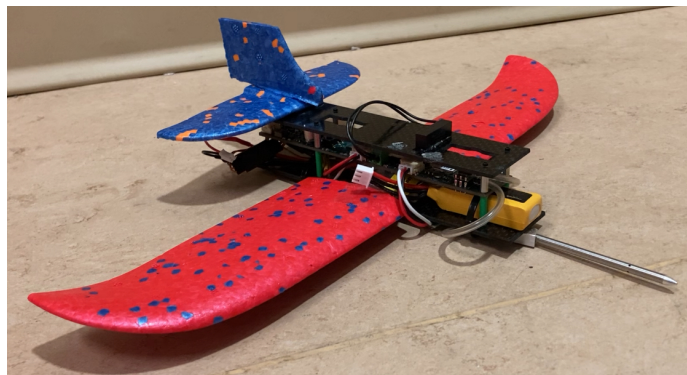


Figure 44: As-Built X-1 Test Vehicle

6.2.10 Fuel Tanks and Pylons

Two 24 fl-oz HydraPak Insulated Polar Sport bottles serve as the fuel tanks and are filled with a combination of steel weights and powdered concrete to achieve the desired payload weight of 8 lb. The two encircling parts and connected bridge piece of the pylon are printed using PLA with CF. This piece is attached to a CF rod by two steel set screws and a PLA with CF spacer. The use of PLA with CF allows the pylons to be custom-made to enclose the 24 fl-oz fuel tanks and minimize any gaps. This is in contrast to other manufacturing processes, which would not allow for customization and a close fit.

6.3 Process Summary

A summary of the selected material and manufacturing processes for each sub system and its components is shown in Table 34. Each aircraft component is divided into its constituent parts, and the materials and manufacturing processes are detailed for each part.

Table 34: Manufacturing Process Summary

Aircraft Component	Parts	Materials	Processes
Fuselage	Firewall	2mm Carbon Fiber	Water Jet Cutting
	Bulkheads	1/8" Lite Plywood	Laser Cutting
	Panels		
	Hatches		
	Sheeting	1/16" Balsa	Purchased (National Balsa)
	Skin	MonoKote	Purchased (Amazon)
Propellor Assembly	Propellor	BA Wood Electric 20x12	Purchased (Innov8tive designs)
	Cap Nut	M10 Stainless Steel Cap Nut	Purchased (McMaster-Carr)
	Nuts	M8 Aluminum Hex Nut	Purchased (Innov8tive designs)
	Washer	M8 Aluminum Washer	
Motor Assembly	Motor	Badass Motors 6245 180KV	Purchased (Innov8tive designs)
	Motor Shaft	Badass M10 Threaded Bolt-on Prop adapter	
	Motor Mount	Badass 62mm Cross Mount	
	Mounting Bolts	Aluminum Socket Head Screw 10-32 1 1/4"	Purchased (McMaster-Carr)
	Nuts	10-32 Locknut	Acquired in Past Year
Fuse Assembly	Fuse Mount	Scosche Maxi Fuse Holder	Purchased (Amazon)
	Fuse	Scosche 100A Maxi Fuse	
Wings	Spars	12mm ID x 15mm OD Twill Weave Carbon Fiber Tube	Purchased (Clearwater Composites)
	Ribs	1/8" Lite Plywood	Laser Cutting
	Sheeting	1/16" Balsa	Hand Sawed
	Servo Mount	0.25" x 0.25" Basswood Spars	
	Control Horn Mount	0.25" x 0.5" Basswood	
	Skin	MonoKote	Purchased (Amazon)
Wing Box Assembly	Wing Box	1 1/4" x 2" x 4" 6061 Aluminium Bar	CNC Milled
	Fastening Bolts	10-32 Stainless Steel Hex Screw 1/2" with Loctite	Purchased (McMaster-Carr)
	Spar Washers	ABS Plastic	3D Printed
Empennage	Boom Tail	0.75 x 0.82 x 72 Unidirectional Carbon Fiber Tube	Purchased (Rock West Composites)
	Boom Tail Housing	10-32 Stainless Steel Hex Screw 3 1/2"	Purchased (McMaster-Carr)
		10-32 Nylon Lock Nut	
		1/8" Lite Plywood	Laser Cutting
Vertical & Horizontal Stabilizers	Ribs	1/8" Lite Plywood	Laser Cutting
	Sheeting	1/16" Balsa	Hand Sawed
	Skin	MonoKote	Purchased (Amazon)
	Stabilizer Mount	ABS Plastic	3D-Printed
		10-24 Stainless Steel 1.5" bolt	Purchased (McMaster-Carr)
		10-24 Nylon Lock Nut	
	Spars	10 x 10 mm Pultruded Carbon Fiber Spars	Band Sawed
		1/4" x 1/2" Basswood Spar	Hand Sawed
Payload	X-1 Test Vehicle Plates	0.079 in Carbon Fiber	Waterjet Cutting
	X-1 Test Vehicle Tail		Purchased (Amazon)
	X-1 Test Vehicle Wings	X08H KST Servo	Purchased (McMaster-Carr)
	X-1 Test Vehicle Construction	3/16" Black-Oxide Steel Head Screws	
		5/8" Black-Oxide Steel Head Screws	
		7/16" Female Threaded Hex Standoff	
	X-1 Test Vehicle Release Mechanism	ABS Plastic Custom Servo Horn	3D-Printed
		KST DS113MG Servos	Acquired in Past Year
	Fuel Tanks	HydraPak Insulated Polar Sport Bottle	Purchased (Amazon)
		1/4 oz Adhesive Steel Wheel Weights	Acquired in Past Year
		1/2 oz Adhesive Steel Wheel Weights	
		Powdered Concrete	
	Pylons	PLA with infused Carbon Fiber	3D-Printed
		3/16" Stainless Steel Clevis Pin	Purchased (McMaster-Carr)
		Zinc-Plated Spring Steel Cotter Pin	
		MECCANIXITY 4.5" Carbon Fiber Rod	Purchased (Amazon)
Landing Gear Assembly	Main Landing Gear	Carbon Fiber Bow	Wet Layup
		2.75" Foam Wheels	Acquired in Past Year
	Mount	10-24 Nylon Bolts	Purchased (McMaster-Carr)
		10-24 Nylon Wing Nuts	
		1/8" Lite Plywood	Laser Cutting
	Rear Landing Gear	Steerable Tail Wheel	Acquired in Past Year

6.4 Manufacturing Milestones

The manufacturing schedule shown in Table 35 shows the manufacturing of all ND DBF aircraft in 2024-25. This includes both full redesigns and small repairs between flight testing days. Omitted components are reused from previous iterations, for example the wings were reused between Mk 1 and 2.

Table 35: Manufacturing Schedule

Manufacturing Schedule 24-25				Nov				Dec			Jan	Feb				Mar				Apr
Task	Start	End	3	10	17	24	1	8	15	26	2	9	16	23	2	9	16	23	W3	
Mark 1	11/4/2024	11/9/2024																		
Fuselage	11/4/2024	11/8/2024																		
Wings	11/5/2024	11/6/2024																		
Electronics	11/5/2024	11/9/2024																		
Vertical and Horizontal Stabilizers	11/6/2024	11/7/2024																		
Assembly	11/8/2024	11/9/2024																		
Mark 1.1	11/11/2024	11/15/2024																		
Landing Gear Repairs	11/11/2024	11/15/2024																		
Mark 2	12/2/2024	12/6/2024																		
Fuselage	12/2/2024	12/5/2024																		
Vertical and Horizontal Stabilizers	12/3/2024	12/6/2024																		
Payload	12/3/2024	12/6/2024																		
Electronics	12/4/2024	12/6/2024																		
Assembly	12/5/2024	12/6/2024																		
Mark 2.1	12/10/2024	12/19/2024																		
Landing Gear Repairs	12/10/2024	12/19/2024																		
Electronics	12/19/2024	12/19/2024																		
Assembly	12/19/2024	12/19/2024																		
Mark 3	1/26/2025	1/31/2025																		
Fuselage	1/26/2025	1/28/2025																		
Payload	1/26/2025	1/30/2025																		
Wings	1/27/2025	1/29/2025																		
Vertical and Horizontal Stabilizers	1/27/2025	1/28/2025																		
Electronics	1/28/2025	1/30/2025																		
Assembly	1/30/2025	1/31/2025																		
Mark 4	2/9/2025	2/16/2025																		
Fuselage	2/9/2025	2/14/2025																		
Vertical and Horizontal Stabilizers	2/10/2025	2/13/2025																		
Wings	2/12/2025	2/14/2025																		
Payload	2/14/2025	2/16/2025																		
Electronics	2/14/2025	2/16/2025																		
Assembly	2/15/2025	2/16/2025																		
Mark 5	3/17/2025	3/21/2025																		
Fuselage	3/17/2025	3/18/2025																		
Payload	3/17/2025	3/19/2025																		
Wings	3/18/2025	3/20/2025																		
Vertical and Horizontal Stabilizers	3/18/2025	3/19/2025																		
Electronics	3/20/2025	3/21/2025																		
Assembly	3/20/2025	3/21/2025																		
Competition	4/10/2025	4/13/2025																		

Winter Break

Legend
Planned Task
Actual Timing
Important Date

7 Testing Plan

Frequent testing is crucial to ND DBF's iterative design approach, ensuring that each iteration allows the team to learn as much as possible about the aircraft as constructed. To accomplish this goal, a schedule was laid out in Table 36. Tests include new manufacturing methods, propulsion, structural, payload, and flight tests. Following all tests, analysis is carried out to maximize information gained and applied to future designs.

Table 36: Test Schedule 2024-2025

Testing 24-25			Sept		Oct				Nov				Dec			Jan	Feb				Mar				Apr		
Task	Start	End	22	29	6	13	20	27	3	10	17	24	1	8	15		19	26	2	9	16	23	2	9	16	23	30
Design Analysis Testing	9/22/2025	3/22/2025																									
Material Testing	9/22/2024	10/26/2024																									
Pylon Testing	11/10/2024	2/1/2025																									
X-1 Vehicle Control Actuation Test	12/5/2024	2/1/2025																									
X-1 Vehicle Control Board Testing	11/1/2024	2/16/2025																									
Propulsion Testing	2/9/2025	2/26/2025																									
Mission Testing	11/10/2024	3/22/2025																									
Mk 1 Flight Test	11/10/2024	11/23/2024																									
Mk 2 Flight Test	12/7/2024	12/21/2024																									
PreFlight Assembly Speed Testing	2/15/2025	2/16/2025																									
Mk 3 Flight Tests	2/1/2025	2/8/2025																									
GM Testing	2/22/2025	3/30/2025																									
Mk 4 Flight Test	2/17/2025	3/1/2025																									
Mk 5 Flight Test	3/22/2025	3/22/2025																									
Competition	4/10/2025	4/13/2025																									

Winter Break

Legend

Planned Task

Actual Timing

Important Date

Legend

Planned Task

Actual Timing

Important Date

7.1 Aircraft Design Testing

7.1.1 Propulsion Testing

Propulsion testing is done to verify calculations based on manufacturer claims and the preliminary design estimates. Components are selected and aircraft are designed based off of manufacturer claims of thrust and current draw for battery and propeller combinations. Propulsion testing is conducted to verify these claims and learn about combinations listed in manufacturer data sheets. Data from propulsion tests is compared to manufacturer provided data to inform future design decisions. This year, the team decided to build a new thrust test stand, shown in Figure 45, since the previous thrust test stand was not rated for the significantly larger propulsion system that had been selected this year. This was used as an opportunity to design a stand that would allow for more precise data collection in addition to the improved structure of the stand.

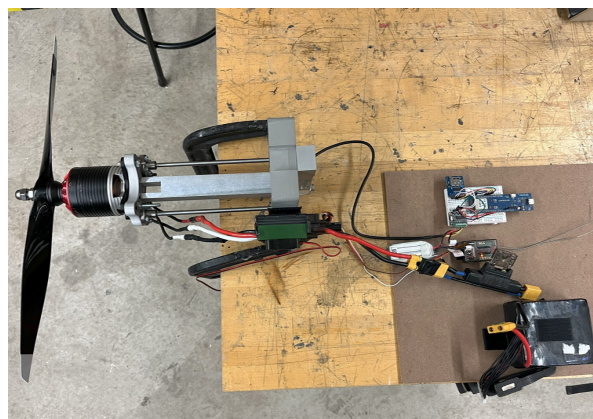


Figure 45: Static Propulsion Test Setup

7.1.2 Composite Layup Testing

Research over the summer months yielded information on wet layup methods of manufacturing carbon fiber and other composite materials. Following the procedure described in Section 6.1.5, a series of 1/2 scale fuselages were manufactured to refine details such as layers of PVA mold release and fiber weave.

7.1.3 Wing Tip Load Testing

Before each flight test, a wing tip load test was performed to verify the location of the CG in relation to the quarter chord. If the aircraft slowly tipped forwards, the aircraft passed the test and was deemed ready to fly, otherwise components were shifted slightly to adjust the CG and the test was repeated.

7.1.4 Fuel Tank Load Testing

For M2 and M3, the fuel tanks will be loaded at the wing tip of the aircraft. To ensure the structural stability the main spar, a load test was conducted in the M2, and M3 configurations along with a control test conducted with only the pylons installed. The

aircraft was placed on the ground and loaded in the above configurations. The maximum vertical displacement of the wing tip was recorded and compared to the theoretical value calculated in Section 5.2.5.1.

7.1.5 Payload Testing

Payload testing was performed by rapidly prototyping and testing designs for the fuel tank pylons and X-1 test vehicle release mechanism. The fuel tank pylons were originally 3D-printed using ABS plastic for the Mk 2 Flight Test, then with PLA plastic with infused CF for the Mk 3 and 4 Flight Tests. The servo horns for the X-1 test vehicle release mechanism were 3D-printed with ABS plastic. The pylons were evaluated by attaching them to the main wing spar and examining their performance under their own weight, while the release mechanism was evaluated by actuating the servos and initiating the release of the vehicle from a height of 3 ft. The X-1 test vehicle was attached to the main aircraft during the Mk 2 and 4 Test Flights, but it was not released.

7.2 Mission Testing

7.2.1 Assembly Testing

The team prepared for flight assembly by isolating the GM from the flight-line assembly for M1, M2, and M3. Team members were timed attaching the pylons to the wings and attaching the X-1 test vehicle to the release mechanism in order to identify the best performing Ground Crew for GM who is then given time to practice. The assembly process was carefully evaluated and adjusted to minimize the time required to mount components securely in accordance with competition scoring parameters and rules.

7.2.2 Flight Testing

The team used four aircraft for testing, each with modifications reflecting design iteration. The flight tests validated expected aircraft performance, simulated the competition missions, and identified areas of improvement for the aircraft design. The purpose of Mk 1 was to practice the manufacturing methods and verify flight performance in takeoff, stability, cruise and maximum speeds, and landing. The purpose of Mk 2 was to further validate flight performance, test an optimized landing gear configuration for improved landing, and simulate M2. The purpose of Mk 3 was to test a new wing and tail configuration to improve landing. Mk 4 was made to solve issues with the CG of Mk 3 that was discovered during manufacturing. The aircraft differences between each test flight are outlined in Table 37.

Table 37: Aircraft Features

Aircraft Features	Mk 1	Mk 2	Mk 3	Mk 4
Landing Gear Configuration	Tricycle	Taildragger	Taildragger	Taildragger
Boom Tail	No	No	Yes	Yes
Wing Span (in)	60	60	64	64
Taper Ratio	0.6	0.6	1	1
Aspect Ratio	6.49	6.49	3.82	3.82
Pylon Design	None	Under wing	Wing tip	Wing tip
X-1 Test Vehicle Control	N/A	Non-operational	Non-operational	Operational
1/4 Chord Location (in from nose)	18.28	18.28	14.6	20.69
Tail Location (in from nose)	48	48	62	51

7.3 Flight and Documentation Checklists

Prior to all flights, the ND DBF team conducted the Flight and Documentation Checklists shown below in Table 38 to ensure the safety of all members during flight testing. After completing the flight day and documentation checklists, the aircraft was ready to test, as displayed in Figure 46.

Table 38: Flight Checklists**(a) Flight Day Checklist**

PREFLIGHT		BEFORE TAKEOFF	
Assembly		Wind & Takeoff Direction	VERIFY/SELECT
Motor	SECURE	Receiver Switch	ON
Servos & Linkages	SECURE	Control Surfaces	FREE AND CORRECT
Vertical & Horizontal Stabilizers	SECURE	Propeller Area	CLEAR
Tail Mount	SECURE	Arming Fuse	ARMED
Boom Tail	SECURE	Throttle Cut Switch	OFF
Wings	SECURE	Propeller Spin (behind aircraft)	CLOCKWISE
Landing Gear	SECURE		
Propeller	SECURE	Camera	RECORDING
Wingtip Test	PASS/FAIL	Data Acquisition System	RECORDING
Avionics		Takeoff Area	CLEAR
Throttle Cut Switch	ON	Takeoff Intentions	ANNOUNCED
Transmitter	ON	Timer	ON
Receiver Connections	SECURE	BEFORE LANDING	
Receiver Battery Voltage	4.2V/Cell	Landing Area	CLEAR
Propulsion Battery Voltage	4.2V/Cell	Landing Intentions	ANNOUNCED
Receiver Battery	CONNECTED	AFTER LANDING	
Propulsion Battery	CONNECTED	Timer	OFF
Receiver Battery	SECURE	Throttle Cut Switch	ON
Propulsion Battery	SECURE	Receiver Switch	OFF
Data Logging Module	SECURE	Arming Fuse	DISARMED
Pitot Tube	SECURE	Propulsion Battery	DISCONNECTED
Avionics Access Hatches	SECURE	Receiver Battery	DISCONNECTED
Range Test	PASS/FAIL	Camera	OFF
Failsafe	PASS/FAIL	Data Acquisition System	OFF
Receiver Switch	OFF	Transmitter	OFF
Camera (if applicable)	SECURE	Receiver Battery Voltage	$\geq 3.5\text{V/Cell}$
Airframe		Propulsion Battery Voltage	$\geq 3.5\text{V/Cell}$
Payload (X-1 and/or Fuel Tanks)	SECURE	Aircraft Structure	DAMAGE FREE
Payload Access Hatches	CLOSED		
Landing Gear Roll	SMOOTH		
Control Inputs	TRIMMED & CENTERED		

(b) Documentation Checklist

DOCUMENTATION
PREFLIGHT DOCUMENTATION
CG Location of Avionics
Aircraft Weight
Receiver Battery Voltage
Propulsion Battery Voltage
Photographs of Internal Components
POSTFLIGHT DOCUMENTATION
Takeoff Distance
Receiver Battery Voltage
Propulsion Battery Voltage
Laptime
Total Mission Time
Structural Damage (if any)
ESC Max Current
ESC Continuous Current
Turn radius
Cruise Airspeed
Max Airspeed
Groundspeed
Comments From Pilot

**(a) Mk 1****(b) Mk 2****(c) Mk 3****(d) Mk 4****Figure 46: Aircraft During Test Flights**

8 Performance Results

Planned tests detailed in Section 7 were conducted and informed ND DBF on the performance of various subsystems of the aircraft as compared to simulated values.

8.1 Aircraft Design Test Results and Analysis

8.1.1 Propulsion Test Results

Propulsion tests completed by ND DBF in previous years have not been suitably accurate to give the team a notable edge in continuing the design of the aircraft. With the increase in expected static thrust of this year's propulsion selections, the team's thrust test stand was deemed a safety liability as there was little confidence it would remain structurally sound throughout a static thrust test. Because of this, a new thrust test stand was designed and built, as seen in Section 7.1.1, and yielded promising results. As seen in Figures 47 and 48, which plot average data points for continuous trials, the selected 20x12 propellor produced the most thrust with a lower diameter than the 21x10, which is critical for landing gear sizing to ensure propellor clearance. Furthermore, these thrust test values match closely with the predicted maximum thrust from the BadAss [11] data sheets, which listed a maximum static thrust of 32.3 lbs compared to the measured maximum thrust being 25.8 lbs. This lower value is likely due to the friction and imperfect alignments present in the thrust test stand, which can cause the thrust values to be reduced.

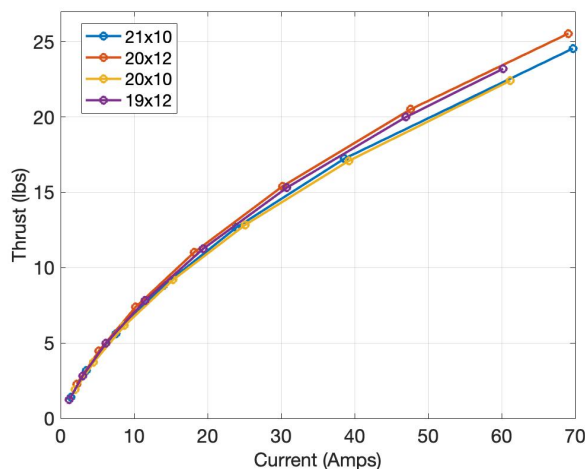


Figure 47: Static Thrust as a Function of Current

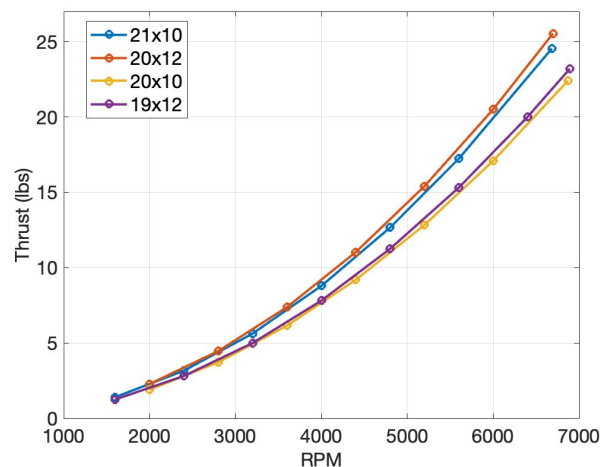


Figure 48: Static Thrust as a Function of RPM

8.1.2 Composite Layup Test Results

Initial tests resulted in high quality composite parts with minor defects such as wrinkles and pockets of epoxy. They were corrected in subsequent attempts by optimizing the mold design process and refining the vacuum bagging procedure. While the original plan for a carbon fiber fuselage was discarded, the process was used to manufacture the front landing gear.

8.1.3 Wing Tip Loading Test Results

Each aircraft passed the wing tip loading test before flight. Mk 1, 2, and 4 all were tested at the quarter chord, as is typical. For these prototypes, favorable balance was attained with minimal adjustments in component placement. However, Mk 3 had a neutral point 3.775 in behind the quarter chord, so the wing tip test was conducted from this location. Additionally, ballast was used for this flight to attain proper balancing after components of the tail were built heavier than expected, which the team used to more accurately predict the weight and balance of the Mk4 aircraft.

8.1.4 Fuel Tank Loading Test Results

Following the Mk3 test flight, the team conducted a fuel tank loading test. The deformation of the main spar at the tip is detailed for the following three configurations: loaded with only pylons, pylons and empty fuel tanks, and pylons with loaded fuel tanks in Table 39. For the pylons attached by themselves and with empty fuel tanks, there is negligible deformation in the main wing. The maximum deflection was achieved in the M2 configuration and was 0.52 in, which is similar to the expected deformation calculated in Section 5.2.5.1. This confirmed that the desired fuel tank weight would not damage the main wing spar.

Table 39: Vertical Deformation of Wing Tip During Fuel Tank Loading

Configuration	Vertical Deformation on Wing Tip (in)
Pylons Only	0.09
Pylon and Empty Fuel Tank	0.13
Pylon and Fully Loaded Fuel Tank	0.52

8.1.5 Pylon and X-1 Test Vehicle Test Results

The team initially designed a 3D-printed ABS plastic fuel tank pylon, consisting of a bridge piece permanently attached to the wing spar and two encircling pieces securing the fuel tank. A truncated conical ABS piece was screwed onto the front encircling piece, while steel pins connected the encircling pieces to the bridge. This design was successful in holding empty fuel tanks, but payload testing was halted when the Mk 2 Flight Test was cut short. After discovering the bridge piece's permanent attachment was not allowed according to the rules, the team redesigned the pylon using PLA plastic reinforced with carbon fiber. The original design featured a 3D-printed part that inserted into the wing spar that broke when loaded fuel tanks were added. An updated version featured a carbon fiber rod inserted into the wing spar and secured with a steel pin, with the 3D-printed pylon attached via a spacer and set screws. However, this design obscured the fuel tank's front and rear fascia, prompting another redesign. The current pylon, similar to the previous iteration, eliminates the screw-in mechanism. The fuel tank now rests in the pylon, secured by a releasable zip tie. The team successfully performed a static attachment test with fully loaded fuel tank on the aircraft's main wings. The X-1 test vehicle attachment mechanism was tested during the Mk 2 and 4 Flight Tests, but its release was not evaluated due to the flight tests being cut short.

8.2 Mission Test Results

8.2.1 Assembly Test Results

Assembly testing revealed an unexpected issue with placing the weighted fuel tanks on the wing tips. When only one loaded fuel tank is attached, the imbalance causes the plane to lean far to the loaded side if not supported. However, the fuselage is able to withstand the large torsion, and the spars and wing box remain sufficiently rigid during this process. Therefore, if the ground crew member supports the unloaded wing before placing the other fuel tank in the pylon, no damage is sustained and the plane is ready for flight. Additionally, this issue does not affect the time-critical GM because the fuel tanks may be empty for the mission. No other results of the assembly test were unexpected, and the team was able to identify the top performing ground crew member for competition.

8.2.2 Flight Test Results

Table 40 summarizes the team's flight testing thus far. Initial flight tests yielded successful performance in the air, determined largely by pilot feedback on aircraft handling and response times. However, these flight tests also faced consistent failures on landing. Though early failures were attributed to legitimate concerns of landing gear integrity, after the third failure on landing,

it was concluded that a major redesign was also necessary to allow for lower approach speeds and safe landings. Since this redesign, flight characteristics have been much improved, especially at low speeds. While explicit testing objectives of the most recent flights have not been met in most cases, clear strides have been made in aircraft handling and consistency. This is especially true of Mk 4, which successfully completed two flights in temperatures well below freezing and high winds that severely impacted propulsion and stability, respectively.

Table 40: Flight Test Results

Flight Test	Date	Airplane	Objectives	Results
1	11/10/2024	Mk 1	Verify aircraft stability and structural integrity of aircraft.	Experimental tricycle landing gear failed prior to takeoff.
2	11/23/2024	Mk 1.1	Verify aircraft stability and structural integrity of aircraft.	Successful flight despite large required trim adjustments. Nose gear failure on landing.
3	12/7/2024	Mk 2	Complete successful landing and test pylons and flight characteristics when loaded.	Plane rolled off runway on landing. Landing gear ripped off in the snow before loaded flight was attempted.
4	12/21/2024	Mk 2.1	Complete successful landing and test the release mechanism of the X-1 test vehicle.	Transmitter issues lead to failure of release mechanism and landing failed due to high stall speed of aircraft.
5	2/1/2025	Mk 3	Verify stability of updated design. Record Top speed.	Stable flight and successful landing achieved with aid of ballast to adjust CG. Sensor failure meant no top speed recorded.
6	2/8/2025	Mk 3.1	Record Top Speed.	High Speed crash due to poor elevator control. SD card with top speed data lost in wreckage.
7	2/17/2025	Mk 4	Test stability of updated fuselage without ballast. Record Top Speed. Test X-1 vehicle drop.	Stability verified. Ground speed recorded, however extreme cold adversely affected propulsion system, making reading of little use. Due to cold, the plane failed to climb to altitude required to drop X-1 test vehicle.
8	3/1/2025	Mk 4.1	Test flight characteristics with loaded pylons. Test X-1 vehicle drop.	Future Test
9	3/22/2025	Mk 5	Test all systems in competition simulation.	Future Test

8.3 Design Modifications

Modifications were made to the initial aerodynamic design of the plane to lower its stall speed to a flyable range. Though the first failure of the landing gear was caused by its weak design, subsequent landings with redesigned gears also failed. Using input from the team's pilot and data acquired from a reevaluation of the plane's flight characteristics, the team determined that the root cause of the gear failure was the high stall speed and extreme weight of the plane. These led to abrupt landings with unreasonably high stress on the landing gear, causing it to fail even when its design was strong. Following this realization, the team increased the plane's wing area and changed its shape from tapered to rectangular for Mk 3. These design changes sparked others: most notably, the relocation of the tail further aft and the introduction of a boom tail to replace superfluous fuselage. Construction of Mk 3 revealed errors made in CG calculations due to inaccurate estimates of tail weight. Before Mk 4, the team moved the wing backward and the tail forward to adjust the aerodynamic center of the plane.

8.4 Performance of Completed Aircraft

Flight testing has demonstrated the safe and successful performance of the main aircraft's most recent design, and fuel tank loading tests have proven that the aircraft can withstand the significant weights being carried in M2. While the X-1 test vehicle has yet to be dropped at full altitude, attachment mechanisms and control surface responses have been successfully demonstrated, and the team has planned multiple tests for the near future. The 2024-2025 University of Notre Dame DBF team is confident that the finalized aircraft design will be highly competitive and achieve a top score in Tucson, Arizona.

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